

Concept 1 | Arithmetic Sequences

English Student Edition • Focused formulas and guided practice

Formula opener

Term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Test

$$a_{n+1} - a_n = d$$

Worked Solutions

Arithmetic Sequences

Q001 Written

When the general term of the sequence $\{a_n\}$ is $a_n = 2n - 3$, find the first term through the 3rd term.

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = -1$, $a_2 = 1$, $a_3 = 3$.
3. Check each value by putting its index back into the formula.

Final answer

$$a_1 = -1, a_2 = 1, a_3 = 3$$

Q002 Written

When the general term of the sequence $\{a_n\}$ is $a_n = n^2 - 7n$, find the first term through the 3rd term.

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = -6$, $a_2 = -10$, $a_3 = -12$.
3. Check each value by putting its index back into the formula.

Final answer

$$a_1 = -6, a_2 = -10, a_3 = -12$$

Worked Solutions

Arithmetic Sequences

Q003 Written

When the general term of the sequence $\{a_n\}$ is $a_n = 5n - 8$, find the first term through the 3rd term.

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = -3$, $a_2 = 2$, $a_3 = 7$.
3. Check each value by putting its index back into the formula.

Final answer

$$a_1 = -3, a_2 = 2, a_3 = 7$$

Q004 Written

When the general term of the sequence $\{a_n\}$ is $a_n = -3n + 1$, find the first term through the 3rd term.

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = -2$, $a_2 = -5$, $a_3 = -8$.
3. Check each value by putting its index back into the formula.

Final answer

$$a_1 = -2, a_2 = -5, a_3 = -8$$

Worked Solutions

Arithmetic Sequences

Q005 Written

When the general term of the sequence $\{a_n\}$ is $a_n = -2n - 4$, find the first term through the 3rd term.

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = -6$, $a_2 = -8$, $a_3 = -10$.
3. Check each value by putting its index back into the formula.

Final answer

$$a_1 = -6, a_2 = -8, a_3 = -10$$

Q006 Written

When the general term of the sequence $\{a_n\}$ is $a_n = n^2 - 1$, find the first term through the 3rd term.

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = 0$, $a_2 = 3$, $a_3 = 8$.
3. Check each value by putting its index back into the formula.

Final answer

$$a_1 = 0, a_2 = 3, a_3 = 8$$

Worked Solutions

Arithmetic Sequences

Q007 Written

When the general term of the sequence $\{a_n\}$ is $a_n = n^2 + n + 1$, find the first term through the 3rd term.

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = 3$, $a_2 = 7$, $a_3 = 13$.
3. Check each value by putting its index back into the formula.

Final answer

$$a_1 = 3, a_2 = 7, a_3 = 13$$

Q008 Written

When the general term of the sequence $\{a_n\}$ is $a_n = n^2 + 2n + 4$, find the first term through the 3rd term.

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = 7$, $a_2 = 12$, $a_3 = 19$.
3. Check each value by putting its index back into the formula.

Final answer

$$a_1 = 7, a_2 = 12, a_3 = 19$$

Worked Solutions

Arithmetic Sequences

Q009 MCQ

For $a_n = 3n - 5$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute $(n = 1, 2, 3)$ into the displayed general-term formula.
2. This gives $a_1 = -2$, $a_2 = 1$, $a_3 = 4$.
3. Check each value by putting its index back into the formula.

Correct option: B

Final answer

$(-2, 1, 4)$

Q010 MCQ

For $a_n = 4n + 7$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute $(n = 1, 2, 3)$ into the displayed general-term formula.
2. This gives $a_1 = 11$, $a_2 = 15$, $a_3 = 19$.
3. Check each value by putting its index back into the formula.

Correct option: C

Final answer

$(11, 15, 19)$

Worked Solutions

Arithmetic Sequences

Q011 MCQ

For $a_n = n^2 - 4n + 6$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = 3$, $a_2 = 2$, $a_3 = 3$.
3. Check each value by putting its index back into the formula.

Correct option: D

Final answer

(3, 2, 3)

Q012 MCQ

For $a_n = 2n^2 + 3$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = 5$, $a_2 = 11$, $a_3 = 21$.
3. Check each value by putting its index back into the formula.

Correct option: A

Final answer

(5, 11, 21)

Worked Solutions

Arithmetic Sequences

Q013 MCQ

For $a_n = n^2 + 5n - 2$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute $(n = 1, 2, 3)$ into the displayed general-term formula.
2. This gives $a_1 = 4$, $a_2 = 10$, $a_3 = 16$.
3. Check each value by putting its index back into the formula.

Correct option: B

Final answer

$(4, 10, 16)$

Q014 MCQ

For $a_n = 5 - 2n$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute $(n = 1, 2, 3)$ into the displayed general-term formula.
2. This gives $a_1 = 3$, $a_2 = 1$, $a_3 = -1$.
3. Check each value by putting its index back into the formula.

Correct option: C

Final answer

$(3, 1, -1)$

Worked Solutions

Arithmetic Sequences

Q015 MCQ

For $a_n = 2n^2 - 7n + 9$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = 4$, $a_2 = 3$, $a_3 = 6$.
3. Check each value by putting its index back into the formula.

Correct option: D

Final answer

(4, 3, 6)

Q016 MCQ

For $a_n = n^2 - 3n - 1$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = -3$, $a_2 = -3$, $a_3 = -1$.
3. Check each value by putting its index back into the formula.

Correct option: A

Final answer

(-3, -3, -1)

Worked Solutions

Arithmetic Sequences

Q017 MCQ

For $a_n = 3n^2 - 2$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = 1$, $a_2 = 10$, $a_3 = 25$.
3. Check each value by putting its index back into the formula.

Correct option: B

Final answer

(1, 10, 25)

Q018 MCQ

For $a_n = n^2 + 2$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = 3$, $a_2 = 6$, $a_3 = 11$.
3. Check each value by putting its index back into the formula.

Correct option: C

Final answer

(3, 6, 11)

Worked Solutions

Arithmetic Sequences

Q019 MCQ

For $a_n = 7n - 10$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute $(n = 1, 2, 3)$ into the displayed general-term formula.
2. This gives $a_1 = -3$, $a_2 = 4$, $a_3 = 11$.
3. Check each value by putting its index back into the formula.

Correct option: D

Final answer

$(-3, 4, 11)$

Q020 MCQ

For $a_n = 2n^2 + n - 1$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute $(n = 1, 2, 3)$ into the displayed general-term formula.
2. This gives $a_1 = 2$, $a_2 = 9$, $a_3 = 20$.
3. Check each value by putting its index back into the formula.

Correct option: A

Final answer

$(2, 9, 20)$

Worked Solutions

Arithmetic Sequences

Q021 Written

When the general term of the sequence $\{a_n\}$ is $a_n = 2 \times 3^n$, find the first term through the 3rd term.

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = 6$, $a_2 = 18$, $a_3 = 54$.
3. Check each value by putting its index back into the formula.

Final answer

$$a_1 = 6, a_2 = 18, a_3 = 54$$

Q022 Written

When the general term of the sequence $\{a_n\}$ is $a_n = 3^n - 1$, find the first term through the 3rd term.

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = 2$, $a_2 = 8$, $a_3 = 26$.
3. Check each value by putting its index back into the formula.

Final answer

$$a_1 = 2, a_2 = 8, a_3 = 26$$

Worked Solutions

Arithmetic Sequences

Q023 Written

When the general term of the sequence $\{a_n\}$ is $a_n = (-3)^n$, find the first term through the 3rd term.

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = -3$, $a_2 = 9$, $a_3 = -27$.
3. Check each value by putting its index back into the formula.

Final answer

$$a_1 = -3, a_2 = 9, a_3 = -27$$

Q024 Written

When the general term of the sequence $\{a_n\}$ is $a_n = 3 \times (-2)^n$, find the first term through the 3rd term.

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = -6$, $a_2 = 12$, $a_3 = -24$.
3. Check each value by putting its index back into the formula.

Final answer

$$a_1 = -6, a_2 = 12, a_3 = -24$$

Worked Solutions

Arithmetic Sequences

Q025 MCQ

For $a_n = 2^n$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute $(n = 1, 2, 3)$ into the displayed general-term formula.
2. This gives $a_1 = 2$, $a_2 = 4$, $a_3 = 8$.
3. Check each value by putting its index back into the formula.

Correct option: A

Final answer

$(2, 4, 8)$

Q026 MCQ

For $a_n = (-2)^n$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute $(n = 1, 2, 3)$ into the displayed general-term formula.
2. This gives $a_1 = -2$, $a_2 = 4$, $a_3 = -8$.
3. Check each value by putting its index back into the formula.

Correct option: B

Final answer

$(-2, 4, -8)$

Worked Solutions

Arithmetic Sequences

Q027 MCQ

For $a_n = 2 \times 3^n$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = 6$, $a_2 = 18$, $a_3 = 54$.
3. Check each value by putting its index back into the formula.

Correct option: C

Final answer

(6, 18, 54)

Q028 MCQ

For $a_n = 5 \times (-3)^n$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = -15$, $a_2 = 45$, $a_3 = -135$.
3. Check each value by putting its index back into the formula.

Correct option: D

Final answer

(-15, 45, -135)

Worked Solutions

Arithmetic Sequences

Q029 MCQ

For $a_n = 4^n - 1$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = 3$, $a_2 = 15$, $a_3 = 63$.
3. Check each value by putting its index back into the formula.

Correct option: A

Final answer

(3, 15, 63)

Q030 MCQ

For $a_n = 2 \times 3^{n-1}$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = 2$, $a_2 = 6$, $a_3 = 18$.
3. Check each value by putting its index back into the formula.

Correct option: B

Final answer

(2, 6, 18)

Worked Solutions

Arithmetic Sequences

Q031 MCQ

For $a_n = (-1)^{n-1}5^n$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute $(n = 1, 2, 3)$ into the displayed general-term formula.
2. This gives $a_1 = 5$, $a_2 = -25$, $a_3 = 125$.
3. Check each value by putting its index back into the formula.

Correct option: C

Final answer

$(5, -25, 125)$

Q032 MCQ

For $a_n = (-3)^n + 2$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute $(n = 1, 2, 3)$ into the displayed general-term formula.
2. This gives $a_1 = -1$, $a_2 = 11$, $a_3 = -25$.
3. Check each value by putting its index back into the formula.

Correct option: D

Final answer

$(-1, 11, -25)$

Worked Solutions

Arithmetic Sequences

Q033 Written

When the general term of the sequence $\{a_n\}$ is $a_n = \frac{n}{n+1}$, find the first term through the 3rd term.

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = \frac{1}{2}$, $a_2 = \frac{2}{3}$, $a_3 = \frac{3}{4}$.
3. Check each value by putting its index back into the formula.

Final answer

$$a_1 = \frac{1}{2}, a_2 = \frac{2}{3}, a_3 = \frac{3}{4}$$

Q034 Written

When the general term of the sequence $\{a_n\}$ is $a_n = \frac{n}{2n+1}$, find the first term through the 3rd term.

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = \frac{1}{3}$, $a_2 = \frac{2}{5}$, $a_3 = \frac{3}{7}$.
3. Check each value by putting its index back into the formula.

Final answer

$$a_1 = \frac{1}{3}, a_2 = \frac{2}{5}, a_3 = \frac{3}{7}$$

Worked Solutions

Arithmetic Sequences

Q035 Written

When the general term of the sequence $\{a_n\}$ is $a_n = \frac{1}{3n-2}$, find the first term through the 3rd term.

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = 1$, $a_2 = \frac{1}{4}$, $a_3 = \frac{1}{7}$.
3. Check each value by putting its index back into the formula.

Final answer

$$a_1 = 1, a_2 = \frac{1}{4}, a_3 = \frac{1}{7}$$

Q036 Written

When the general term of the sequence $\{a_n\}$ is $a_n = \frac{2n}{n+3}$, find the first term through the 3rd term.

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = \frac{1}{2}$, $a_2 = \frac{4}{5}$, $a_3 = 1$.
3. Check each value by putting its index back into the formula.

Final answer

$$a_1 = \frac{1}{2}, a_2 = \frac{4}{5}, a_3 = 1$$

Worked Solutions

Arithmetic Sequences

Q037 Written

When the general term of the sequence $\{a_n\}$ is $a_n = \frac{n+1}{2n-1}$, find the first term through the 3rd term.

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = 2$, $a_2 = 1$, $a_3 = \frac{4}{5}$.
3. Check each value by putting its index back into the formula.

Final answer

$$a_1 = 2, a_2 = 1, a_3 = \frac{4}{5}$$

Q038 MCQ

For $a_n = \frac{n+1}{n+2}$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = \frac{2}{3}$, $a_2 = \frac{3}{4}$, $a_3 = \frac{4}{5}$.
3. Check each value by putting its index back into the formula.

Correct option: C

Final answer

$$\left(\frac{2}{3}, \frac{3}{4}, \frac{4}{5}\right)$$

Worked Solutions

Arithmetic Sequences

Q039 MCQ

For $a_n = \frac{2n-1}{3n+1}$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = \frac{1}{4}$, $a_2 = \frac{3}{7}$, $a_3 = \frac{1}{2}$.
3. Check each value by putting its index back into the formula.

Correct option: D**Final answer**

$$\left(\frac{1}{4}, \frac{3}{7}, \frac{1}{2}\right)$$

Q040 MCQ

For $a_n = \frac{n^2+1}{n+1}$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = 1$, $a_2 = \frac{5}{3}$, $a_3 = \frac{5}{2}$.
3. Check each value by putting its index back into the formula.

Correct option: A**Final answer**

$$\left(1, \frac{5}{3}, \frac{5}{2}\right)$$

Worked Solutions

Arithmetic Sequences

Q041 MCQ

For $a_n = \frac{3n}{2n-1}$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = 3$, $a_2 = 2$, $a_3 = \frac{9}{5}$.
3. Check each value by putting its index back into the formula.

Correct option: B**Final answer** $(3, 2, \frac{9}{5})$

Q042 MCQ

For $a_n = \frac{2n+3}{n+4}$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = 1$, $a_2 = \frac{7}{6}$, $a_3 = \frac{9}{7}$.
3. Check each value by putting its index back into the formula.

Correct option: C**Final answer** $(1, \frac{7}{6}, \frac{9}{7})$

Worked Solutions

Arithmetic Sequences

Q043 MCQ

For $a_n = \frac{n}{3n-2}$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = 1$, $a_2 = \frac{1}{4}$, $a_3 = \frac{3}{7}$.
3. Check each value by putting its index back into the formula.

Correct option: D**Final answer**

$$(1, \frac{1}{4}, \frac{3}{7})$$

Q044 Written

For the sequence $\{a_n\} : 0, 1, 4, 9, 16, \dots$, deduce the general term.

Solution

1. The terms are (term number $- 1$)², so $a_n = (n - 1)^2$.
2. Checking $n = 1, 2, 3$ gives 0,1,4.

Final answer

$$a_n = (n - 1)^2$$

Worked Solutions

Arithmetic Sequences

Q045 **Written****For the sequence $\{a_n\} : 0, 1, 8, 27, 64, \dots$, deduce the general term.****Solution**

1. The terms are successive cubes starting with 0: $(n - 1)^3$.
2. The formula gives 0,1,8,27,64 for $n = 1, 2, 3, 4, 5$.

Final answer

$$a_n = (n - 1)^3$$

Q046 **Written****For the sequence $\{a_n\} : 1, 4, 9, 16, 25, \dots$, deduce the general term.****Solution**

1. These are the squares $1^2, 2^2, 3^2, \dots$, so $a_n = n^2$.

Final answer

$$a_n = n^2$$

Worked Solutions

Arithmetic Sequences

Q047 **Written**

For the sequence $\{a_n\} : 7^2, 8^2, 9^2, 10^2, 11^2, \dots$, deduce the general term.

Solution

1. The bases are $n + 6$, so $a_n = (n + 6)^2$.

Final answer

$$a_n = (n + 6)^2$$

Q048 **MCQ**

Which formula gives the general term of $\{a_n\} : 1, 8, 27, 64, \dots$?:

Solution

1. Separate the visible pattern: the rule is $a_n = n^3$.
2. Check $n = 1, 2, 3$ against the displayed terms.

Correct option: B

Final answer

$$n^3$$

Worked Solutions

Arithmetic Sequences

Q049 MCQ

Which formula gives the general term of $\{a_n\}$: 9, 16, 25, 36, ...?:**Solution**

1. Separate the visible pattern: the rule is $a_n = (n + 2)^2$.
2. Check $n = 1, 2, 3$ against the displayed terms.

Correct option: C**Final answer**

$$(n + 2)^2$$

Q050 MCQ

Which formula gives the general term of $\{a_n\}$: -1, 0, 1, 8, ...?:**Solution**

1. Separate the visible pattern: the rule is $a_n = (n - 2)^3$.
2. Check $n = 1, 2, 3$ against the displayed terms.

Correct option: D**Final answer**

$$(n - 2)^3$$

Worked Solutions

Arithmetic Sequences

Q051 MCQ

Which formula gives the general term of $\{a_n\}$: 36, 49, 64, 81, ...?:**Solution**

1. Separate the visible pattern: the rule is $a_n = (n + 5)^2$.
2. Check $n = 1, 2, 3$ against the displayed terms.

Correct option: A**Final answer**

$$(n + 5)^2$$

Q052 MCQ

Which formula gives the general term of $\{a_n\}$: 1, 9, 25, 49, ...?:**Solution**

1. Separate the visible pattern: the rule is $a_n = (2n - 1)^2$.
2. Check $n = 1, 2, 3$ against the displayed terms.

Correct option: B**Final answer**

$$(2n - 1)^2$$

Worked Solutions

Arithmetic Sequences

Q053 MCQ

Which formula gives the general term of $\{a_n\} : 0, 1, 16, 81, \dots$?:**Solution**

1. Separate the visible pattern: the rule is $a_n = (n - 1)^4$.
2. Check $n = 1, 2, 3$ against the displayed terms.

Correct option: C**Final answer**

$$(n - 1)^4$$

Q054 Written

For the sequence $\{a_n\} : 2, 5, 8, 11, 14, \dots$, deduce the general term.**Solution**

1. The common increase is 3 and the first term is 2, so $a_n = 2 + 3(n - 1) = 3n - 1$.

Final answer

$$a_n = 3n - 1$$

Worked Solutions

Arithmetic Sequences

Q055 Written

For the sequence $\{a_n\} : 0, 2, 4, 6, 8, \dots$, deduce the general term.

Solution

1. The common difference is 2 and $a_1=0$, so $a_n=2(n-1)=2n-2$.

Final answer

$$a_n = 2n - 2$$

Q056 Written

For the sequence $\{a_n\} : 0, 3, 6, 9, 12, \dots$, deduce the general term.

Solution

1. The common difference is 3 and $a_1=0$, so $a_n=3(n-1)=3n-3$.

Final answer

$$a_n = 3n - 3$$

Worked Solutions

Arithmetic Sequences

Q057 **Written****For the sequence $\{a_n\}$: 4, 9, 14, 19, 24, ..., deduce the general term.****Solution**

1. The common difference is 5 and $a_1=4$, so $a_n=4+5(n-1)=5n-1$.

Final answer

$$a_n = 5n - 1$$

Q058 **Written****For the sequence $\{a_n\}$: 6, 13, 20, 27, 34, ..., deduce the general term.****Solution**

1. The common difference is 7 and $a_1=6$, so $a_n=6+7(n-1)=7n-1$.

Final answer

$$a_n = 7n - 1$$

Worked Solutions

Arithmetic Sequences

Q059 MCQ

Which formula gives the general term of $\{a_n\}$: 7, 12, 17, 22, ...?:**Solution**

1. Separate the visible pattern: the rule is $a_n = 5n + 2$.
2. Check $n = 1, 2, 3$ against the displayed terms.

Correct option: A**Final answer**

$$5n + 2$$

Q060 MCQ

Which formula gives the general term of $\{a_n\}$: -4, -1, 2, 5, ...?:**Solution**

1. Separate the visible pattern: the rule is $a_n = 3n - 7$.
2. Check $n = 1, 2, 3$ against the displayed terms.

Correct option: B**Final answer**

$$3n - 7$$

Worked Solutions

Arithmetic Sequences

Q061 MCQ

Which formula gives the general term of $\{a_n\}$: 10, 6, 2, -2, ...?:**Solution**

1. Separate the visible pattern: the rule is $a_n = 14 - 4n$.
2. Check $n = 1, 2, 3$ against the displayed terms.

Correct option: C**Final answer**

$$14 - 4n$$

Q062 MCQ

Which formula gives the general term of $\{a_n\}$: $\frac{1}{2}$, 1, $\frac{3}{2}$, 2, ...?:**Solution**

1. Separate the visible pattern: the rule is $a_n = \frac{n}{2}$.
2. Check $n = 1, 2, 3$ against the displayed terms.

Correct option: D**Final answer**

$$\frac{n}{2}$$

Worked Solutions

Arithmetic Sequences

Q063 Written

For the sequence $\{a_n\} : 2, -4, 6, -8, 10, \dots$, deduce the general term.

Solution

1. The magnitude is $2n$ and the first sign is positive with alternation, so the sign factor is $(-1)^{n-1}$.
2. Therefore $a_n = (-1)^{n-1} 2n$.

Final answer

$$a_n = (-1)^{n-1} 2n$$

Q064 Written

For the sequence $\{a_n\} : -3, 6, -9, 12, -15, \dots$, deduce the general term.

Solution

1. The magnitude is $3n$; the first term is negative and signs alternate, giving $(-1)^n$.
2. Therefore $a_n = (-1)^n 3n$.

Final answer

$$a_n = (-1)^n 3n$$

Worked Solutions

Arithmetic Sequences

Q065 Written

For the sequence $\{a_n\} : -5, 10, -15, 20, -25, \dots$, deduce the general term.

Solution

1. The magnitude is $5n$ and the first sign is negative, so $a_n = (-1)^n 5n$.

Final answer

$$a_n = (-1)^n 5n$$

Q066 Written

For the sequence $\{a_n\} : -1, 4, -7, 10, -13, \dots$, deduce the general term.

Solution

1. The magnitudes 1, 4, 7, 10, 13 are $3n - 2$; the first sign is negative, so $a_n = (-1)^n (3n - 2)$.

Final answer

$$a_n = (-1)^n (3n - 2)$$

Worked Solutions

Arithmetic Sequences

Q067 Written

For the sequence $\{a_n\} : 1, \frac{2}{3}, \frac{3}{5}, \frac{4}{7}, \frac{5}{9}, \dots$, deduce the general term.

Solution

1. The numerator is n and the denominator is the odd number $2n-1$.
2. Therefore $a_n = n/(2n-1)$.

Final answer

$$a_n = \frac{n}{2n-1}$$

Q068 Written

For the sequence $\{a_n\} : 1, \frac{3}{4}, \frac{5}{9}, \frac{7}{16}, \frac{9}{25}, \dots$, deduce the general term.

Solution

1. The numerator is the odd number $2n-1$ and the denominator is n^2 .
2. Therefore $a_n = (2n-1)/n^2$.

Final answer

$$a_n = \frac{2n-1}{n^2}$$

Worked Solutions

Arithmetic Sequences

Q069 Written

For the sequence $\{a_n\} : 1, -\frac{1}{3}, \frac{1}{5}, -\frac{1}{7}, \frac{1}{9}, \dots$, deduce the general term.

Solution

1. The denominator is $2n - 1$ and the signs alternate from a positive first term, so $a_n = (-1)^{n-1}/(2n - 1)$.

Final answer

$$a_n = \frac{(-1)^{n-1}}{2n - 1}$$

Q070 Written

For the sequence $\{a_n\} : \frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \frac{5}{6}, \dots$, deduce the general term.

Solution

1. The numerator is n and the denominator is $n+1$, so $a_n = n/(n+1)$.

Final answer

$$a_n = \frac{n}{n + 1}$$

Worked Solutions

Arithmetic Sequences

Q071 Written

For the sequence $\{a_n\} : 1 \times 3, 2 \times 9, 3 \times 27, 4 \times 81, 5 \times 243, \dots$, deduce the general term.

Solution

1. The first factor is n and the second factor is 3^n , so $a_n = n3^n$.

Final answer

$$a_n = n 3^n$$

Q072 Written

For the sequence $\{a_n\} : 1 \times \frac{1}{2}, 2 \times \frac{1}{3}, 3 \times \frac{1}{4}, 4 \times \frac{1}{5}, 5 \times \frac{1}{6}, \dots$, deduce the general term then find the 20th term.

Solution

1. The first factor is the term number n and the second factor is $1/(n+1)$.
2. Hence the general term is $a_n = n/(n+1)$.
3. Substitute $n = 20$: $a_{20} = 20/(20+1) = 20/21$.

Final answer

$$a_n = \frac{n}{n+1}; \quad a_{20} = \frac{20}{21}$$

Worked Solutions

Arithmetic Sequences

Q073 Written

Find the first term and the common difference of the arithmetic sequences: (a) 2, 5, 8, 11, ...; (b) 3, -2, -7, -12, -17,

Solution

1. For (a), $5 - 2 = 8 - 5 = 11 - 8 = 3$, so $a = 2, d = 3$.
2. For (b), each successive difference is -5 , so $a = 3, d = -5$.

Final answer

(a) $a = 2, d = 3$ (b) $a = 3, d = -5$

Q074 Written

Find the first term and common difference of: (a) 1, 5, 9, 13, 17, ...; (b) 23, 19, 15, 11, 7,

Solution

1. (a) The constant increase is (4), so $a = 1, d = 4$.
2. (b) The constant change is -4 , so $a = 23, d = -4$.

Final answer

(a) $a = 1, d = 4$ (b) $a = 23, d = -4$

Worked Solutions

Arithmetic Sequences

Q075 MCQ

The listed terms $\{a_n\} = (4, 9, 14, 19, \dots)$ form an arithmetic sequence. Find the ordered pair (a, d) of first term and common difference select the correct ordered pair a, d :

Solution

1. The first term is $a = 4$.
2. Subtract consecutive terms: $d = 5$.

Correct option: B

Final answer

$$a = 4, d = 5$$

Q076 MCQ

The listed terms $\{a_n\} = (-7, -3, 1, 5, \dots)$ form an arithmetic sequence. Find the ordered pair (a, d) of first term and common difference select the correct ordered pair a, d :

Solution

1. The first term is $a = -7$.
2. Subtract consecutive terms: $d = 4$.

Correct option: A

Final answer

$$a = -7, d = 4$$

Worked Solutions

Arithmetic Sequences

Q077 MCQ

The listed terms $\{a_n\} = (18, 13, 8, 3, \dots)$ form an arithmetic sequence. Find the ordered pair (a, d) of first term and common difference select the correct ordered pair a, d :

Solution

1. The first term is $a = 18$.
2. Subtract consecutive terms: $d = -5$.

Correct option: C**Final answer**

$$a = 18, d = -5$$

Q078 MCQ

The listed terms $\{a_n\} = (12, 8, 4, 0, \dots)$ form an arithmetic sequence. Find the ordered pair (a, d) of first term and common difference select the correct ordered pair a, d :

Solution

1. The first term is $a = 12$.
2. Subtract consecutive terms: $d = -4$.

Correct option: D**Final answer**

$$a = 12, d = -4$$

Worked Solutions

Arithmetic Sequences

Q079 Written

Find the first term and common difference of: (a) 3, 5, 7, 9, 11, ...; (b) $-13, -9, -5, -1, 3, \dots$; (c) $-2, 1, 4, 7, 10, \dots$; (d) $10, 7, 4, 1, -2, \dots$; (e) $6, \frac{16}{3}, \frac{14}{3}, 4, \frac{10}{3}, \dots$; (f) $-4, -\frac{9}{2}, -5, -\frac{11}{2}, -6, \dots$.

Solution

1. Subtract consecutive terms in each list. The six constant differences are 2, 4, 3, $-3, -\frac{2}{3}, -\frac{1}{2}$, respectively; the first terms are the first entries.

Final answer

(a)(3, 2); (b)($-13, 4$); (c)($-2, 3$); (d)($10, -3$); (e)($6, -\frac{2}{3}$); (f)($-4, -\frac{1}{2}$).

Worked Solutions

Arithmetic Sequences

Q080 MCQ

The listed terms $\{a_n\} = (\frac{1}{2}, \frac{5}{6}, \frac{7}{6}, \frac{3}{2}, \dots)$ form an arithmetic sequence. Find the ordered pair (a, d) of first term and common difference select the correct ordered pair a, d :

Solution

1. The first term is $a = \frac{1}{2}$.
2. Subtract consecutive terms: $d = \frac{1}{3}$.

Correct option: B**Final answer**

$$a = \frac{1}{2}, d = \frac{1}{3}$$

Q081 MCQ

The listed terms $\{a_n\} = (-\frac{3}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{3}{4}, \dots)$ form an arithmetic sequence. Find the ordered pair (a, d) of first term and common difference select the correct ordered pair a, d :

Solution

1. The first term is $a = -\frac{3}{4}$.
2. Subtract consecutive terms: $d = \frac{1}{2}$.

Correct option: A**Final answer**

$$a = -\frac{3}{4}, d = \frac{1}{2}$$

Worked Solutions

Arithmetic Sequences

Q082 MCQ

The listed terms $\{a_n\} = (\frac{5}{3}, 2, \frac{7}{3}, \frac{8}{3}, \dots)$ form an arithmetic sequence. Find the ordered pair (a, d) of first term and common difference select the correct ordered pair a, d :

Solution

1. The first term is $a = \frac{5}{3}$.
2. Subtract consecutive terms: $d = \frac{1}{3}$.

Correct option: C

Final answer

$$a = \frac{5}{3}, d = \frac{1}{3}$$

Q083 MCQ

The listed terms $\{a_n\} = (-\frac{5}{2}, -1, \frac{1}{2}, 2, \dots)$ form an arithmetic sequence. Find the ordered pair (a, d) of first term and common difference select the correct ordered pair a, d :

Solution

1. The first term is $a = -\frac{5}{2}$.
2. Subtract consecutive terms: $d = \frac{3}{2}$.

Correct option: B

Final answer

$$a = -\frac{5}{2}, d = \frac{3}{2}$$

Worked Solutions

Arithmetic Sequences

Q084 Written

The sequences are arithmetic. Find the common difference and the numbers that fit in (A) and (B): (a) $A, 2, -1, B, \dots$; (b) $13, A, B, -2, \dots$

Solution

- (a) $d = -1 - 2 = -3$; hence $A = 2 - d = 5$ and $B = -1 + d = -4$.
- (b) The change from 13 to -2 covers three equal steps: $3d = -15$, so $d = -5$; therefore $A = 8, B = 3$.

Final answer

$$(a) \quad d = -3, A = 5, B = -4 \quad (b) \quad d = -5, A = 8, B = 3$$

Q085 Written

The sequences are arithmetic. Find the common difference and the missing numbers: (a) $2, 9, A, B, \dots$; (b) $A, 17, B, 5, \dots$

Solution

- (a) $d = 9 - 2 = 7$, so $A = 16, B = 23$.
- (b) $5 - 17 = 2d$, hence $d = -6$; then $A = 23, B = 11$.

Final answer

$$(a) \quad d = 7, A = 16, B = 23 \quad (b) \quad d = -6, A = 23, B = 11$$

Worked Solutions

Arithmetic Sequences

Q086 MCQ

The sequence $\{a_n\} = (6, A, 14, B, \dots)$ is arithmetic. Determine the common difference and every missing letter select the correct values of d and the missing terms:

Solution

1. Set the equal-step difference (d).
2. Substitute successively to obtain $d = 4, A = 10, B = 18$.

Correct option: A

Final answer

$$d = 4, A = 10, B = 18$$

Q087 MCQ

The sequence $\{a_n\} = (A, 5, 9, B, \dots)$ is arithmetic. Determine the common difference and every missing letter select the correct values of d and the missing terms:

Solution

1. Set the equal-step difference (d).
2. Substitute successively to obtain $d = 4, A = 1, B = 13$.

Correct option: C

Final answer

$$d = 4, A = 1, B = 13$$

Worked Solutions

Arithmetic Sequences

Q088 MCQ

The sequence $\{a_n\} = (11, A, B, 23, \dots)$ is arithmetic. Determine the common difference and every missing letter select the correct values of d and the missing terms:

Solution

1. Set the equal-step difference (d).
2. Substitute successively to obtain $d = 4, A = 15, B = 19$.

Correct option: B**Final answer**

$$d = 4, A = 15, B = 19$$

Worked Solutions

Arithmetic Sequences

Q089 Written

Find the common difference and missing terms: (a) $8, 15, A, B, \dots$; (b) $7, 13, A, B, \dots$; (c) $A, 9, 4, B, \dots$; (d) $A, 15, B, 9, \dots$; (e) $11, A, B, 20, \dots$; (f) $17, A, B, 5, C, \dots$

Solution

1. Use the equal-step condition. For example, in (d), $9 - 15 = 2d$, so $d = -2$, giving $A = 17, B = 13$. Applying the same calculation to (a)–(f) gives the listed values.

Final answer

(a) $d = 7, A = 22, B = 29$; (b) $d = 6, A = 19, B = 25$; (c) $d = -5, A = 14, B = -1$; (d) $d = -2, A = 17, B = 13$; (e) $d = 3, A = 14, B = 17$; (f) $d = -4, A = 13, B = 9, C = 1$.

Worked Solutions

Arithmetic Sequences

Q090 Written

For each arithmetic sequence, find the general term and the 7th term: (a) first term (4), common difference (3); (b) first term (2), common difference -3 .

Solution

1. (a) $a_n = 4 + 3(n - 1) = 3n + 1$; $a_7 = 22$.
2. (b) $a_n = 2 - 3(n - 1) = 5 - 3n$; $a_7 = -16$.

Final answer

$$(a) \quad a_n = 3n + 1, a_7 = 22 \quad (b) \quad a_n = 5 - 3n, a_7 = -16$$

Q091 Written

For each arithmetic sequence, find the general term and the 7th term: (a) $a = 5, d = 4$; (b) $a = -2, d = 3$; (c) $a = 10, d = -3$; (d) $a = -\frac{1}{2}, d = -\frac{1}{2}$.

Solution

1. Substitute each pair into $a_n = a + (n - 1)d$, then put $n = 7$. The results are $4n + 1, 3n - 5, 13 - 3n, -\frac{n}{2}$, with 7th terms 29, 16, $-8, -\frac{7}{2}$.

Final answer

$$(a) a_n = 4n + 1, a_7 = 29; (b) a_n = 3n - 5, a_7 = 16; (c) a_n = 13 - 3n, a_7 = -8; (d) a_n = -\frac{n}{2}, a_7 = -\frac{7}{2}.$$

Worked Solutions

Arithmetic Sequences

Q092 Written

For the arithmetic sequence a_n with first term (8) and common difference -6 , find the general term and the 20th term.

Solution

1. Use $a_n = a + (n - 1)d$: $a_n = 8 - 6(n - 1) = 14 - 6n$.
2. Substitute $n = 20$: $a_{20} = 14 - 120 = -106$.

Final answer

$$a_n = 8 + (n - 1)(-6) = 14 - 6n; \quad a_{20} = -106$$

Q093 Written

For each arithmetic sequence, find the general term and the 10th term: (a) $a = -7, d = 3$; (b) $a = -2, d = -3$; (c) $a = 7, d = \frac{1}{2}$; (d) $a = \frac{2}{3}, d = -\frac{1}{3}$.

Solution

1. Use $a_n = a + (n - 1)d$ and substitute $n = 10$. Simplification gives the four displayed formulas and terms.

Final answer

$$(a)a_n = 3n - 10, a_{10} = 20; (b)a_n = 1 - 3n, a_{10} = -29; (c)a_n = \frac{n + 13}{2}, a_{10} = \frac{23}{2}; (d)a_n = \frac{3 - n}{3}, a_{10} = -\frac{7}{3}.$$

Worked Solutions

Arithmetic Sequences

Q094 Written

In an arithmetic sequence, $a_3 + a_7 = 38$ and $\frac{a_4}{a_{12}} = \frac{2}{5}$. Find the first term and common difference.

Solution

1. $a_3 + a_7 = (a + 2d) + (a + 6d) = 2a + 8d = 38$, so $a + 4d = 19$.
2. The ratio gives $5(a + 3d) = 2(a + 11d)$, hence $3a = 7d$.
3. Solving yields $d = 3$ and $a = 7$.

Final answer

$$a = 7, d = 3.$$

Q095 Written

Find the general term of the arithmetic sequence whose first term is -5 and whose 17th term is 75.

Solution

1. $75 = -5 + 16d$, so $d = 5$.
2. $a_n = -5 + 5(n - 1) = 5n - 10$.

Final answer

$$a_n = 5n - 10$$

Worked Solutions

Arithmetic Sequences

Q096 Written

The first term is 6, and the 4th term is 18. Find a_n .

Solution

1. $18 = 6 + 3d$, so $d = 4$.
2. $a_n = 6 + 4(n - 1) = 4n + 2$.

Final answer

$$a_n = 4n + 2$$

Q097 Written

The first term is 20, and the 8th term is -1 . Find a_n .

Solution

1. $-1 = 20 + 7d$, so $d = -3$.
2. $a_n = 20 - 3(n - 1) = 23 - 3n$.

Final answer

$$a_n = 23 - 3n$$

Worked Solutions

Arithmetic Sequences

Q098 **Written****The first term is -8 , and the 8th term is 55 . Find a_n .****Solution**

1. $55 = -8 + 7d$, so $d = 9$.
2. $a_n = -8 + 9(n - 1) = 9n - 17$.

Final answer

$$a_n = 9n - 17$$

Q099 **Written****The first term is -18 , and the 10th term is -63 . Find a_n .****Solution**

1. $-63 = -18 + 9d$, so $d = -5$.
2. $a_n = -18 - 5(n - 1) = -5n - 13$.

Final answer

$$a_n = -5n - 13$$

Worked Solutions

Arithmetic Sequences

Q100 MCQ

An arithmetic sequence has $a = 4$ and $a_7 = 22$. Select the correct pair (a_n, d) :

Solution

1. $22 = 4 + 6d$, so $d = 3$.
2. $a_n = 4 + 3(n - 1) = 3n + 1$.

Correct option: D

Final answer

$$(a_n, d) = (3n + 1, 3)$$

Q101 MCQ

An arithmetic sequence has $a = -3$ and $a_9 = 29$. Select the correct pair (a_n, d) :

Solution

1. $29 = -3 + 8d$, so $d = 4$.
2. $a_n = -3 + 4(n - 1) = 4n - 7$.

Correct option: B

Final answer

$$(a_n, d) = (4n - 7, 4)$$

Worked Solutions

Arithmetic Sequences

Q102 MCQ

An arithmetic sequence has $a = 12$ and $a_5 = -4$. Select the correct pair (a_n, d) :

Solution

1. $-4 = 12 + 4d$, so $d = -4$.
2. $a_n = 12 - 4(n - 1) = 16 - 4n$.

Correct option: A

Final answer

$$(a_n, d) = (16 - 4n, -4)$$

Q103 MCQ

An arithmetic sequence has $a = 5$ and $a_{11} = 45$. Select the correct pair (a_n, d) :

Solution

1. $45 = 5 + 10d$, so $d = 4$.
2. $a_n = 5 + 4(n - 1) = 4n + 1$.

Correct option: B

Final answer

$$(a_n, d) = (4n + 1, 4)$$

Worked Solutions

Arithmetic Sequences

Q104 Written

Find the general term of the arithmetic sequence whose common difference is 3 and whose 16th term is 81.

Solution

1. $81 = a + 15(3)$, so $a = 36$.
2. $a_n = 36 + 3(n - 1) = 3n + 33$.

Final answer

$$a_n = 3n + 33$$

Q105 Written

The common difference is 2, and the 8th term is 4. Find a_n .

Solution

1. $4 = a + 7(2)$, so $a = -10$.
2. $a_n = -10 + 2(n - 1) = 2n - 12$.

Final answer

$$a_n = 2n - 12$$

Worked Solutions

Arithmetic Sequences

Q106 Written

The common difference is -4 , and the 5th term is -7 . Find a_n .

Solution

1. $-7 = a + 4(-4)$, so $a = 9$.
2. $a_n = 9 - 4(n - 1) = 13 - 4n$.

Final answer

$$a_n = 13 - 4n$$

Q107 Written

The common difference is -7 , and the 6th term is -24 . Find a_n .

Solution

1. $-24 = a + 5(-7)$, so $a = 11$.
2. $a_n = 11 - 7(n - 1) = 18 - 7n$.

Final answer

$$a_n = 18 - 7n$$

Worked Solutions

Arithmetic Sequences

Q108 MCQ

An arithmetic sequence has common difference -2 and $a_4 = 1$. Select the correct pair (a_n, a) :

Solution

1. $1 = a + 3(-2)$, so $a = 7$.

2. $a_n = 7 - 2(n - 1) = 9 - 2n$.

Correct option: A

Final answer

$$(a_n, a) = (9 - 2n, 7)$$

Q109 MCQ

An arithmetic sequence has common difference 5 and $a_6 = 17$. Select the correct pair (a_n, a) :

Solution

1. $17 = a + 5(5)$, so $a = -8$.

2. $a_n = -8 + 5(n - 1) = 5n - 13$.

Correct option: D

Final answer

$$(a_n, a) = (5n - 13, -8)$$

Worked Solutions

Arithmetic Sequences

Q110 MCQ

An arithmetic sequence has common difference 4 and $a_{10} = 38$. Select the correct pair (a_n, a) :

Solution

1. $38 = a + 9(4)$, so $a = 2$.
2. $a_n = 2 + 4(n - 1) = 4n - 2$.

Correct option: C

Final answer

$$(a_n, a) = (4n - 2, 2)$$

Q111 MCQ

An arithmetic sequence has common difference -3 and $a_8 = -18$. Select the correct pair (a_n, a) :

Solution

1. $-18 = a + 7(-3)$, so $a = 3$.
2. $a_n = 3 - 3(n - 1) = 6 - 3n$.

Correct option: D

Final answer

$$(a_n, a) = (6 - 3n, 3)$$

Worked Solutions

Arithmetic Sequences

Q112 Written

Find the general term of the arithmetic sequence $\{a_n\}$ whose 3rd term is 10 and the 8th term is 50.

Solution

1. Use $a_n = a + (n - 1)d$. The data give $10 = a + 2d$ and $50 = a + 7d$.
2. Subtracting gives $5d = 40$, so $d = 8$ and $a = -6$.
3. Therefore $a_n = -6 + 8(n - 1) = 8n - 14$.

Final answer

$$a_n = 8n - 14$$

Q113 Written

Find the general term of the arithmetic sequence whose 6th term is 2 and whose 15th term is -25 .

Solution

1. $2 = a + 5d$ and $-25 = a + 14d$, so $d = -3$, $a = 17$.
2. $a_n = 17 - 3(n - 1) = 20 - 3n$.

Final answer

$$a_n = 20 - 3n$$

Worked Solutions

Arithmetic Sequences

Q114 **Written****The 4th term is 14, and the 10th term is 62. Find a_n .****Solution**

1. $14 = a + 3d$ and $62 = a + 9d$, so $d = 8$, $a = -10$.
2. $a_n = 8n - 18$.

Final answer

$$a_n = 8n - 18$$

Q115 **Written****The 3rd term is 30, and the 10th term is 2. Find a_n .****Solution**

1. $30 = a + 2d$ and $2 = a + 9d$, so $d = -4$, $a = 38$.
2. $a_n = 42 - 4n$.

Final answer

$$a_n = 42 - 4n$$

Worked Solutions

Arithmetic Sequences

Q116 Written

The 10th term is 49, and the 23rd term is 10. Find a_n .

Solution

1. $49 = a + 9d$ and $10 = a + 22d$, so $d = -3$, $a = 76$.
2. $a_n = 79 - 3n$.

Final answer

$$a_n = 79 - 3n$$

Q117 Written

The 3rd term is 6, and the 10th term is 62. Find a_n .

Solution

1. $6 = a + 2d$ and $62 = a + 9d$, so $d = 8$, $a = -10$.
2. $a_n = 8n - 18$.

Final answer

$$a_n = 8n - 18$$

Worked Solutions

Arithmetic Sequences

Q118 MCQ

An arithmetic sequence has $a_3 = 5$ and $a_9 = 17$. Select the correct pair (a_n, a) :

Solution

1. $5 = a + 2d$ and $17 = a + 8d$, so $d = 2$, $a = 1$.
2. $a_n = 1 + 2(n - 1) = 2n - 1$.

Correct option: C

Final answer

$$(a_n, a) = (2n - 1, 1)$$

Q119 MCQ

An arithmetic sequence has $a_4 = 7$ and $a_{10} = -5$. Select the correct pair (a_n, a) :

Solution

1. $7 = a + 3d$ and $-5 = a + 9d$, so $d = -2$, $a = 13$.
2. $a_n = 13 - 2(n - 1) = 15 - 2n$.

Correct option: C

Final answer

$$(a_n, a) = (15 - 2n, 13)$$

Worked Solutions

Arithmetic Sequences

Q120 MCQ

An arithmetic sequence has $a_7 = 31$ and $a_{12} = 56$. Select the correct pair (a_n, a) :

Solution

1. $31 = a + 6d$ and $56 = a + 11d$, so $d = 5$, $a = 1$.

2. $a_n = 1 + 5(n - 1) = 5n - 4$.

Correct option: B

Final answer

$$(a_n, a) = (5n - 4, 1)$$

Q121 MCQ

An arithmetic sequence has $a_2 = -4$ and $a_8 = 14$. Select the correct pair (a_n, a) :

Solution

1. $-4 = a + d$ and $14 = a + 7d$, so $d = 3$, $a = -7$.

2. $a_n = -7 + 3(n - 1) = 3n - 10$.

Correct option: A

Final answer

$$(a_n, a) = (3n - 10, -7)$$

Worked Solutions

Arithmetic Sequences

Q122 MCQ

An arithmetic sequence has $a_5 = 3$ and $a_{11} = 9$. Select the correct pair (a_n, a) :

Solution

1. $3 = a + 4d$ and $9 = a + 10d$, so $d = \frac{1}{2}$, $a = -\frac{1}{2}$.

2. $a_n = -\frac{1}{2} + \frac{1}{2}(n-1) = \frac{n-2}{2}$.

Correct option: A

Final answer

$$\left(a_n = \frac{n-2}{2}, -\frac{1}{2}\right)$$

Q123 MCQ

An arithmetic sequence has $a_2 = 6$ and $a_9 = 20$. Select the correct pair (a_n, a) :

Solution

1. $6 = a + d$ and $20 = a + 8d$, so $d = 2$, $a = 4$.

2. $a_n = 4 + 2(n-1) = 2n + 2$.

Correct option: B

Final answer

$$(a_n = 2n + 2, 4)$$

Worked Solutions

Arithmetic Sequences

Q124 Written

The common difference is $\frac{2}{5}$, and the 30th term is $\frac{23}{5}$. Find a_n .

Solution

1. $\frac{23}{5} = a + 29(\frac{2}{5})$, so $a = -7$.

2. $a_n = -7 + \frac{2}{5}(n - 1) = \frac{2n - 37}{5}$.

Final answer

$$a_n = \frac{2n - 37}{5}$$

Q125 MCQ

An arithmetic sequence has $d = -\frac{3}{2}$ and $a_5 = -9$. Select the correct pair (a_n, a) :

Solution

1. $-9 = a + 4(-\frac{3}{2})$, so $a = -3$.

2. $a_n = -3 - \frac{3}{2}(n - 1) = -\frac{3(n+1)}{2}$.

Correct option: A

Final answer

$$\left(a_n = -\frac{3(n+1)}{2}, -3 \right)$$

Worked Solutions

Arithmetic Sequences

Q126 MCQ

An arithmetic sequence has $d = \frac{2}{3}$ and $a_{12} = 10$. Select the correct pair (a_n, a) :

Solution

1. $10 = a + 11(\frac{2}{3})$, so $a = \frac{8}{3}$.
2. $a_n = \frac{8}{3} + \frac{2}{3}(n - 1) = \frac{2n+6}{3}$.

Correct option: D

Final answer

$$\left(a_n = \frac{2n+6}{3}, \frac{8}{3}\right)$$

Q127 Written

For the arithmetic sequence $\{a_n\}$ with common difference 4 and $a_8 = 33$, which term is 113?

Solution

1. First find a : $33 = a + 7(4)$, so $a = 5$.
2. Then $a_n = 5 + 4(n - 1) = 4n + 1$.
3. Solve $4n + 1 = 113$: $n = 28$.

Final answer

$n = 28$, the 28th term

Worked Solutions

Arithmetic Sequences

Q128 Written

For the arithmetic sequence with first term 1 and common difference 3, determine which term is 40.

Solution

1. $a_n = 1 + 3(n - 1) = 3n - 2$.
2. $3n - 2 = 40$ gives $n = 14$.

Final answer

$n = 14$, the 14th term

Q129 Written

For the arithmetic sequence with common difference -5 and 13th term -7 , determine which term is -52 .

Solution

1. $-7 = a + 12(-5)$, so $a = 53$ and $a_n = 58 - 5n$.
2. $58 - 5n = -52$ gives $n = 22$.

Final answer

$n = 22$, the 22nd term

Worked Solutions

Arithmetic Sequences

Q130 Written

For the arithmetic sequence whose 10th term is -20 and 20th term is -50 , determine which term is -110 .

Solution

1. $-20 = a + 9d$ and $-50 = a + 19d$, so $d = -3$, $a = 7$.
2. $a_n = 10 - 3n$, and $10 - 3n = -110$ gives $n = 40$.

Final answer

$n = 40$, the 40th term

Q131 MCQ

For $a = 3$, $d = 4$, which term of the arithmetic sequence is 59?:

Solution

1. $a_n = 3 + 4(n - 1) = 4n - 1$.
2. $4n - 1 = 59$ gives $n = 15$.

Correct option: C

Final answer

$n = 15$

Worked Solutions

Arithmetic Sequences

Q132 MCQ

For $a = -5, d = 6$, which term of the arithmetic sequence is 55?:

Solution

1. $a_n = -5 + 6(n - 1) = 6n - 11$.
2. $6n - 11 = 55$ gives $n = 11$.

Correct option: D

Final answer

$$n = 11$$

Q133 MCQ

For $a = 20, d = -3$, which term of the arithmetic sequence is -16 ?:

Solution

1. $a_n = 20 - 3(n - 1) = 23 - 3n$.
2. $23 - 3n = -16$ gives $n = 13$.

Correct option: A

Final answer

$$n = 13$$

Worked Solutions

Arithmetic Sequences

Q134 MCQ

For $a = 1$, $d = 5$, which term of the arithmetic sequence is 86?:

Solution

1. $a_n = 1 + 5(n - 1) = 5n - 4$.
2. $5n - 4 = 86$ gives $n = 18$.

Correct option: B

Final answer

$$n = 18$$

Q135 Written

For the arithmetic sequence $\{a_n\}$ with common difference 4 and $a_8 = 33$, find the first term greater than 500.

Solution

1. From the given data, $a_n = 4n + 1$.
2. Require $4n + 1 > 500$, hence $n > 124.75$.
3. The least natural index is $n = 125$.

Final answer

$n = 125$, the 125th term

Worked Solutions

Arithmetic Sequences

Q136 Written

For the arithmetic sequence with first term 1 and common difference 3, find the first term greater than 100.

Solution

1. $a_n = 3n - 2$.
2. $3n - 2 > 100$ gives $n > 34$.
3. The first valid index is $n = 35$.

Final answer

$n = 35$, the 35th term

Q137 MCQ

For $a = 2, d = 7$, find the first term greater than 100:

Solution

1. $a_n = 7n - 5$.
2. $7n - 5 > 100$ gives $n > 15$, so $n = 16$.

Correct option: A

Final answer

$n = 16$

Worked Solutions

Arithmetic Sequences

Q138 MCQ

For $a = -9, d = 4$, find the first term greater than 50:**Solution**

1. $a_n = 4n - 13$.
2. $4n - 13 > 50$ gives $n > 15.75$, so $n = 16$.

Correct option: D**Final answer**

$$n = 16$$

Q139 MCQ

For $a = 6, d = 3$, find the first term greater than 200:**Solution**

1. $a_n = 3n + 3$.
2. $3n + 3 > 200$ gives $n > 65\frac{2}{3}$, so $n = 66$.

Correct option: D**Final answer**

$$n = 66$$

Worked Solutions

Arithmetic Sequences

Q140 MCQ

For $a = -1$, $d = 9$, find the first term greater than 300:

Solution

1. $a_n = 9n - 10$.
2. $9n - 10 > 300$ gives $n > 34\frac{4}{9}$, so $n = 35$.

Correct option: B

Final answer

$$n = 35$$

Q141 Written

For the arithmetic sequence with common difference -5 and 13th term -7 , find the first term smaller than -300 .

Solution

1. $a_n = 58 - 5n$.
2. $58 - 5n < -300$ gives $n > 71.6$.
3. The first valid index is $n = 72$.

Final answer

$n = 72$, the 72nd term

Worked Solutions

Arithmetic Sequences

Q142 Written

For the arithmetic sequence whose 10th term is -20 and 20th term is -50 , find the first term smaller than -500 .

Solution

1. $a_n = 10 - 3n$.
2. $10 - 3n < -500$ gives $n > 170$.
3. The first valid index is $n = 171$.

Final answer

$n = 171$, the 171st term

Q143 MCQ

For $a = 10$, $d = -4$, find the first term smaller than -100 :

Solution

1. $a_n = 14 - 4n$.
2. $14 - 4n < -100$ gives $n > 28.5$, so $n = 29$.

Correct option: B

Final answer

$n = 29$

Worked Solutions

Arithmetic Sequences

Q144 MCQ

For $a = 25, d = -6$, find the first term smaller than -250 :**Solution**

1. $a_n = 25 - 6n$.
2. $25 - 6n < -250$ gives $n > 46\frac{5}{6}$, so $n = 47$.

Correct option: B**Final answer**

$$n = 47$$

Q145 MCQ

For $a = -3, d = -5$, find the first term smaller than -200 :**Solution**

1. $a_n = -3 - 5n$.
2. $-3 - 5n < -200$ gives $n > 40.4$, so $n = 41$.

Correct option: C**Final answer**

$$n = 41$$

Worked Solutions

Arithmetic Sequences

Q146 MCQ

For $a = 4$, $d = 6$, find the first term that is at least 100:

Solution

1. $a_n = 6n - 2$.
2. $6n - 2 \geq 100$ gives $n \geq 17$, so $n = 17$.

Correct option: A

Final answer

$$n = 17$$

Q147 MCQ

For the expression For $a = 5$, $d = 4$, does the sequence contain the term 42?, the required value is:

Solution

1. $a_n = 4n + 1$.
2. $4n + 1 = 42$ gives $n = \frac{41}{4}$, not a natural index.
3. Therefore 42 is not a term.

Correct option: B

Final answer

No term

Worked Solutions

Arithmetic Sequences

Q148 **Written**

When $a, 6, 2a$ form an arithmetic sequence in this order, find a .

Solution

1. $2(6) = a + 2a$.
2. $3a = 12$, so $a = 4$.

Final answer

$$a = 4$$

Q149 **Written**

When the three numbers $x, 6, x^2$ form an arithmetic sequence in this order, find x .

Solution

1. Use $2b = a + c$: $2(6) = x + x^2$.
2. Thus $x^2 + x - 12 = 0 = (x + 4)(x - 3)$.
3. Therefore $x = -4$ or $x = 3$.

Final answer

$$x = -4 \text{ or } x = 3$$

Worked Solutions

Arithmetic Sequences

Q150 **Written****When $x^2, x, -8$ form an arithmetic sequence in this order, find x .****Solution**

1. $2x = x^2 - 8$.
2. $x^2 - 2x - 8 = (x - 4)(x + 2) = 0$.
3. $x = -2$ or $x = 4$.

Final answer

$$x = -2 \text{ or } x = 4$$

Q151 **Written****When $-3, x, x^2$ form an arithmetic sequence in this order, find x .****Solution**

1. $2x = -3 + x^2$.
2. $x^2 - 2x - 3 = (x - 3)(x + 1) = 0$.
3. $x = -1$ or $x = 3$.

Final answer

$$x = -1 \text{ or } x = 3$$

Worked Solutions

Arithmetic Sequences

Q152 Challenge

When $2, x, y, z$ form an arithmetic sequence in this order, and $2x, y + 1, z - y$ also form an arithmetic sequence in this order, find x, y, z .

Solution

1. Let the common difference of $2, x, y, z$ be d : $x = 2 + d$, $y = 2 + 2d$, $z = 2 + 3d$.
2. $2(y + 1) = 2x + (z - y)$ gives $6 + 4d = 4 + 3d$, hence $d = -2$.
3. $(x, y, z) = (0, -2, -4)$.

Final answer

$$x = 0, \quad y = -2, \quad z = -4$$

Worked Solutions

Arithmetic Sequences

Q153 Written

Find the sum (S_7) of a sequence with first term (8), last term (22), and (7) terms.

Solution

1. Use ($S_n = \frac{n}{2}(a + l)$).
2. ($S_7 = \frac{7}{2}(8 + 22) = 105$).

Final answer

105

Q154 Written

Find (S_{10}) for a sequence with first term (7), last term (13), and (10) terms.

Solution

1. ($S_{10} = \frac{10}{2}(7 + 13) = 100$).

Final answer

100

Worked Solutions

Arithmetic Sequences

Q155 **Written****Find (S_{10}) for a sequence with first term (3), last term (21), and (10) terms.****Solution**

1. $(S_{10} = \frac{10}{2}(3 + 21) = 120).$

Final answer

120

Q156 **Written****Find (S_8) for a sequence with first term (5), last term (33), and (8) terms.****Solution**

1. $(S_8 = \frac{8}{2}(5 + 33) = 152).$

Final answer

152

Worked Solutions

Arithmetic Sequences

Q157 **Written****Find (S_{10}) for a sequence with first term (2), last term (29), and (10) terms.****Solution**

1. $(S_{10} = \frac{10}{2}(2 + 29) = 155).$

Final answer

155

Q158 **MCQ****A sequence has first term 6, last term 30, and 7 terms. Find its sum:****Solution**

1. Use $S_7 = \frac{7}{2}(6 + 30).$
2. Thus $S_7 = 126.$

Correct option: B**Final answer**

126

Worked Solutions

Arithmetic Sequences

Q159 MCQ

A sequence has first term -4 , last term 20 , and 9 terms. Find its sum:

Solution

1. $S_9 = \frac{9}{2}(-4 + 20) = 72.$

Correct option: B

Final answer

72

Q160 MCQ

Find the sum of 10 terms whose first term is 15 and last term is 3 :

Solution

1. $S_{10} = \frac{10}{2}(15 + 3) = 90.$

Correct option: B

Final answer

90

Worked Solutions

Arithmetic Sequences

Q161 MCQ

The first term is 7, the last term is 42, and there are 8 terms. Find S_8 :

Solution

1. $S_8 = \frac{8}{2}(7 + 42) = 196.$

Correct option: B

Final answer

196

Q162 MCQ

Find the sum of 6 terms with first term -12 and last term 8:

Solution

1. $S_6 = \frac{6}{2}(-12 + 8) = -12.$

Correct option: B

Final answer

-12

Worked Solutions

Arithmetic Sequences

Q163 MCQ

A sequence has $S_9 = 72$ and first term -4 . Find its last term:

Solution

1. From $72 = \frac{9}{2}(-4 + l)$, $l = 20$.

Correct option: C

Final answer

20

Q164 Written

Find (S_{13}) when the first term is (20) , the common difference is (-5) , and the number of terms is (13) .

Solution

1. Use $(S_n = \frac{n}{2}\{2a + (n - 1)d\})$.

2. $(S_{13} = \frac{13}{2}\{40 + 12(-5)\} = -130)$.

Final answer

-130

Worked Solutions

Arithmetic Sequences

Q165 **Written****Find (S_{12}) when the first term is (-6) , the common difference is (3) , and there are (12) terms.****Solution**

$$1. (S_{12} = \frac{12}{2}\{-12 + 11(3)\} = 144).$$

Final answer

144

Q166 **Written****Find (S_{15}) when the first term is (10) , the common difference is (-3) , and there are (15) terms.****Solution**

$$1. (S_{15} = \frac{15}{2}\{20 + 14(-3)\} = -165).$$

Final answer

-165

Worked Solutions

Arithmetic Sequences

Q167 Written

Find (S_{11}) when the first term is (-4) , the common difference is (4) , and there are (11) terms.

Solution

1. $(S_{11} = \frac{11}{2}\{-8 + 10(4)\} = 176).$

Final answer

176

Q168 Written

Find (S_{18}) when the first term is (3) , the common difference is (2) , and there are (18) terms.

Solution

1. $(S_{18} = \frac{18}{2}\{6 + 17(2)\} = 360).$

Final answer

360

Worked Solutions

Arithmetic Sequences

Q169 MCQ

For $a = 4, d = 3, n = 12$, find S_{12} :**Solution**

$$1. S_{12} = \frac{12}{2}\{8 + 11(3)\} = 246.$$

Correct option: C**Final answer**

246

Q170 MCQ

For $a = 9, d = -2, n = 10$, find S_{10} :**Solution**

$$1. S_{10} = 5\{18 + 9(-2)\} = 0.$$

Correct option: B**Final answer**

0

Worked Solutions

Arithmetic Sequences

Q171 MCQ

For $a = -5, d = 4, n = 9$, find S_9 :**Solution**

$$1. S_9 = \frac{9}{2}\{-10 + 8(4)\} = 88.$$

Correct option: C**Final answer**

88

Q172 MCQ

For $a = 2, d = -3, n = 15$, find S_{15} :**Solution**

$$1. S_{15} = \frac{15}{2}\{4 + 14(-3)\} = -285.$$

Correct option: B**Final answer**

-285

Worked Solutions

Arithmetic Sequences

Q173 MCQ

For $a = 11, d = 0, n = 8$, find S_8 :**Solution**1. All terms equal 11, so $S_8 = 88$.**Correct option:** C**Final answer**

88

Q174 MCQ

For $a = -3, d = -2, n = 7$, find S_7 :**Solution**1. $S_7 = \frac{7}{2}\{-6 + 6(-2)\} = -54$.**Correct option:** B**Final answer**

-54

Worked Solutions

Arithmetic Sequences

Q175 Written

Find the sum (S) of a sequence with first term (2), common difference (3), and last term (47).

Solution

1. Find (n) from $(47 = 2 + (n - 1)3)$, giving ($n = 16$).
2. Then $(S = \frac{16}{2}(2 + 47) = 392)$.

Final answer

392

Q176 Written

Find (S) for a sequence with first term (5), common difference (3), and last term (44).

Solution

1. Solve $(44 = 5 + (n - 1)3)$, so ($n = 14$).
2. $(S = \frac{14}{2}(5 + 44) = 343)$.

Final answer

343

Worked Solutions

Arithmetic Sequences

Q177 Written

Find (S) for a sequence with first term (3), common difference (4), and last term (75).

Solution

1. $(75 = 3 + (n - 1)4)$ gives $(n = 19)$.
2. $(S = \frac{19}{2}(3 + 75) = 741)$.

Final answer

741

Q178 Written

Find (S) for a sequence with first term (2), common difference (5), and last term (57).

Solution

1. $(57 = 2 + (n - 1)5)$ gives $(n = 12)$.
2. $(S = \frac{12}{2}(2 + 57) = 354)$.

Final answer

354

Worked Solutions

Arithmetic Sequences

Q179 Written

Find (S) for a sequence with first term (4), common difference (3), and last term (100).

Solution

1. $100 = 4 + (n - 1)3$ gives $(n = 33)$.
2. $S = \frac{33}{2}(4 + 100) = 1716$.

Final answer

1716

Q180 MCQ

A sequence has $a = 5, d = 4, l = 41$. Find its sum:

Solution

1. $41 = 5 + (n - 1)4$ gives $n = 10$.
2. $S = \frac{10}{2}(5 + 41) = 230$.

Correct option: C

Final answer

230

Worked Solutions

Arithmetic Sequences

Q181 MCQ

A sequence has $a = -7, d = 5, l = 48$. Find its sum:**Solution**

1. $48 = -7 + (n - 1)5$ gives $n = 12$.
2. $S = 6(-7 + 48) = 246$.

Correct option: C**Final answer**

246

Q182 MCQ

A sequence has $a = 12, d = 2, l = 50$. Find its sum:**Solution**

1. $50 = 12 + (n - 1)2$ gives $n = 20$.
2. $S = 10(12 + 50) = 620$.

Correct option: C**Final answer**

620

Worked Solutions

Arithmetic Sequences

Q183 MCQ

A sequence has $a = -10, d = 6, l = 50$. Find its sum:**Solution**

1. $50 = -10 + (n - 1)6$ gives $n = 11$.

2. $S = \frac{11}{2}(-10 + 50) = 220$.

Correct option: C**Final answer**

220

Q184 MCQ

A sequence has $a = 18, d = -3, l = -9$. Find its sum:**Solution**

1. Solve $-9 = 18 + (n - 1)(-3)$, giving $n = 10$.

2. $S = \frac{10}{2}(18 - 9) = 45$.

Correct option: C**Final answer**

45

Worked Solutions

Arithmetic Sequences

Q185 MCQ

A sequence has $a = 40, d = -4, l = -12$. Find its sum:

Solution

1. $n = 14$ from $-12 = 40 + (n - 1)(-4)$.
2. $S = 7(40 - 12) = 196$.

Correct option: B

Final answer

196

Q186 Written

For the arithmetic sequence with first term (48) and common difference (-2) , find the sum from the 20th term to the 30th term.

Solution

1. Use $(S_{30} - S_{19})$, since the range is (a_{20}) through (a_{30}) .
2. $(a_{20} = 10, a_{30} = -10)$, so $(11 \times \frac{10 + (-10)}{2} = 0)$.

Final answer

0

Worked Solutions

Arithmetic Sequences

Q187 Written

For the arithmetic sequence with first term (-53) and common difference (4) , find the sum from the 10th term to the 25th term.

Solution

1. $(a_{10} = -53 + 9(4) = -17)$ and $(a_{25} = -53 + 24(4) = 43)$.
2. There are (16) terms, so the sum is $(16 \times \frac{-17+43}{2} = 208)$.

Final answer

208

Q188 MCQ

For $a = 3, d = 4$, find the sum from a_5 to a_{12} :

Solution

1. $a_5 = 19, a_{12} = 47$, with 8 terms.
2. Sum = $8(19 + 47)/2 = 264$.

Correct option: B

Final answer

264

Worked Solutions

Arithmetic Sequences

Q189 MCQ

For $a = 20$, $d = -3$, find the sum from a_4 to a_{10} :**Solution**

1. $a_4 = 11$, $a_{10} = -7$, with 7 terms.
2. Sum = $7(11 - 7)/2 = 14$.

Correct option: C**Final answer**

14

Q190 MCQ

For $a = 60$, $d = -3$, find the sum from a_8 to a_{16} :**Solution**

1. $a_8 = 39$, $a_{16} = 15$, with 9 terms.
2. Sum = $9(39 + 15)/2 = 243$.

Correct option: C**Final answer**

243

Worked Solutions

Arithmetic Sequences

Q191 MCQ

For $a = 1, d = 7$, find the sum from a_3 to a_9 :**Solution**

1. $a_3 = 15, a_9 = 57$, with 7 terms.
2. $\text{Sum} = 7(15 + 57)/2 = 252$.

Correct option: C**Final answer**

252

Q192 MCQ

For $a = 60, d = -3$, find the sum from a_9 to a_{17} :**Solution**

1. $a_9 = 36, a_{17} = 12$, with 9 terms.
2. $\text{Sum} = 9(36 + 12)/2 = 216$.

Correct option: B**Final answer**

216

Worked Solutions

Arithmetic Sequences

Q193 MCQ

For $a = 12$, $d = 5$, find the sum from a_2 to a_8 :

Solution

1. $a_2 = 17$, $a_8 = 47$, with 7 terms.
2. Sum $= 7(17 + 47)/2 = 224$.

Correct option: C

Final answer

224

Q194 Written

For the arithmetic sequence $(90, 84, 78, \dots)$, find the sum from the 10th term to the 20th term.

Solution

1. Here $(a = 90, d = -6)$, so the required sum is $(S_{20} - S_9)$.
2. $(S_{20} = 660)$ and $(S_9 = 594)$, hence the sum is $(660 - 594 = 66)$.

Final answer

66

Worked Solutions

Arithmetic Sequences

Q195 Written

For the arithmetic sequence $(-9, -3, 3, \dots)$, find the sum from the 11th term to the 20th term.

Solution

1. Here $a = -9$, $d = 6$, so use $S_{20} - S_{10}$.
2. $a_{10} = 45$ and $a_{20} = 105$; there are 10 terms.
3. Required sum = $\frac{10(45+105)}{2} = 750$.

Final answer

750

Q196 Written

For the arithmetic sequence $(74, 69, 64, \dots)$, find the sum from the 8th term to the 18th term.

Solution

1. Here $(a = 74, d = -5)$, so $(a_8 = 39)$ and $(a_{18} = -11)$.
2. There are (11) terms; the sum is $(11 \times \frac{39-11}{2} = 154)$.

Final answer

154

Worked Solutions

Arithmetic Sequences

Q197 MCQ

For $a = 4, d = 5$, find the sum from a_6 to a_{13} :**Solution**

1. $a_6 = 29, a_{13} = 64$, with 8 terms.
2. $\text{Sum} = 8(29 + 64)/2 = 372$.

Correct option: C**Final answer**

372

Q198 MCQ

For $a = -25, d = 6$, find the sum from a_5 to a_{11} :**Solution**

1. $a_5 = -1, a_{11} = 35$, with 7 terms.
2. $\text{Sum} = 7(-1 + 35)/2 = 119$.

Correct option: C**Final answer**

119

Worked Solutions

Arithmetic Sequences

Q199 Written

Find (S_n) for a sequence with first term (1) and common difference (4).**Solution**

1. Substitute $(a = 1, d = 4)$ into $(S_n = \frac{n}{2}\{2a + (n - 1)d\})$.
2. $(S_n = \frac{n}{2}(2 + 4n - 4) = 2n^2 - n)$.

Final answer

$$2n^2 - n$$

Q200 Written

Find (S_n) for a sequence with first term (2) and common difference (-3) .**Solution**

1. $(S_n = \frac{n}{2}\{4 + (n - 1)(-3)\})$.
2. Therefore $(S_n = \frac{n(7-3n)}{2})$.

Final answer

$$\frac{n(7 - 3n)}{2}$$

Worked Solutions

Arithmetic Sequences

Q201 Written

Find (S_n) for a sequence with first term (0) and common difference (5).**Solution**

1. $(S_n = \frac{n}{2}\{0 + 5(n - 1)\})$.
2. Hence $(S_n = \frac{5n(n-1)}{2})$.

Final answer

$$\frac{5n(n-1)}{2}$$

Q202 Written

Find (S_n) for a sequence with first term (3) and common difference (-4) .**Solution**

1. $(S_n = \frac{n}{2}\{6 - 4(n - 1)\})$.
2. Therefore $(S_n = 5n - 2n^2)$.

Final answer

$$5n - 2n^2$$

Worked Solutions

Arithmetic Sequences

Q203 **Written****Find (S_n) for a sequence with first term (-4) and common difference (-2) .****Solution**

1. $(S_n = \frac{n}{2}\{-8 - 2(n - 1)\})$.
2. Therefore $(S_n = -n^2 - 3n)$.

Final answer

$$-n^2 - 3n$$

Worked Solutions

Arithmetic Sequences

Q204 Challenge

For the arithmetic sequence $(94, 89, 84, \dots)$, find the sum of the odd-numbered terms from the first term to the 11th term.

Solution

1. The odd-numbered terms are $(a_1, a_3, a_5, a_7, a_9, a_{11})$, a new arithmetic sequence with first term (94), common difference $(2(-5) = -10)$, and (6) terms.
2. Its last term is (44), so the sum is $(\frac{6}{2}(94 + 44) = 414)$.

Final answer

414

Worked Solutions

Arithmetic Sequences

Q205 Written

Find the sum of the natural numbers from 1 to 100.**Solution**

1. There are $n = 100$ terms, with $a = 1$ and $l = 100$.
2. $S_{100} = \frac{100}{2}(1 + 100) = 5050$.

Final answer

5050

Q206 Written

Find the sum of the natural numbers from 1 to 200.**Solution**

1. Use $n = 200$, $a = 1$, $l = 200$.
2. $S = \frac{200}{2}(1 + 200) = 20100$.

Final answer

20100

Worked Solutions

Arithmetic Sequences

Q207 **Written****Find the sum of the natural numbers from 1 to 150.****Solution**

1. Use $S_n = \frac{n}{2}(1 + n)$ with $n = 150$.
2. *Therefore* $S = 11325$.

Final answer

11325

Q208 **Written****Find the sum of the natural numbers from 1 to 70.****Solution**

1. Use $S_n = \frac{n}{2}(1 + n)$ with $n = 70$.
2. *Therefore* $S = 2485$.

Final answer

2485

Worked Solutions

Arithmetic Sequences

Q209 Written

Find the sum of the natural numbers from 1 to 100.**Solution**

1. Use $S_n = \frac{n}{2}(1 + n)$ with $n = 100$.
2. *Therefore* $S = 5050$.

Final answer

5050

Q210 Written

Find the sum of the natural numbers from 1 to 400.**Solution**

1. Use $S_n = \frac{n}{2}(1 + n)$ with $n = 400$.
2. *Therefore* $S = 80200$.

Final answer

80200

Worked Solutions

Arithmetic Sequences

Q211 Written

Find the sum of the natural numbers from 1 to 250.**Solution**

1. Use $S_n = \frac{n}{2}(1 + n)$ with $n = 250$.
2. *Therefore* $S = 31375$.

Final answer

31375

Q212 Written

Find the sum of the natural numbers from 1 to 81.**Solution**

1. Use $S_n = \frac{n}{2}(1 + n)$ with $n = 81$.
2. *Therefore* $S = 8085$.

Final answer

8085

Worked Solutions

Arithmetic Sequences

Q213 Written

Find the sum of the natural numbers from 1 to 301.

Solution

1. Use $S_n = \frac{n}{2}(1 + n)$ with $n = 301$.
2. Therefore $S = 24535$.

Final answer

24535

Q214 Written

Find the sum of the natural numbers from 1 to 99.

Solution

1. Use $S_n = \frac{n}{2}(1 + n)$ with $n = 99$.
2. Therefore $S = 4950$.

Final answer

4950

Worked Solutions

Arithmetic Sequences

Q215 MCQ

Find the sum of the natural numbers from 1 to 30:**Solution**

$$1. S = \frac{30}{2}(1 + 30) = 465.$$

Correct option: B**Final answer**

465

Q216 MCQ

Find the sum of the natural numbers from 1 to 45:**Solution**

$$1. S = \frac{45}{2}(1 + 45) = 1035.$$

Correct option: B**Final answer**

1035

Worked Solutions

Arithmetic Sequences

Q217 MCQ

Find the sum of the natural numbers from 1 to 60:**Solution**

$$1. S = \frac{60}{2}(1 + 60) = 1830.$$

Correct option: B

Final answer

1830

Q218 MCQ

Find the sum of the natural numbers from 1 to 75:**Solution**

$$1. S = \frac{75}{2}(1 + 75) = 2850.$$

Correct option: B

Final answer

2850

Worked Solutions

Arithmetic Sequences

Q219 MCQ

Find the sum of the natural numbers from 1 to 90:**Solution**

$$1. S = \frac{90}{2}(1 + 90) = 4095.$$

Correct option: B**Final answer**

4095

Q220 MCQ

Find the sum of the natural numbers from 1 to 125:**Solution**

$$1. S = \frac{125}{2}(1 + 125) = 7875.$$

Correct option: B**Final answer**

7875

Worked Solutions

Arithmetic Sequences

Q221 MCQ

Find the sum of the natural numbers from 1 to 180:**Solution**

$$1. S = \frac{180}{2}(1 + 180) = 16290.$$

Correct option: B

Final answer

16290

Q222 MCQ

Find the sum of the natural numbers from 1 to 220:**Solution**

$$1. S = \frac{220}{2}(1 + 220) = 24310.$$

Correct option: B

Final answer

24310

Worked Solutions

Arithmetic Sequences

Q223 MCQ

Find the sum of the natural numbers from 1 to 275:**Solution**

$$1. S = \frac{275}{2}(1 + 275) = 37950.$$

Correct option: B**Final answer**

37950

Q224 MCQ

Find the sum of the natural numbers from 1 to 350:**Solution**

$$1. S = \frac{350}{2}(1 + 350) = 61425.$$

Correct option: B**Final answer**

61425

Worked Solutions

Arithmetic Sequences

Q225 **Written****Find the sum of the natural numbers from 101 to 200.****Solution**

1. There are $n = 100$ terms, with $a = 101$ and $l = 200$.
2. $S = \frac{100}{2}(101 + 200) = 15050$.

Final answer

15050

Q226 **Written****Find the sum of the natural numbers from 201 to 300.****Solution**

1. There are 100 terms from 201 to 300.
2. $S = \frac{100}{2}(201 + 300) = 25050$.

Final answer

25050

Worked Solutions

Arithmetic Sequences

Q227 **Written****Find the sum of all two-digit natural numbers.****Solution**

1. Use $10 + 11 + \cdots + 99 : n = 90$.
2. $S = \frac{90}{2}(10 + 99) = 4905$.

Final answer

4905

Q228 **MCQ****A sequence has first term 5, last term 35, and 7 terms. Select its sum:****Solution**

1. $S = \frac{7}{2}(5 + 35) = 140$.

Correct option: B

Final answer

140

Worked Solutions

Arithmetic Sequences

Q229 MCQ

A sequence has first term 12, last term 68, and 9 terms. Select its sum:

Solution

$$1. S = \frac{9}{2}(12 + 68) = 360.$$

Correct option: B

Final answer

360

Q230 MCQ

A sequence has first term -4, last term 20, and 9 terms. Select its sum:

Solution

$$1. S = \frac{9}{2}(-4 + 20) = 72.$$

Correct option: B

Final answer

72

Worked Solutions

Arithmetic Sequences

Q231 MCQ

A sequence has first term 7, last term 42, and 8 terms. Select its sum:

Solution

1. $S = \frac{8}{2}(7 + 42) = 196.$

Correct option: B

Final answer

196

Q232 MCQ

A sequence has first term 15, last term 3, and 10 terms. Select its sum:

Solution

1. $S = \frac{10}{2}(15 + 3) = 90.$

Correct option: B

Final answer

90

Worked Solutions

Arithmetic Sequences

Q233 MCQ

A sequence has first term 10, last term 70, and 13 terms. Select its sum:

Solution

$$1. S = \frac{13}{2}(10 + 70) = 520.$$

Correct option: B

Final answer

520

Q234 MCQ

A sequence has first term 2, last term 57, and 12 terms. Select its sum:

Solution

$$1. S = \frac{12}{2}(2 + 57) = 354.$$

Correct option: B

Final answer

354

Worked Solutions

Arithmetic Sequences

Q235 MCQ

A sequence has first term 25, last term 100, and 16 terms. Select its sum:

Solution

1. $S = \frac{16}{2}(25 + 100) = 1000.$

Correct option: B

Final answer

1000

Q236 MCQ

A sequence has first term 101, last term 200, and 100 terms. Select its sum:

Solution

1. $S = \frac{100}{2}(101 + 200) = 15050.$

Correct option: B

Final answer

15050

Worked Solutions

Arithmetic Sequences

Q237 MCQ

A sequence has first term 31, last term 91, and 16 terms. Select its sum:

Solution

1. $S = \frac{16}{2}(31 + 91) = 976.$

Correct option: B

Final answer

976

Q238 Written

Find the sum of the multiples of 3 among the two-digit natural numbers.

Solution

1. The sequence is 12, 15, 18, ..., 99, so $a = 12, d = 3, l = 99.$

2. $99 = 12 + (n - 1)3$ gives $n = 30.$

3. $S = \frac{30}{2}(12 + 99) = 1665.$

Final answer

1665

Worked Solutions

Arithmetic Sequences

Q239 Written

Find the sum of all two-digit natural numbers that are multiples of 5.**Solution**

1. The sequence is 10, 15, ..., 95; $n = (95 - 10)/5 + 1 = 18$.
2. $S = \frac{18}{2}(10 + 95) = 945$.

Final answer

945

Q240 Written

Find the sum of the multiples of 4 from 1 to 100.**Solution**

1. The sequence is 4, 8, ..., 100; $n = 25$.
2. $S = \frac{25}{2}(4 + 100) = 1300$.

Final answer

1300

Worked Solutions

Arithmetic Sequences

Q241 **Written****Find the sum of all multiples of 6 from 10 to 100.****Solution**

1. The sequence is 12, 18, ..., 96; $n = 15$.
2. $S = \frac{15}{2}(12 + 96) = 810$.

Final answer

810

Q242 **MCQ****Select the sum of the multiples of 3 from 3 to 99:****Solution**

1. $n = (99 - 3)/3 + 1 = 33$; $S = \frac{33}{2}(3 + 99) = 1683$.

Correct option: B**Final answer**

1683

Worked Solutions

Arithmetic Sequences

Q243 MCQ

Select the sum of the multiples of 4 from 4 to 100:**Solution**

$$1. \ n = (100 - 4)/4 + 1 = 25; S = \frac{25}{2}(4 + 100) = 1300.$$

Correct option: B**Final answer**

1300

Q244 Written

Find the sum of the natural numbers from 1 to 100 that leave a remainder of 1 when divided by 4.**Solution**

$$1. \ \text{The sequence is } 1, 5, \dots, 97; n = 25.$$

$$2. \ S = \frac{25}{2}(1 + 97) = 1225.$$

Final answer

1225

Worked Solutions

Arithmetic Sequences

Q245 Written

Find the sum of all two-digit natural numbers that leave a remainder of 3 when divided by 5.

Solution

1. The sequence is 13, 18, ..., 98; $n = (98 - 13)/5 + 1 = 18$.
2. $S = \frac{18}{2}(13 + 98) = 999$.

Final answer

999

Q246 Written

Find the sum of all two-digit odd natural numbers.

Solution

1. The sequence is 11, 13, ..., 99; $n = 45$.
2. $S = \frac{45}{2}(11 + 99) = 2475$.

Final answer

2475

Worked Solutions**Arithmetic Sequences****Q247** Written**Find the sum of all natural numbers from 10 to 100 that are not divisible by 6.****Solution**

1. Total 10 to 100 is 5005; multiples of 6 are 12 to 96 with sum 810.
2. Required sum = $5005 - 810 = 4195$.

Final answer

4195

Worked Solutions

Arithmetic Sequences

Q248 Challenge

Find the sum of the numbers from 1 to 100 which are not divisible by 4.

Solution

1. Total $S_{100} = 5050$. Multiples of 4 have sum 1300.
2. Required sum = $5050 - 1300 = 3750$.

Final answer

3750

Worked Solutions

Arithmetic Sequences

Q249 **Written**

In an arithmetic sequence with first term 4 and common difference 3, find how many terms from the first term have sum 50.

Solution

1. $50 = \frac{n}{2}\{2(4) + (n-1)3\} \Rightarrow 3n^2 + 5n - 100 = 0.$
2. The natural root is $n = 5$; $S_5 = 50.$

Final answer

5

Q250 **Written**

Find n when $a = 3, d = 2$ and $S_n = 63.$

Solution

1. $S_n = n(n+2) = 63 \Rightarrow n = 7.$

Final answer

7

Worked Solutions

Arithmetic Sequences

Q251 MCQ

Select the correct option: Find n if $S_n = 48$ for $a = 3, d = 2$:

Solution

1. $S_n = n(n + 2) = 48 \Rightarrow n = 6$.

Correct option: B

Final answer

6

Q252 MCQ

Select the correct option: Find n if $S_n = 80$ for $a = 3, d = 2$:

Solution

1. $S_n = n(n + 2) = 80 \Rightarrow n = 8$.

Correct option: B

Final answer

8

Worked Solutions

Arithmetic Sequences

Q253 MCQ

Select the correct option: Find n if $S_n = 120$ for $a = 3, d = 2$:**Solution**

$$1. S_n = n(n + 2) = 120 \Rightarrow n = 10.$$

Correct option: B**Final answer**

10

Q254 MCQ

Select the correct option: Find n if $S_n = 168$ for $a = 3, d = 2$:**Solution**

$$1. S_n = n(n + 2) = 168 \Rightarrow n = 12.$$

Correct option: B**Final answer**

12

Worked Solutions

Arithmetic Sequences

Q255 MCQ

Select the correct option: Find n if $S_n = 255$ for $a = 3, d = 2$:**Solution**

$$1. S_n = n(n + 2) = 255 \Rightarrow n = 15.$$

Correct option: B

Final answer

15

Q256 MCQ

Select the correct option: Find n if $S_n = 360$ for $a = 3, d = 2$:**Solution**

$$1. S_n = n(n + 2) = 360 \Rightarrow n = 18.$$

Correct option: B

Final answer

18

Worked Solutions

Arithmetic Sequences

Q257 MCQ

Select the correct option: Find n if $S_n = 440$ for $a = 3, d = 2$:**Solution**

$$1. S_n = n(n + 2) = 440 \Rightarrow n = 20.$$

Correct option: B**Final answer**

20

Q258 MCQ

Select the correct option: Find n if $S_n = 624$ for $a = 3, d = 2$:**Solution**

$$1. S_n = n(n + 2) = 624 \Rightarrow n = 24.$$

Correct option: B**Final answer**

24

Worked Solutions

Arithmetic Sequences

Q259 MCQ

Select the correct option: Find n if $S_n = 840$ for $a = 3, d = 2$:**Solution**

$$1. S_n = n(n + 2) = 840 \Rightarrow n = 28.$$

Correct option: B

Final answer

28

Q260 MCQ

Select the correct option: Find n if $S_n = 1088$ for $a = 3, d = 2$:**Solution**

$$1. S_n = n(n + 2) = 1088 \Rightarrow n = 32.$$

Correct option: B

Final answer

32

Worked Solutions

Arithmetic Sequences

Q261 MCQ

Select the correct option: Find n if $S_n = 1368$ for $a = 3, d = 2$:

Solution

1. $S_n = n(n + 2) = 1368 \Rightarrow n = 36.$

Correct option: B

Final answer

36

Q262 MCQ

Select the correct option: Find n if $S_n = 1680$ for $a = 3, d = 2$:

Solution

1. $S_n = n(n + 2) = 1680 \Rightarrow n = 40.$

Correct option: B

Final answer

40

Worked Solutions

Arithmetic Sequences

Q263 Written

For first term 17 and common difference -2 , find how many terms have sum -63 .

Solution

1. $S_n = n(18 - n)$.
2. $n(18 - n) = -63 \Rightarrow n^2 - 18n - 63 = 0$; hence $n = 21$.

Final answer

21

Q264 Written

Find n when $a = 24$, $d = -4$ and $S_n = -28$.

Solution

1. $S_n = n(26 - 2n) = -28 \Rightarrow n^2 - 13n - 14 = 0$; $n = 14$.

Final answer

14

Worked Solutions

Arithmetic Sequences

Q265 MCQ

Select the correct option: For $a = 10, d = 3$, find n when $S_n = 80$:**Solution**

1. $S_n = \frac{n}{2}(17 + 3n) = 80$; positive root $n = 5$.

Correct option: C**Final answer**

5

Q266 MCQ

Select the correct option: For $a = 10, d = 3$, find n when $S_n = 133$:**Solution**

1. $S_n = \frac{n}{2}(17 + 3n) = 133$; positive root $n = 7$.

Correct option: C**Final answer**

7

Worked Solutions

Arithmetic Sequences

Q267 MCQ

Select the correct option: For $a = 10, d = 3$, find n when $S_n = 198$:**Solution**

1. $S_n = \frac{n}{2}(17 + 3n) = 198$; positive root $n = 9$.

Correct option: C

Final answer

9

Q268 MCQ

Select the correct option: For $a = 10, d = 3$, find n when $S_n = 275$:**Solution**

1. $S_n = \frac{n}{2}(17 + 3n) = 275$; positive root $n = 11$.

Correct option: C

Final answer

11

Worked Solutions

Arithmetic Sequences

Q269 Written

Find the general term of $\{a_n\}$ with first term -60 and sum of the first 15 terms -60 .

Solution

1. $-60 = \frac{15}{2}\{2(-60) + 14d\} \Rightarrow d = 8.$
2. $a_n = -60 + 8(n - 1) = 8n - 68.$

Final answer

$$a_n = 8n - 68$$

Q270 Written

Find the general term with first term 8 and sum of the first 13 terms -130 .

Solution

1. $-130 = \frac{13}{2}(16 + 12d) \Rightarrow d = -3.$
2. $a_n = 8 - 3(n - 1) = 11 - 3n.$

Final answer

$$a_n = 11 - 3n$$

Worked Solutions

Arithmetic Sequences

Q271 Written

Find the general term with first term 4 and $S_6 = 114$.**Solution**

$$1. \frac{6}{2}(8 + 5d) = 114 \Rightarrow d = 6.$$

$$2. a_n = 4 + 6(n - 1) = 6n - 2.$$

Final answer

$$a_n = 6n - 2$$

Q272 Written

Find the general term when $a_3 = 10$ and $S_{10} = 175$.**Solution**

$$1. a + 2d = 10, 2a + 9d = 35 \Rightarrow d = 3, a = 4.$$

$$2. a_n = 4 + 3(n - 1) = 3n + 1.$$

Final answer

$$a_n = 3n + 1$$

Worked Solutions

Arithmetic Sequences

Q273 Written

For first term 65 and common difference -4 , find the number of terms giving the maximum sum and the sum.

Solution

1. $a_n = 65 - 4n$; $a_n > 0 \Rightarrow n < 17.25$.
2. The last positive term is $a_{17} = 1$; $S_{17} = \frac{17}{2}(65 + 1) = 561$.

Final answer

$$n = 17, S = 561$$

Q274 Written

For first term 70 and common difference -4 , find the number of terms giving the maximum sum and the sum.

Solution

1. $a_n = 70 - 4n$; $a_{18} = 2$ is the last positive term.
2. $S_{18} = \frac{18}{2}(70 + 2) = 648$.

Final answer

$$n = 18, S = 648$$

Worked Solutions

Arithmetic Sequences

Q275 Written

For $a = 20, d = -6$, find n giving the maximum sum and the sum.

Solution

1. $a_n = 26 - 6n$; the last positive term is $a_4 = 2$.
2. $S_4 = \frac{4}{2}(20 + 2) = 44$.

Final answer

$$n = 4, S = 44$$

Q276 Written

For $a = 126, d = -13$, find n giving the maximum sum and the sum.

Solution

1. $a_n = 139 - 13n$; the last positive term is $a_{10} = 9$.
2. $S_{10} = \frac{10}{2}(126 + 9) = 675$.

Final answer

$$n = 10, S = 675$$

Worked Solutions

Arithmetic Sequences

Q277 Written

Find the general term where $S_{10} = 10$ and $S_{20} = 40$.**Solution**

1. $2a + 9d = 2, 2a + 19d = 4 \Rightarrow d = \frac{1}{5}, a = \frac{1}{10}.$
2. $a_n = \frac{1}{10} + (n-1)\frac{1}{5} = \frac{2n-1}{10}.$

Final answer

$$a_n = \frac{2n-1}{10}$$

Q278 Written

Find the general term where $S_6 = 3$ and $S_{10} = 65$.**Solution**

1. $2a + 5d = 1, 2a + 9d = 13 \Rightarrow d = 3, a = -7.$
2. $a_n = -7 + 3(n-1) = 3n - 10.$

Final answer

$$a_n = 3n - 10$$

Worked Solutions

Arithmetic Sequences

Q279 **Written****Find the general term where $S_{20} = 400$ and $S_{30} = 900$.****Solution**

$$1. \ 2a + 19d = 40, \ 2a + 29d = 60 \Rightarrow d = 2, \ a = 1.$$

$$2. \ a_n = 1 + 2(n - 1) = 2n - 1.$$

Final answer

$$a_n = 2n - 1$$

Worked Solutions

Arithmetic Sequences

Q280 Challenge

The 15th term is 33 and the 50th term is 103. Find the sum of the first 79 terms.

Solution

1. $35d = 70 \Rightarrow d = 2, a = 5.$
2. $S_{79} = \frac{79}{2}\{2(5) + 78(2)\} = 6557.$

Final answer

6557

Worked Solutions

Arithmetic Sequences

Quiz MCQ 1 **Concept Quiz · MCQ**

For $a_n = \frac{2n+1}{n+2}$, what is a_5 ?:

Solution

1. Substitute $n = 5$: $a_5 = (2(5)+1)/(5+2) = 11/7$.

Correct option: A

Final answer

$$\frac{11}{7}$$

Quiz MCQ 2 **Concept Quiz · MCQ**

An arithmetic sequence has $a_3 = 10$ and $a_8 = 30$. Select its general term:

Solution

1. $10 = a + 2d$ and $30 = a + 7d$, so $d = 4$, $a = -2$.

2. $a_n = 4n - 2$.

Correct option: B

Final answer

$$a_n = 4n - 2$$

Worked Solutions

Arithmetic Sequences

Quiz MCQ 3 **Concept Quiz · MCQ**

For $a = 12$, $d = -2$, select the sum of odd-numbered terms through a_9 :

Solution

1. Odd terms are 12, 8, 4, 0, -4 ; their sum is 20.

Correct option: C

Final answer

20

Quiz MCQ 4 **Concept Quiz · MCQ**

Select the correct option: If $S_{10} = 10$ and $S_{20} = 40$, find d :

Solution

1. $d = \frac{1}{5}$.

Correct option: B

Final answer

$\frac{1}{5}$

Worked Solutions

Arithmetic Sequences

Quiz WRITTEN 5 Concept Quiz · Written

If $a_n = 3n^2 - 2n + 1$, find a_1, a_2, a_3 .

Solution

1. Substitution gives $a_1=2$, $a_2=9$, and $a_3=22$.

Final answer

$$a_1 = 2, a_2 = 9, a_3 = 22$$

Quiz WRITTEN 6 Concept Quiz · Written

Find a_n for an arithmetic sequence with $a_4 = 14$ and $a_{10} = 62$.

Solution

1. $14 = a + 3d$ and $62 = a + 9d$, so $d = 8$, $a = -10$.

2. $a_n = 8n - 18$.

Final answer

$$a_n = 8n - 18$$

Worked Solutions

Arithmetic Sequences

Quiz WRITTEN 7 Concept Quiz · Written

For $a = 9, d = 4$, find the sum of odd-numbered terms from a_1 to a_{15} .

Solution

1. Selected terms have 8 terms, first 9, last 65.
2. $\text{Sum} = 8(9 + 65)/2 = 296$.

Final answer

296

Quiz WRITTEN 8 Concept Quiz · Written

Find the general term when $S_{20} = 400$ and $S_{30} = 900$.

Solution

1. $d = 2, a = 1$; therefore $a_n = 2n - 1$.

Final answer

$a_n = 2n - 1$

Answer key

Use the question number shown on each page.

Q001 $a_1 = -1, a_2 = 1, a_3 = 3$

Q002 $a_1 = -6, a_2 = -10, a_3 = -12$

Q003 $a_1 = -3, a_2 = 2, a_3 = 7$

Q004 $a_1 = -2, a_2 = -5, a_3 = -8$

Q005 $a_1 = -6, a_2 = -8, a_3 = -10$

Q006 $a_1 = 0, a_2 = 3, a_3 = 8$

Q007 $a_1 = 3, a_2 = 7, a_3 = 13$

Q008 $a_1 = 7, a_2 = 12, a_3 = 19$

Q009 B

Q010 C

Q011 D

Q012 A

Q013 B

Q014 C

Q015 D

Q016 A

Q017 B

Q018 C

Q019 D

Q020 A

Q021 $a_1 = 6, a_2 = 18, a_3 = 54$

Q022 $a_1 = 2, a_2 = 8, a_3 = 26$

Q023 $a_1 = -3, a_2 = 9, a_3 = -27$

Q024 $a_1 = -6, a_2 = 12, a_3 = -24$

Q025 A

Q026 B

Q027 C

Q028 D

Q029 A

Q030 B

Q031 C

Q032 D

Q033 $a_1 = \frac{1}{2}, a_2 = \frac{2}{3}, a_3 = \frac{3}{4}$

Q034 $a_1 = \frac{1}{3}, a_2 = \frac{2}{5}, a_3 = \frac{3}{7}$

Q035 $a_1 = 1, a_2 = \frac{1}{4}, a_3 = \frac{1}{7}$

Q036 $a_1 = \frac{1}{2}, a_2 = \frac{4}{5}, a_3 = 1$

Q037 $a_1 = 2, a_2 = 1, a_3 = \frac{4}{5}$

Q038 C

Q039 D

Q040 A

Q041 B

Q042 C

Q043 D

Q044 $a_n = (n-1)^2$

Q045 $a_n = (n-1)^3$

Q046 $a_n = n^2$

Q047 $a_n = (n+6)^2$

Q048 B

Q049 C

Q050 D

Q051 A

Q052 B

Q053 C

Q054 $a_n = 3n - 1$

Q055 $a_n = 2n - 2$

Q056 $a_n = 3n - 3$

Q057 $a_n = 5n - 1$

Q058 $a_n = 7n - 1$

Q059 A

Q060 B

Q061 C

Q062 D

Q063 $a_n = (-1)^{n-1} 2n$

Q064 $a_n = (-1)^n 3n$

Q065 $a_n = (-1)^n 5n$

Q066 $a_n = (-1)^n (3n - 2)$

Q067 $a_n = \frac{n}{2n-1}$

Q068 $a_n = \frac{2n-1}{n^2}$

Q069 $a_n = \frac{(-1)^{n-1}}{2n-1}$

Q070 $a_n = \frac{n}{n+1}$

Q071 $a_n = n 3^n$

Q072 $a_n = \frac{n}{n+1}; a_{20} = \frac{20}{21}$

Q073 (a) $a = 2, d = 3(b) a = 3, d = -5$

Q074 (a) $a = 1, d = 4(b) a = 23, d = -4$

Q075 B

Q076 A

Q077 C

Q078 D

Q079 (a)(3, 2); (b)(-13, 4); (c)(-2, 3); (d)(10, -3); (e)(6, $-\frac{2}{3}$); (f)(-4, $-\frac{1}{2}$).

Q080 B

Q081 A

Q082 C

Q083 B

Q084 (a) $d = -3, A = 5, B = -4(b) d = -5, A = 8, B = 3$

Q085 (a) $d = 7, A = 16, B = 23(b) d = -6, A = 23, B = 11$

Q086 A

Q087 C

Q088 B

Q089 (a)(d) = 7, A = 22, B = 20; (b)(d) = 6, A = 19, B = 21; (c)(d) = -5, A = 14, B = -1; (d)(d) = -2, A = 17, B = 13; (e)(d) = 3, A = 14, B = 17; (f)(d) = -4, A = 13, B = 9, C = 1.

Q090 (a) $a_n = 3n + 1, a_7 = 22(b) a_n = 5 - 3n, a_7 = -16$

Q091 (a) $a_n = 4n + 1, a_7 = 29$; (b) $a_n = 3n - 5, a_7 = 16$; (c) $a_n = 13 - 3n, a_7 = -8$; (d) $a_n = -\frac{n}{2}, a_7 = -\frac{7}{2}$.

Q092 $a_n = 8 + (n-1)(-6) = 14 - 6n; a_{20} = -106$

Q093 (a) $a_n = 3n - 10, a_{10} = 20$; (b) $a_n = 1 - 3n, a_{10} = -29$; (c) $a_n = \frac{n+13}{2}, a_{10} = \frac{23}{2}$; (d) $a_n = \frac{3-n}{3}, a_{10} = -\frac{7}{3}$.

Q094 $a = 7, d = 3$.

Q095 $a_n = 5n - 10$

Q096 $a_n = 4n + 2$

Q097 $a_n = 23 - 3n$

Q098 $a_n = 9n - 17$

Q099 $a_n = -5n - 13$

Q100 D

Q101 B	Q133 A	Q165 144	Q197 C
Q102 A	Q134 B	Q166 -165	Q198 C
Q103 B	Q135 $n = 125$, the 125th term	Q167 176	Q199 $2n^2 - n$
Q104 $a_n = 3n + 33$	Q136 $n = 35$, the 35th term	Q168 360	Q200 $\frac{n(7-3n)}{2}$
Q105 $a_n = 2n - 12$	Q137 A	Q169 C	Q201 $\frac{5n(n-1)}{2}$
Q106 $a_n = 13 - 4n$	Q138 D	Q170 B	Q202 $5n - 2n^2$
Q107 $a_n = 18 - 7n$	Q139 D	Q171 C	Q203 $-n^2 - 3n$
Q108 A	Q140 B	Q172 B	Q204 414
Q109 D	Q141 $n = 72$, the 72nd term	Q173 C	Q205 5050
Q110 C	Q142 $n = 171$, the 171st term	Q174 B	Q206 20100
Q111 D	Q143 B	Q175 392	Q207 11325
Q112 $a_n = 8n - 14$	Q144 B	Q176 343	Q208 2485
Q113 $a_n = 20 - 3n$	Q145 C	Q177 741	Q209 5050
Q114 $a_n = 8n - 18$	Q146 A	Q178 354	Q210 80200
Q115 $a_n = 42 - 4n$	Q147 B	Q179 1716	Q211 31375
Q116 $a_n = 79 - 3n$	Q148 $a = 4$	Q180 C	Q212 8085
Q117 $a_n = 8n - 18$	Q149 $x = -4$ or $x = 3$	Q181 C	Q213 24535
Q118 C	Q150 $x = -2$ or $x = 4$	Q182 C	Q214 4950
Q119 C	Q151 $x = -1$ or $x = 3$	Q183 C	Q215 B
Q120 B	Q152 $x = 0$, $y = -2$, $z = -4$	Q184 C	Q216 B
Q121 A	Q153 105	Q185 B	Q217 B
Q122 A	Q154 100	Q186 0	Q218 B
Q123 B	Q155 120	Q187 208	Q219 B
Q124 $a_n = \frac{2n-37}{5}$	Q156 152	Q188 B	Q220 B
Q125 A	Q157 155	Q189 C	Q221 B
Q126 D	Q158 B	Q190 C	Q222 B
Q127 $n = 28$, the 28th term	Q159 B	Q191 C	Q223 B
Q128 $n = 14$, the 14th term	Q160 B	Q192 B	Q224 B
Q129 $n = 22$, the 22nd term	Q161 B	Q193 C	Q225 15050
Q130 $n = 40$, the 40th term	Q162 B	Q194 66	Q226 25050
Q131 C	Q163 C	Q195 750	Q227 4905
Q132 D	Q164 -130	Q196 154	

Q228 B	Q244 1225	Q260 B	Q276 $n = 10, S = 675$
Q229 B	Q245 999	Q261 B	Q277 $a_n = \frac{2n-1}{10}$
Q230 B	Q246 2475	Q262 B	Q278 $a_n = 3n - 10$
Q231 B	Q247 4195	Q263 21	Q279 $a_n = 2n - 1$
Q232 B	Q248 3750	Q264 14	Q280 6557
Q233 B	Q249 5	Q265 C	Quiz MCQ 1 A
Q234 B	Q250 7	Q266 C	Quiz MCQ 2 B
Q235 B	Q251 B	Q267 C	Quiz MCQ 3 C
Q236 B	Q252 B	Q268 C	Quiz MCQ 4 B
Q237 B	Q253 B	Q269 $a_n = 8n - 68$	Quiz WRITTEN 5 $a_1 = 2, a_2 = 9, a_3 = 22$
Q238 1665	Q254 B	Q270 $a_n = 11 - 3n$	Quiz WRITTEN 6 $a_n = 8n - 18$
Q239 945	Q255 B	Q271 $a_n = 6n - 2$	Quiz WRITTEN 7 296
Q240 1300	Q256 B	Q272 $a_n = 3n + 1$	Quiz WRITTEN 8 $a_n = 2n - 1$
Q241 810	Q257 B	Q273 $n = 17, S = 561$	
Q242 B	Q258 B	Q274 $n = 18, S = 648$	
Q243 B	Q259 B	Q275 $n = 4, S = 44$	

Concept 2 | Geometric Sequences

English Student Edition • Focused formulas and guided practice

Formula opener

Term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1}$$

Infinite sum

$$S_{\infty} = \frac{a}{1-r} \quad (|r| < 1)$$

Worked Solutions

Geometric Sequences

Q001 Written

Find the general term of a geometric sequence whose first term is 2 and common ratio is 3.

Solution

1. Here $a = 2$ and $r = 3$.
2. $a_n = 2 \times 3^{n-1}$

Final answer

$$2 \times 3^{n-1}$$

Q002 Written

Find the general term when the first term is 3 and the common ratio is 4.

Solution

1. $a_n = 3 \times 4^{n-1}$

Final answer

$$3 \times 4^{n-1}$$

Worked Solutions

Geometric Sequences

Q003 Written

Find the general term of the geometric sequence in part a. The first term is 5, common ratio 2.

Solution

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $5 \times 2^{n-1}$.

Final answer

$$5 \times 2^{n-1}$$

Q004 Written

Find the general term of the geometric sequence in part b. The first term is 2, common ratio 5.

Solution

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $2 \times 5^{n-1}$.

Final answer

$$2 \times 5^{n-1}$$

Worked Solutions

Geometric Sequences

Q005 **Written**

Find the general term of the geometric sequence in part c. The first term is 7, common ratio 4.

Solution

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $7 \times 4^{n-1}$.

Final answer

$$7 \times 4^{n-1}$$

Q006 **MCQ**

Select the correct option: For first term $a = 2$, common ratio $r = 3$, and $n = 5$, which expression gives a_n ?:

Solution

1. Use $a_n = ar^{n-1}$; substitute $a = 2$ and $r = 3$.

Correct option: A

Final answer

$$a_n = 2(3)^4$$

Worked Solutions

Geometric Sequences

Q007 MCQ

Select the correct option: For first term $a = 3$, common ratio $r = 2$, and $n = 6$, which expression gives a_n ?:

Solution

1. Use $a_n = ar^{n-1}$; substitute $a = 3$ and $r = 2$.

Correct option: A

Final answer

$$a_n = 3(2)^5$$

Q008 MCQ

Select the correct option: For first term $a = 5$, common ratio $r = 4$, and $n = 4$, which expression gives a_n ?:

Solution

1. Use $a_n = ar^{n-1}$; substitute $a = 5$ and $r = 4$.

Correct option: A

Final answer

$$a_n = 5(4)^3$$

Worked Solutions

Geometric Sequences

Q009 MCQ

Select the correct option: For first term $a = 4$, common ratio $r = 5$, and $n = 3$, which expression gives a_n ?:

Solution

1. Use $a_n = ar^{n-1}$; substitute $a = 4$ and $r = 5$.

Correct option: A

Final answer

$$a_n = 4(5)^2$$

Q010 MCQ

Select the correct option: For first term $a = 5$, common ratio $r = 2$, and $n = 7$, which expression gives a_n ?:

Solution

1. Use $a_n = ar^{n-1}$; substitute $a = 5$ and $r = 2$.

Correct option: A

Final answer

$$a_n = 5(2)^6$$

Worked Solutions

Geometric Sequences

Q011 MCQ

Select the correct option: For first term $a = 7$, common ratio $r = 4$, and $n = 3$, which expression gives a_n ?:

Solution

1. Use $a_n = ar^{n-1}$; substitute $a = 7$ and $r = 4$.

Correct option: A

Final answer

$$a_n = 7(4)^2$$

Q012 Written

Find the general term when the first term is 5 and the common ratio is -2 .

Solution

1. $a_n = 5(-2)^{n-1}$

Final answer

$$5(-2)^{n-1}$$

Worked Solutions

Geometric Sequences

Q013 Written

Find the general term of the geometric sequence in part d. The first term is 2, common ratio -3 .

Solution

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $2(-3)^{n-1}$.

Final answer

$$2(-3)^{n-1}$$

Q014 Written

Find the general term of the geometric sequence in part e. The first term is 3, common ratio -5 .

Solution

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $3(-5)^{n-1}$.

Final answer

$$3(-5)^{n-1}$$

Worked Solutions

Geometric Sequences

Q015 Written

Find the general term of the geometric sequence in part f. The first term is 4, common ratio -2 .

Solution

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $4(-2)^{n-1}$.

Final answer

$$4(-2)^{n-1}$$

Q016 MCQ

Select the correct option: For first term $a = 2$, common ratio $r = -3$, and $n = 5$, which expression gives a_n ?:

Solution

1. Use $a_n = ar^{n-1}$; substitute $a = 2$ and $r = -3$.

Correct option: A

Final answer

$$a_n = 2(-3)^4$$

Worked Solutions

Geometric Sequences

Q017 MCQ

Select the correct option: For first term $a = 3$, common ratio $r = -2$, and $n = 6$, which expression gives a_n ?:

Solution

1. Use $a_n = ar^{n-1}$; substitute $a = 3$ and $r = -2$.

Correct option: A

Final answer

$$a_n = 3(-2)^5$$

Q018 MCQ

Select the correct option: For first term $a = 6$, common ratio $r = -1$, and $n = 4$, which expression gives a_n ?:

Solution

1. Use $a_n = ar^{n-1}$; substitute $a = 6$ and $r = -1$.

Correct option: A

Final answer

$$a_n = 6(-1)^3$$

Worked Solutions

Geometric Sequences

Q019 MCQ

Select the correct option: For first term $a = 8$, common ratio $r = -2$, and $n = 3$, which expression gives a_n ?:

Solution

1. Use $a_n = ar^{n-1}$; substitute $a = 8$ and $r = -2$.

Correct option: A

Final answer

$$a_n = 8(-2)^2$$

Q020 MCQ

Select the correct option: For first term $a = 9$, common ratio $r = -2$, and $n = 5$, which expression gives a_n ?:

Solution

1. Use $a_n = ar^{n-1}$; substitute $a = 9$ and $r = -2$.

Correct option: A

Final answer

$$a_n = 9(-2)^4$$

Worked Solutions

Geometric Sequences

Q021 Written

Find the general term when the first term is 8 and the common ratio is $-\frac{1}{3}$.

Solution

1. $a_n = 8\left(-\frac{1}{3}\right)^{n-1}$

Final answer

$$8\left(-\frac{1}{3}\right)^{n-1}$$

Q022 Written

Find the general term of the geometric sequence in part g. The first term is 4, common ratio $\frac{1}{3}$.

Solution

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $4\left(\frac{1}{3}\right)^{n-1}$.

Final answer

$$4\left(\frac{1}{3}\right)^{n-1}$$

Worked Solutions

Geometric Sequences

Q023 Written

Find the general term of the geometric sequence in part h. The first term is 3, common ratio $-\frac{1}{2}$.

Solution

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $3\left(-\frac{1}{2}\right)^{n-1}$.

Final answer

$$3\left(-\frac{1}{2}\right)^{n-1}$$

Q024 Written

Find the general term of the geometric sequence in part i. The first term is 6, common ratio $\frac{1}{2}$.

Solution

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $6\left(\frac{1}{2}\right)^{n-1}$.

Final answer

$$6\left(\frac{1}{2}\right)^{n-1}$$

Worked Solutions

Geometric Sequences

Q025 MCQ

Select the correct option: For first term $a = 4$, common ratio $r = \frac{1}{3}$, and $n = 5$, which expression gives a_n ?:

Solution

1. Use $a_n = ar^{n-1}$; substitute $a = 4$ and $r = \frac{1}{3}$.

Correct option: A

Final answer

$$a_n = 4 \left(\frac{1}{3} \right)^4$$

Q026 MCQ

Select the correct option: For first term $a = 3$, common ratio $r = -\frac{1}{2}$, and $n = 6$, which expression gives a_n ?:

Solution

1. Use $a_n = ar^{n-1}$; substitute $a = 3$ and $r = -\frac{1}{2}$.

Correct option: A

Final answer

$$a_n = 3 \left(-\frac{1}{2} \right)^5$$

Worked Solutions

Geometric Sequences

Q027 MCQ

Select the correct option: For first term $a = 6$, common ratio $r = \frac{1}{2}$, and $n = 4$, which expression gives a_n ?

Solution

1. Use $a_n = ar^{n-1}$; substitute $a = 6$ and $r = \frac{1}{2}$.

Correct option: A

Final answer

$$a_n = 6 \left(\frac{1}{2} \right)^3$$

Q028 MCQ

Select the correct option: For first term $a = 2$, common ratio $r = \frac{3}{2}$, and $n = 5$, which expression gives a_n ?

Solution

1. Use $a_n = ar^{n-1}$; substitute $a = 2$ and $r = \frac{3}{2}$.

Correct option: A

Final answer

$$a_n = 2 \left(\frac{3}{2} \right)^4$$

Worked Solutions

Geometric Sequences

Q029 MCQ

Select the correct option: For first term $a = 8$, common ratio $r = -\frac{3}{2}$, and $n = 4$, which expression gives a_n ?:

Solution

1. Use $a_n = ar^{n-1}$; substitute $a = 8$ and $r = -\frac{3}{2}$.

Correct option: A

Final answer

$$a_n = 8 \left(-\frac{3}{2} \right)^3$$

Q030 Written

Find the general term of 5, 10, 20, 40,

Solution

1. The common ratio is $r = 10/5 = 2$.

2. $a_n = 5 \times 2^{n-1}$

Final answer

$$5 \times 2^{n-1}$$

Worked Solutions

Geometric Sequences

Q031 **Written****Find the general term of 2, 6, 18, 54,****Solution**

1. The common ratio is 3.
2. $a_n = 2 \times 3^{n-1}$

Final answer

$$2 \times 3^{n-1}$$

Q032 **Written****Find the general term of the geometric sequence in part j. The sequence is 3, 6, 12, 24,****Solution**

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $3 \times 2^{n-1}$.

Final answer

$$3 \times 2^{n-1}$$

Worked Solutions

Geometric Sequences

Q033 MCQ

Select the correct option: Find the general term of the geometric sequence 3, 6, 12, 24, ...:**Solution**

1. Divide consecutive terms to obtain r , then use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$3 \times 2^{n-1}$$

Q034 MCQ

Select the correct option: Find the general term of the geometric sequence 2, 10, 50, 250, ...:**Solution**

1. Divide consecutive terms to obtain r , then use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$2 \times 5^{n-1}$$

Worked Solutions

Geometric Sequences

Q035 MCQ

Select the correct option: Find the general term of the geometric sequence 4, 8, 16, 32, ...:**Solution**1. Divide consecutive terms to obtain r , then use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$4 \times 2^{n-1}$$

Q036 MCQ

Select the correct option: Find the general term of the geometric sequence 7, 21, 63, 189, ...:**Solution**1. Divide consecutive terms to obtain r , then use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$7 \times 3^{n-1}$$

Worked Solutions

Geometric Sequences

Q037 MCQ

Select the correct option: Find the general term of the geometric sequence 1, 4, 16, 64, ...:**Solution**

1. Divide consecutive terms to obtain r , then use $a_n = ar^{n-1}$.

Correct option: A**Final answer**

$$4^{n-1}$$

Q038 Written

Find the general term of the geometric sequence in part k. The sequence is 2, -6, 18, -54,**Solution**

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $2(-3)^{n-1}$.

Final answer

$$2(-3)^{n-1}$$

Worked Solutions

Geometric Sequences

Q039 Written

Find the general term of the geometric sequence in part I. The sequence is 5, -10, 20, -40, ...

Solution

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $5(-2)^{n-1}$.

Final answer

$$5(-2)^{n-1}$$

Q040 MCQ

Select the correct option: Find the general term of the geometric sequence 5, -10, 20, -40, ...:

Solution

1. Divide consecutive terms to obtain r , then use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$5(-2)^{n-1}$$

Worked Solutions

Geometric Sequences

Q041 MCQ

Select the correct option: Find the general term of the geometric sequence $2, -6, 18, -54, \dots$:**Solution**1. Divide consecutive terms to obtain r , then use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$2(-3)^{n-1}$$

Q042 MCQ

Select the correct option: Find the general term of the geometric sequence $-4, 8, -16, 32, \dots$:**Solution**1. Divide consecutive terms to obtain r , then use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$-4(-2)^{n-1}$$

Worked Solutions

Geometric Sequences

Q043 MCQ

Select the correct option: Find the general term of the geometric sequence 6, −18, 54, −162, ...:**Solution**1. Divide consecutive terms to obtain r , then use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$6(-3)^{n-1}$$

Q044 Written

Find the general term of $-4, 6, -9, \frac{27}{2}, \dots$ **Solution**1. The common ratio is $r = 6/(-4) = -3/2$.

2. $a_n = -4\left(-\frac{3}{2}\right)^{n-1}$

Final answer

$$-4\left(-\frac{3}{2}\right)^{n-1}$$

Worked Solutions

Geometric Sequences

Q045 Written

Find the general term of $-8, 12, -18, 27, \dots$ **Solution**

1. The common ratio is $-3/2$.
2. $a_n = -8\left(-\frac{3}{2}\right)^{n-1}$

Final answer

$$-8\left(-\frac{3}{2}\right)^{n-1}$$

Q046 Written

Find the general term of $2, \frac{2}{3}, \frac{2}{9}, \frac{2}{27}, \dots$ **Solution**

1. The common ratio is $1/3$.
2. $a_n = 2\left(\frac{1}{3}\right)^{n-1}$

Final answer

$$2\left(\frac{1}{3}\right)^{n-1}$$

Worked Solutions

Geometric Sequences

Q047 Written

Find the general term of the geometric sequence in part m. The sequence is 48, 24, 12, 6,

Solution

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $48\left(\frac{1}{2}\right)^{n-1}$.

Final answer

$$48\left(\frac{1}{2}\right)^{n-1}$$

Q048 Written

Find the general term of the geometric sequence in part n. The sequence is $\frac{1}{6}, \frac{1}{4}, \frac{3}{8}, \frac{9}{16}, \dots$

Solution

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $\frac{1}{6}\left(\frac{3}{2}\right)^{n-1}$.

Final answer

$$\frac{1}{6}\left(\frac{3}{2}\right)^{n-1}$$

Worked Solutions

Geometric Sequences

Q049 Written

Find the general term of the geometric sequence in part o. The sequence is $8, 16\sqrt{2}, 64, 128\sqrt{2}, \dots$

Solution

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $8(2\sqrt{2})^{n-1}$.

Final answer

$$8(2\sqrt{2})^{n-1}$$

Q050 Written

Find the general term of the geometric sequence in part p. The sequence is $3, -\frac{15}{2}, \frac{75}{4}, -\frac{375}{8}, \dots$

Solution

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $3\left(-\frac{5}{2}\right)^{n-1}$.

Final answer

$$3\left(-\frac{5}{2}\right)^{n-1}$$

Worked Solutions

Geometric Sequences

Q051 Written

Find the general term of the geometric sequence in part q. The sequence is $9, -3\sqrt{3}, 3, -\sqrt{3}, \dots$

Solution

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $9\left(-\frac{\sqrt{3}}{3}\right)^{n-1}$.

Final answer

$$9\left(-\frac{\sqrt{3}}{3}\right)^{n-1}$$

Q052 Written

Find the general term of the geometric sequence in part r. The sequence is $5, -\frac{10}{3}, \frac{20}{9}, -\frac{40}{27}, \dots$

Solution

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $5\left(-\frac{2}{3}\right)^{n-1}$.

Final answer

$$5\left(-\frac{2}{3}\right)^{n-1}$$

Worked Solutions

Geometric Sequences

Q053 MCQ

Select the correct option: Find the general term of the geometric sequence $-8, 12, -18, 27, \dots$:**Solution**1. Divide consecutive terms to obtain r , then use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$-8\left(-\frac{3}{2}\right)^{n-1}$$

Q054 MCQ

Select the correct option: Find the general term of the geometric sequence $48, 24, 12, 6, \dots$:**Solution**1. Divide consecutive terms to obtain r , then use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$48\left(\frac{1}{2}\right)^{n-1}$$

Worked Solutions

Geometric Sequences

Q055 MCQ

Select the correct option: Find the general term of the geometric sequence $\frac{1}{6}, \frac{1}{4}, \frac{3}{8}, \dots$:

Solution

1. Divide consecutive terms to obtain r , then use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$\frac{1}{6} \left(\frac{3}{2} \right)^{n-1}$$

Q056 MCQ

Select the correct option: Find the general term of the geometric sequence $8, 16\sqrt{2}, 64, \dots$:

Solution

1. Divide consecutive terms to obtain r , then use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$8 \left(2\sqrt{2} \right)^{n-1}$$

Worked Solutions

Geometric Sequences

Q057 MCQ

Select the correct option: Find the general term of the geometric sequence $9, -3\sqrt{3}, 3, \dots$:

Solution

1. Divide consecutive terms to obtain r , then use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$9\left(-\frac{\sqrt{3}}{3}\right)^{n-1}$$

Q058 MCQ

Select the correct option: Find the general term of the geometric sequence $3, -\frac{3}{2}, \frac{3}{4}, \dots$:

Solution

1. Divide consecutive terms to obtain r , then use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$3\left(-\frac{1}{2}\right)^{n-1}$$

Worked Solutions

Geometric Sequences

Q059 MCQ

Select the correct option: Find the general term of the geometric sequence $10, 5, \frac{5}{2}, \dots$:

Solution

1. Divide consecutive terms to obtain r , then use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$10\left(\frac{1}{2}\right)^{n-1}$$

Q060 Written

Find the common ratio and missing values in $A, -18, 6, B, \dots$

Solution

1. $r = 6/(-18) = -\frac{1}{3}$

2. $A = 54, \quad B = 6\left(-\frac{1}{3}\right) = -2$

Final answer

$$r = -\frac{1}{3}, A = 54, B = -2$$

Worked Solutions

Geometric Sequences

Q061 MCQ

Select the correct option: The sequence $A, 12, 6, B, \dots$ is geometric. Find r, A, B :

Solution

1. Divide adjacent known terms to find r , then extend one step in each direction.

Correct option: A

Final answer

$$r = \frac{1}{2}, A = 24, B = 3$$

Q062 Written

Find the common ratio and the missing values in $A, 24, 12, B, \dots$

Solution

1. $24r = 12 \Rightarrow r = \frac{1}{2}$

2. $A\left(\frac{1}{2}\right) = 24 \Rightarrow A = 48, \quad B = 12\left(\frac{1}{2}\right) = 6$

Final answer

$$r = \frac{1}{2}, A = 48, B = 6$$

Worked Solutions

Geometric Sequences

Q063 Written

Find the common ratio and missing values in $A, -3, 9, B$.

Solution

1. $r = -3, \quad A = 1, \quad B = -27$

Final answer

$$r = -3, A = 1, B = -27$$

Q064 Written

Find the common ratio and missing values in $A, \frac{1}{3}, \frac{1}{9}, B$.

Solution

1. $r = \frac{1}{3}, \quad A = 1, \quad B = \frac{1}{27}$

Final answer

$$r = \frac{1}{3}, A = 1, B = \frac{1}{27}$$

Worked Solutions

Geometric Sequences

Q065 Written

Find the common ratio and missing values in $A, -\frac{1}{4}, \frac{1}{8}, B$.

Solution

$$1. \ r = -\frac{1}{2}, \quad A = \frac{1}{2}, \quad B = -\frac{1}{16}$$

Final answer

$$r = -\frac{1}{2}, \quad A = \frac{1}{2}, \quad B = -\frac{1}{16}$$

Worked Solutions

Geometric Sequences

Q066 Challenge

In an increasing geometric sequence, if $a_3 + a_5 = 80$ and $a_4 = 24$, find a_7 .

Solution

1. $a_3 = \frac{24}{r}$, $a_5 = 24r$, $24\left(r + \frac{1}{r}\right) = 80$
2. $3r^2 - 10r + 3 = 0 \Rightarrow r = 3$ or $\frac{1}{3}$
3. The sequence is increasing, so $r = 3$; hence $a_7 = 24(3)^3 = 648$.

Final answer

$$a_7 = 648$$

Worked Solutions

Geometric Sequences

Q067 Written

Find the general term of a geometric sequence whose 4th term is 24 and 6th term is 96.

Solution

1. $r^2 = 96/24 = 4 \Rightarrow r = \pm 2$
2. $(a, r) = (3, 2)$ or $(-3, -2)$
3. Therefore both real branches are retained.

Final answer

$$a_n = 3(2)^{n-1} \text{ or } -3(-2)^{n-1}$$

Q068 Written

Find the general term when the 3rd term is 18 and the 5th term is 162.

Solution

1. $r^2 = 162/18 = 9 \Rightarrow r = \pm 3$
2. $a = 18/r^2 = 2$

Final answer

$$a_n = 2(3)^{n-1} \text{ or } 2(-3)^{n-1}$$

Worked Solutions

Geometric Sequences

Q069 Written

Find the requested values. The 2nd term is 6 and the 4th term is 54.

Solution

1. $r^2 = 54/6 = 9 \Rightarrow r = \pm 3$

2. $a = 6/r$

Final answer

$$a_n = 2(3)^{n-1} \text{ or } -2(-3)^{n-1}$$

Q070 Written

Find the requested values. The 3rd term is 27 and the 5th term is 243.

Solution

1. $r^2 = 243/27 = 9 \Rightarrow r = \pm 3$

2. $a = 3$

Final answer

$$a_n = 3(3)^{n-1} \text{ or } 3(-3)^{n-1}$$

Worked Solutions

Geometric Sequences

Q071 **Written****Find the requested values. The 2nd term is 14 and the 5th term is 112.****Solution**

1. $r^3 = 112/14 = 8 \Rightarrow r = 2$
2. $a = 7$

Final answer

$$a_n = 7(2)^{n-1}$$

Q072 **Written****Find the general term when the 3rd term is 12 and the 6th term is -96 .****Solution**

1. $r^3 = -96/12 = -8 \Rightarrow r = -2$
2. $a = 12/r^2 = 3$
3. $a_n = 3(-2)^{n-1}$

Final answer

$$a_n = 3(-2)^{n-1}$$

Worked Solutions

Geometric Sequences

Q073 Written

Find the general term when the 2nd term is -6 and the 5th term is 48 .

Solution

$$1. r^3 = 48/(-6) = -8 \Rightarrow r = -2$$

$$2. a = -6/r = 3$$

Final answer

$$a_n = 3(-2)^{n-1}$$

Q074 Written

Find the general term when the 3rd term is 12 and the 7th term is 192 .

Solution

$$1. r^4 = 192/12 = 16 \Rightarrow r = \pm 2$$

$$2. a = 12/r^2 = 3$$

Final answer

$$a_n = 3(2)^{n-1} \text{ or } 3(-2)^{n-1}$$

Worked Solutions

Geometric Sequences

Q075 Written

Find the requested values. The 3rd term is -28 and the 5th term is -112 .

Solution

1. $r^2 = 4 \Rightarrow r = \pm 2$
2. $a = -28/4 = -7$

Final answer

$$a_n = -7(2)^{n-1} \text{ or } -7(-2)^{n-1}$$

Q076 Written

Find the requested values. The 4th term is 16 and the 7th term is -128 .

Solution

1. $r^3 = -8 \Rightarrow r = -2$
2. $a = -2$

Final answer

$$a_n = -2(-2)^{n-1}$$

Worked Solutions

Geometric Sequences

Q077 Written

Find the requested values. The 3rd term is -20 and the 6th term is 160 .

Solution

1. $r^3 = -8 \Rightarrow r = -2$
2. $a = -5$

Final answer

$$a_n = -5(-2)^{n-1}$$

Q078 Written

Find the requested values. The 2nd term is 3 and the 6th term is 768 .

Solution

1. $r^4 = 768/3 = 256 \Rightarrow r = \pm 4$
2. $a = 3/r$

Final answer

$$a_n = \frac{3}{4}(4)^{n-1} \text{ or } -\frac{3}{4}(-4)^{n-1}$$

Worked Solutions

Geometric Sequences

Q079 Written

Find the requested values. The 3rd term is 9 and the 7th term is $729/16$.

Solution

1. $r^4 = 81/16 \Rightarrow r = \pm \frac{3}{2}$
2. $a = 4$

Final answer

$$a_n = 4\left(\frac{3}{2}\right)^{n-1} \text{ or } 4\left(-\frac{3}{2}\right)^{n-1}$$

Q080 Written

Find the requested values. The 4th term is $2/9$ and the 8th term is 18.

Solution

1. $r^4 = 81 \Rightarrow r = \pm 3$
2. $a = \frac{2/9}{r^3} = \pm \frac{2}{243}$

Final answer

$$a_n = \frac{2}{243}(3)^{n-1} \text{ or } -\frac{2}{243}(-3)^{n-1}$$

Worked Solutions

Geometric Sequences

Q081 MCQ

Select the correct option: For $a = 2$, $r = 3$, which expression gives the general term a_n :

Solution

1. Use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$a_n = 2(3)^{n-1}$$

Q082 MCQ

Select the correct option: For $a = 3$, $r = 2$, which expression gives the general term a_n :

Solution

1. Use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$a_n = 3(2)^{n-1}$$

Worked Solutions

Geometric Sequences

Q083 MCQ

Select the correct option: For $a = 5$, $r = 4$, which expression gives the general term a_n :

Solution

1. Use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$a_n = 5(4)^{n-1}$$

Q084 MCQ

Select the correct option: For $a = 4$, $r = 5$, which expression gives the general term a_n :

Solution

1. Use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$a_n = 4(5)^{n-1}$$

Worked Solutions

Geometric Sequences

Q085 MCQ

Select the correct option: For $a = 2$, $r = -3$, which expression gives the general term a_n :

Solution

1. Use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$a_n = 2(-3)^{n-1}$$

Q086 MCQ

Select the correct option: For $a = 3$, $r = -2$, which expression gives the general term a_n :

Solution

1. Use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$a_n = 3(-2)^{n-1}$$

Worked Solutions

Geometric Sequences

Q087 MCQ

Select the correct option: For $a = 6$, $r = -1$, which expression gives the general term a_n :

Solution

1. Use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$a_n = 6(-1)^{n-1}$$

Q088 MCQ

Select the correct option: For $a = 8$, $r = -2$, which expression gives the general term a_n :

Solution

1. Use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$a_n = 8(-2)^{n-1}$$

Worked Solutions

Geometric Sequences

Q089 MCQ

Select the correct option: For $a = 4$, $r = \frac{1}{3}$, which expression gives the general term a_n :

Solution

1. Use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$a_n = 4 \left(\frac{1}{3} \right)^{n-1}$$

Q090 MCQ

Select the correct option: For $a = 3$, $r = -\frac{1}{2}$, which expression gives the general term a_n :

Solution

1. Use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$a_n = 3 \left(-\frac{1}{2} \right)^{n-1}$$

Worked Solutions

Geometric Sequences

Q091 MCQ

Select the correct option: For $a = 6$, $r = \frac{1}{2}$, which expression gives the general term a_n :

Solution

1. Use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$a_n = 6 \left(\frac{1}{2} \right)^{n-1}$$

Q092 MCQ

Select the correct option: For $a = 2$, $r = \frac{3}{2}$, which expression gives the general term a_n :

Solution

1. Use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$a_n = 2 \left(\frac{3}{2} \right)^{n-1}$$

Worked Solutions

Geometric Sequences

Q093 MCQ

Select the correct option: For $a = 8$, $r = -\frac{3}{2}$, which expression gives the general term a_n :

Solution

1. Use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$a_n = 8 \left(-\frac{3}{2}\right)^{n-1}$$

Q094 MCQ

Select the correct option: For $a = 5$, $r = 2$, which expression gives the general term a_n :

Solution

1. Use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$a_n = 5(2)^{n-1}$$

Worked Solutions

Geometric Sequences

Q095 MCQ

Select the correct option: For $a = 7, r = 4$, which expression gives the general term a_n :

Solution

1. Use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$a_n = 7(4)^{n-1}$$

Q096 MCQ

Select the correct option: For $a = 9, r = -2$, which expression gives the general term a_n :

Solution

1. Use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$a_n = 9(-2)^{n-1}$$

Worked Solutions

Geometric Sequences

Q097 Written

When 2, x , 8 form a geometric sequence in this order, find x .

Solution

1. $x^2 = 2 \times 8 = 16$

2. $x = \pm 4$

Final answer

$x = \pm 4$

Q098 Written

Find the requested values. When 1, x , 25 form a geometric sequence, find x .

Solution

1. $x^2 = 25 \Rightarrow x = \pm 5$

Final answer

$x = \pm 5$

Worked Solutions

Geometric Sequences

Q099 Written

Find the requested values. When $1/2, x, 8$ form a geometric sequence, find x .

Solution

$$1. x^2 = \frac{1}{2} \times 8 = 4 \Rightarrow x = \pm 2$$

Final answer

$$x = \pm 2$$

Q100 Written

When $1, x, 12$ form a geometric sequence in this order, find x .

Solution

$$1. x^2 = 1 \times 12 = 12$$

$$2. x = \pm 2\sqrt{3}$$

Final answer

$$x = \pm 2\sqrt{3}$$

Worked Solutions

Geometric Sequences

Q101 MCQ

Select the correct option: If x is the middle term of the geometric triple $2, x, 8$, find x :

Solution

1. Use $x^2 = (2)(8)$.

Correct option: A

Final answer

± 4

Q102 MCQ

Select the correct option: If x is the middle term of the geometric triple $1, x, 25$, find x :

Solution

1. Use $x^2 = (1)(25)$.

Correct option: A

Final answer

± 5

Worked Solutions

Geometric Sequences

Q103 MCQ

Select the correct option: If x is the middle term of the geometric triple $\frac{1}{2}, x, 8$, find x :

Solution

1. Use $x^2 = (\frac{1}{2})(8)$.

Correct option: A

Final answer

± 2

Q104 MCQ

Select the correct option: If x is the middle term of the geometric triple $3, x, 12$, find x :

Solution

1. Use $x^2 = (3)(12)$.

Correct option: A

Final answer

± 6

Worked Solutions

Geometric Sequences

Q105 MCQ

Select the correct option: If x is the middle term of the geometric triple 4, x , 9, find x :

Solution

1. Use $x^2 = (4)(9)$.

Correct option: A

Final answer

± 6

Q106 MCQ

Select the correct option: If x is the middle term of the geometric triple 2, x , 18, find x :

Solution

1. Use $x^2 = (2)(18)$.

Correct option: A

Final answer

± 6

Worked Solutions

Geometric Sequences

Q107 Written

The numbers $6, a, b$ form an arithmetic sequence and $a, b, 16$ form a geometric sequence. Find a, b .

Solution

1. $2a = 6 + b \Rightarrow b = 2a - 6$
2. $b^2 = 16a \Rightarrow 4a^2 - 40a + 36 = 0$
3. $(a, b) = (9, 12)$ or $(1, -4)$

Final answer

$$(a, b) = (9, 12) \text{ or } (1, -4)$$

Q108 Written

Find the requested values. The numbers $a, b, 2$ form an arithmetic sequence and $a, 2, b$ form a geometric sequence. Find a, b .

Solution

1. $2b = a + 2$
2. $b^2 = 2a$
3. $(a, b) = (2, 2)$ or $(-4, -1)$

Final answer

$$(a, b) = (2, 2) \text{ or } (-4, -1)$$

Worked Solutions

Geometric Sequences

Q109 Written

The numbers $-2, a, b$ form an arithmetic sequence and $a, b, 18$ form a geometric sequence. Find a, b .

Solution

1. $b = 2a + 2$

2. $b^2 = 18a \Rightarrow (a, b) = (2, 6) \text{ or } (\frac{1}{2}, 3)$

Final answer

$$(a, b) = (2, 6) \text{ or } (\frac{1}{2}, 3)$$

Worked Solutions

Geometric Sequences

Q110 Challenge

A planning model gives $-5, a, b$ as an arithmetic sequence, while $a, b, 45$ form a geometric sequence. Find a, b .

Solution

1. $b = 2a + 5$
2. $b^2 = 45a \Rightarrow 4a^2 - 25a + 25 = 0$
3. $(a, b) = (5, 15)$ or $(\frac{5}{4}, \frac{15}{2})$

Final answer

$$(a, b) = (5, 15) \text{ or } (\frac{5}{4}, \frac{15}{2})$$

Worked Solutions

Geometric Sequences

Q111 Written

Find the sum S of a geometric sequence with first term 4, common ratio 3, and 5 terms.

Solution

1. Use $S_5 = \frac{4(3^5 - 1)}{3 - 1} = 2(3^5 - 1)$.
2. Therefore $S_5 = 484$.

Final answer

484

Q112 Written

Find the sum S of a geometric sequence with first term 3, common ratio 2, and 5 terms.

Solution

1. $S_5 = \frac{3(2^5 - 1)}{2 - 1} = 3(31)$.
2. Therefore $S_5 = 93$.

Final answer

93

Worked Solutions

Geometric Sequences

Q113 Written

Find the sum S of a geometric sequence with first term 5, common ratio 3, and 4 terms.

Solution

1. $S_4 = \frac{5(3^4 - 1)}{3 - 1} = 5(40).$
2. Therefore $S_4 = 200.$

Final answer

200

Q114 Written

Find the sum S of a geometric sequence with first term 3, common ratio 2, and 10 terms.

Solution

1. $S_{10} = \frac{3(2^{10} - 1)}{2 - 1} = 3(1023).$
2. Therefore $S_{10} = 3069.$

Final answer

3069

Worked Solutions

Geometric Sequences

Q115 MCQ

For a geometric sequence with $a = 4$, $r = 3$, and $n = 4$, calculate S_n :

Solution

1. $S_n = \frac{4(1 - (3)^4)}{1 - (3)}$.
2. Therefore $S_4 = 160$.

Correct option: D

Final answer

160

Q116 MCQ

For a geometric sequence with $a = 5$, $r = 2$, and $n = 6$, calculate S_n :

Solution

1. $S_n = \frac{5(1 - (2)^6)}{1 - (2)}$.
2. Therefore $S_6 = 315$.

Correct option: C

Final answer

315

Worked Solutions

Geometric Sequences

Q117 MCQ

For a geometric sequence with $a = 2$, $r = 4$, and $n = 5$, calculate S_n :

Solution

1. $S_n = \frac{2(1 - (4)^5)}{1 - (4)}$.

2. Therefore $S_5 = 682$.

Correct option: B

Final answer

682

Q118 MCQ

For a geometric sequence with $a = 3$, $r = 5$, and $n = 4$, calculate S_n :

Solution

1. $S_n = \frac{3(1 - (5)^4)}{1 - (5)}$.

2. Therefore $S_4 = 468$.

Correct option: A

Final answer

468

Worked Solutions

Geometric Sequences

Q119 MCQ

For a geometric sequence with $a = 7$, $r = 2$, and $n = 5$, calculate S_n :

Solution

1. $S_n = \frac{7(1 - (2)^5)}{1 - (2)}$.
2. Therefore $S_5 = 217$.

Correct option: D

Final answer

217

Q120 MCQ

For a geometric sequence with $a = 6$, $r = 3$, and $n = 3$, calculate S_n :

Solution

1. $S_n = \frac{6(1 - (3)^3)}{1 - (3)}$.
2. Therefore $S_3 = 78$.

Correct option: C

Final answer

78

Worked Solutions

Geometric Sequences

Q121 Written

Find the sum S of a geometric sequence with first term 8, common ratio $\frac{1}{2}$, and 6 terms.

Solution

1. Use $S_6 = \frac{8(1 - (1/2)^6)}{1 - 1/2} = 16\{1 - (1/2)^6\}$.
2. Therefore $S_6 = \frac{63}{4}$.

Final answer

$$\frac{63}{4}$$

Q122 Written

Find the sum S of a geometric sequence with first term 6, common ratio $\frac{1}{2}$, and 6 terms.

Solution

1. $S_6 = \frac{6(1 - (1/2)^6)}{1 - 1/2} = 12\{1 - (1/2)^6\}$.
2. Therefore $S_6 = \frac{189}{16}$.

Final answer

$$\frac{189}{16}$$

Worked Solutions

Geometric Sequences

Q123 Written

Find the sum S of a geometric sequence with first term 1, common ratio $\frac{1}{3}$, and 3 terms.

Solution

$$1. S_3 = \frac{1 - (1/3)^3}{1 - 1/3} = \frac{26}{27} \cdot \frac{3}{2}.$$

$$2. \text{ Therefore } S_3 = \frac{13}{9}.$$

Final answer

$$\frac{13}{9}$$

Q124 Written

Find the sum S of a geometric sequence with first term 1, common ratio $\frac{1}{2}$, and 8 terms.

Solution

$$1. S_8 = \frac{1 - (1/2)^8}{1 - 1/2} = 2\{1 - (1/2)^8\}.$$

$$2. \text{ Therefore } S_8 = \frac{255}{128}.$$

Final answer

$$\frac{255}{128}$$

Worked Solutions

Geometric Sequences

Q125 MCQ

For a geometric sequence with $a = 8$, $r = \frac{1}{2}$, and $n = 4$, calculate S_n :

Solution

1. $S_n = \frac{8(1 - (\frac{1}{2})^4)}{1 - (\frac{1}{2})}$.
2. Therefore $S_4 = 15$.

Correct option: B

Final answer

15

Q126 MCQ

For a geometric sequence with $a = 9$, $r = \frac{1}{3}$, and $n = 3$, calculate S_n :

Solution

1. $S_n = \frac{9(1 - (\frac{1}{3})^3)}{1 - (\frac{1}{3})}$.
2. Therefore $S_3 = 13$.

Correct option: A

Final answer

13

Worked Solutions

Geometric Sequences

Q127 MCQ

For a geometric sequence with $a = 12$, $r = \frac{1}{2}$, and $n = 5$, calculate S_n :

Solution

$$1. S_n = \frac{12(1 - (\frac{1}{2})^5)}{1 - (\frac{1}{2})}.$$

$$2. \text{ Therefore } S_5 = \frac{93}{4}.$$

Correct option: D

Final answer

$$\frac{93}{4}$$

Q128 MCQ

For a geometric sequence with $a = 5$, $r = \frac{1}{4}$, and $n = 3$, calculate S_n :

Solution

$$1. S_n = \frac{5(1 - (\frac{1}{4})^3)}{1 - (\frac{1}{4})}.$$

$$2. \text{ Therefore } S_3 = \frac{105}{16}.$$

Correct option: C

Final answer

$$\frac{105}{16}$$

Worked Solutions

Geometric Sequences

Q129 MCQ

For a geometric sequence with $a = 10$, $r = \frac{1}{2}$, and $n = 6$, calculate S_n :

Solution

$$1. S_n = \frac{10(1 - (\frac{1}{2})^6)}{1 - (\frac{1}{2})}.$$

$$2. \text{ Therefore } S_6 = \frac{315}{16}.$$

Correct option: B

Final answer

$$\frac{315}{16}$$

Q130 MCQ

For a geometric sequence with $a = 4$, $r = \frac{1}{3}$, and $n = 4$, calculate S_n :

Solution

$$1. S_n = \frac{4(1 - (\frac{1}{3})^4)}{1 - (\frac{1}{3})}.$$

$$2. \text{ Therefore } S_4 = \frac{160}{27}.$$

Correct option: A

Final answer

$$\frac{160}{27}$$

Worked Solutions

Geometric Sequences

Q131 Written

Find the sum S of a geometric sequence with first term 6, common ratio -2 , and 4 terms.

Solution

1. $S_4 = \frac{6\{1 - (-2)^4\}}{1 - (-2)} = \frac{6(-15)}{3}.$
2. Therefore $S_4 = -30.$

Final answer

-30

Q132 Written

Find the sum S of a geometric sequence with first term 1, common ratio -3 , and 4 terms.

Solution

1. $S_4 = \frac{1 - (-3)^4}{1 - (-3)} = \frac{-80}{4}.$
2. Therefore $S_4 = -20.$

Final answer

-20

Worked Solutions

Geometric Sequences

Q133 Written

Find the sum S of a geometric sequence with first term 2, common ratio -3 , and 6 terms.

Solution

$$1. S_6 = \frac{2\{1 - (-3)^6\}}{1 - (-3)} = \frac{2(-728)}{4}.$$

$$2. \text{ Therefore } S_6 = -364.$$

Final answer

 -364

Q134 MCQ

For a geometric sequence with $a = 4$, $r = -2$, and $n = 5$, calculate S_n :

Solution

$$1. S_n = \frac{4(1 - (-2)^5)}{1 - (-2)}.$$

$$2. \text{ Therefore } S_5 = 44.$$

Correct option: D

Final answer

44

Worked Solutions

Geometric Sequences

Q135 MCQ

For a geometric sequence with $a = 3$, $r = -3$, and $n = 4$, calculate S_n :

Solution

1. $S_n = \frac{3(1 - (-3)^4)}{1 - (-3)}$.
2. Therefore $S_4 = -60$.

Correct option: C

Final answer

-60

Q136 MCQ

For a geometric sequence with $a = 2$, $r = -1$, and $n = 6$, calculate S_n :

Solution

1. $S_n = \frac{2(1 - (-1)^6)}{1 - (-1)}$.
2. Therefore $S_6 = 0$.

Correct option: B

Final answer

0

Worked Solutions

Geometric Sequences

Q137 MCQ

For a geometric sequence with $a = 5$, $r = -2$, and $n = 6$, calculate S_n :

Solution

1. $S_n = \frac{5(1 - (-2)^6)}{1 - (-2)}$.
2. Therefore $S_6 = -105$.

Correct option: A

Final answer

-105

Q138 MCQ

For a geometric sequence with $a = 7$, $r = -1$, and $n = 5$, calculate S_n :

Solution

1. $S_n = \frac{7(1 - (-1)^5)}{1 - (-1)}$.
2. Therefore $S_5 = 7$.

Correct option: D

Final answer

7

Worked Solutions

Geometric Sequences

Q139 MCQ

For a geometric sequence with $a = 6$, $r = -3$, and $n = 3$, calculate S_n :

Solution

$$1. S_n = \frac{6(1 - (-3)^3)}{1 - (-3)}.$$

$$2. \text{ Therefore } S_3 = 42.$$

Correct option: C

Final answer

42

Q140 Written

Find S_n from the first term to the n -th term when the first term is 3 and the common ratio is -2 .

Solution

$$1. \text{ Here } a = 3 \text{ and } r = -2.$$

$$2. S_n = \frac{3\{1 - (-2)^n\}}{1 - (-2)} = 1 - (-2)^n.$$

Final answer $1 - (-2)^n$

Worked Solutions

Geometric Sequences

Q141 Written

Find S_n from the first term to the n -th term when the first term is 7 and the common ratio is 2.

Solution

1. $S_n = \frac{7(2^n - 1)}{2 - 1}$.
2. Hence $S_n = 7(2^n - 1)$.

Final answer

$$7(2^n - 1)$$

Q142 Written

Find S_n from the first term to the n -th term when the first term is 2 and the common ratio is $\frac{1}{3}$.

Solution

1. $S_n = \frac{2\{1 - (1/3)^n\}}{1 - 1/3}$.
2. Hence $S_n = 3\{1 - (1/3)^n\}$.

Final answer

$$3\{1 - (\frac{1}{3})^n\}$$

Worked Solutions

Geometric Sequences

Q143 Written

Find S_n from the first term to the n -th term when the first term is 4 and the common ratio is 2.

Solution

1. $S_n = \frac{4(2^n - 1)}{2 - 1}$.
2. Hence $S_n = 4(2^n - 1)$.

Final answer

$$4(2^n - 1)$$

Q144 MCQ

For the geometric sequence with $a = 3, r = 2$, find the sum S_n from the first term to the n -th term:

Solution

1. Identify a and r from the stated parameters.
2. Substitute into $S_n = \frac{a(1 - r^n)}{1 - r}$ to obtain $S_n = 3(2^n - 1)$.

Correct option: C

Final answer

$$3(2^n - 1)$$

Worked Solutions

Geometric Sequences

Q145 MCQ

For the geometric sequence with $a = 5, r = \frac{1}{2}$, find the sum S_n from the first term to the n -th term:

Solution

1. Identify a and r from the stated parameters.
2. Substitute into $S_n = \frac{a(1 - r^n)}{1 - r}$ to obtain $S_n = 10\{1 - (\frac{1}{2})^n\}$.

Correct option: D**Final answer**

$$10\{1 - (\frac{1}{2})^n\}$$

Q146 MCQ

For the geometric sequence with $a = 4, r = -2$, find the sum S_n from the first term to the n -th term:

Solution

1. Identify a and r from the stated parameters.
2. Substitute into $S_n = \frac{a(1 - r^n)}{1 - r}$ to obtain $S_n = \frac{4\{1 - (-2)^n\}}{3}$.

Correct option: A**Final answer**

$$\frac{4\{1 - (-2)^n\}}{3}$$

Worked Solutions

Geometric Sequences

Q147 **Written****Find S_n from the first term to the n -th term of 2, 6, 18, 54,****Solution**

1. The sequence has $a = 2$ and $r = 3$.
2. $S_n = \frac{2(3^n - 1)}{3 - 1} = 3^n - 1$.

Final answer

$$3^n - 1$$

Q148 **Written****Find S_n from the first term to the n -th term of 2, -6, 18, -54,****Solution**

1. The sequence has $a = 2$ and $r = -3$.
2. $S_n = \frac{2\{1 - (-3)^n\}}{1 - (-3)} = \frac{1 - (-3)^n}{2}$.

Final answer

$$\frac{1 - (-3)^n}{2}$$

Worked Solutions

Geometric Sequences

Q149 Written

Find S_n from the first term to the n -th term of $8, -4, 2, -1, \dots$

Solution

1. The sequence has $a = 8$ and $r = -\frac{1}{2}$.
2. $S_n = \frac{8\{1 - (-1/2)^n\}}{1 + 1/2} = \frac{16}{3}\{1 - (-1/2)^n\}$.

Final answer

$$\frac{16}{3}\{1 - (-\frac{1}{2})^n\}$$

Q150 Written

Find S_n from the first term to the n -th term of $1, -3, 9, -27, \dots$

Solution

1. The sequence has $a = 1$ and $r = -3$.
2. $S_n = \frac{1 - (-3)^n}{1 + 3} = \frac{1 - (-3)^n}{4}$.

Final answer

$$\frac{1 - (-3)^n}{4}$$

Worked Solutions

Geometric Sequences

Q151 Written

Find S_n from the first term to the n -th term when the first term is 16 and the common ratio is $-\frac{1}{2}$.

Solution

1. $S_n = \frac{16\{1 - (-1/2)^n\}}{1 + 1/2}$.
2. Hence $S_n = \frac{32}{3}\{1 - (-1/2)^n\}$.

Final answer

$$\frac{32}{3}\{1 - (-\frac{1}{2})^n\}$$

Q152 Written

Find S_n from the first term to the n -th term of 1, -2, 4, -8,

Solution

1. The sequence has $a = 1$ and $r = -2$.
2. $S_n = \frac{1 - (-2)^n}{1 + 2} = \frac{1 - (-2)^n}{3}$.

Final answer

$$\frac{1 - (-2)^n}{3}$$

Worked Solutions

Geometric Sequences

Q153 MCQ

A culture grows by 3, 9, 27, 81 units. Find the four-stage total:

Solution

1. Recognise the listed stages as a finite geometric sequence and identify its first term and ratio.
2. Apply the finite-GP formula to obtain $S_n = 120$.

Correct option: A

Final answer

120

Worked Solutions

Geometric Sequences

Q154 Challenge

A filtration scale is geometric: $3, 1, \frac{1}{3}, \frac{1}{9}, \dots$ mg/L removed per stage. Calculate the total removed in 4 stages.

Solution

1. Here $a = 3$, $r = \frac{1}{3}$, and $n = 4$.
2. $S_4 = \frac{3\{1 - (1/3)^4\}}{1 - 1/3} = \frac{40}{9}$ mg/L.

Final answer

$$\frac{40}{9}$$

Worked Solutions

Geometric Sequences

Q155 MCQ

Given $S_3 = 7$ and the sum from the 3rd to the 5th terms is 28, find all real pairs (a, r) :

Solution

1. The shifted block is $r^2 S_3 = 28$, so $r^2 = 4$; then $a(1 + r + r^2) = 7$.
2. Solving in the real numbers gives $r = 2$ or $r = -2$.
3. Back-substitution gives $a = 1$ or $a = \frac{7}{3}$, respectively.

Correct option: D

Final answer

$$(a, r) = (1, 2) \text{ or } (a, r) = \left(\frac{7}{3}, -2\right)$$

Worked Solutions

Geometric Sequences

Q156 MCQ

Given $S_3 = 26$ and the sum from the 3rd to the 5th terms is 234, find all real pairs (a, r) :

Solution

1. The shifted block is $r^2 S_3 = 234$, so $r^2 = 9$; then $a(1 + r + r^2) = 26$.
2. Solving in the real numbers gives $r = 3$ or $r = -3$.
3. Back-substitution gives $a = 2$ or $a = \frac{26}{7}$, respectively.

Correct option: C

Final answer

$$(a, r) = (2, 3) \text{ or } (a, r) = \left(\frac{26}{7}, -3\right)$$

Worked Solutions

Geometric Sequences

Q157 MCQ

Given $S_3 = 21$ and the sum from the 3rd to the 5th terms is 84, find all real pairs (a, r) :

Solution

1. The shifted block is $r^2 S_3 = 84$, so $r^2 = 4$; then $a(1 + r + r^2) = 21$.
2. Solving in the real numbers gives $r = 2$ or $r = -2$.
3. Back-substitution gives $a = 3$ or $a = 7$, respectively.

Correct option: B

Final answer

$$(a, r) = (3, 2) \text{ or } (a, r) = (7, -2)$$

Worked Solutions

Geometric Sequences

Q158 MCQ

Given $S_3 = 28$ and the sum from the 3rd to the 5th terms is 112, find all real pairs (a, r) :

Solution

1. The shifted block is $r^2 S_3 = 112$, so $r^2 = 4$; then $a(1 + r + r^2) = 28$.
2. Solving in the real numbers gives $r = 2$ or $r = -2$.
3. Back-substitution gives $a = 4$ or $a = \frac{28}{3}$, respectively.

Correct option: A

Final answer

$$(a, r) = (4, 2) \text{ or } (a, r) = \left(\frac{28}{3}, -2\right)$$

Worked Solutions

Geometric Sequences

Q159 MCQ

Given $S_3 = 13$ and the sum from the 3rd to the 5th terms is 117, find all real pairs (a, r) :

Solution

1. The shifted block is $r^2 S_3 = 117$, so $r^2 = 9$; then $a(1 + r + r^2) = 13$.
2. Solving in the real numbers gives $r = 3$ or $r = -3$.
3. Back-substitution gives $a = 1$ or $a = \frac{13}{7}$, respectively.

Correct option: D

Final answer

$$(a, r) = (1, 3) \text{ or } (a, r) = \left(\frac{13}{7}, -3\right)$$

Worked Solutions

Geometric Sequences

Q160 MCQ

Given $S_3 = 35$ and the sum from the 3rd to the 5th terms is 140, find all real pairs (a, r) :

Solution

1. The shifted block is $r^2 S_3 = 140$, so $r^2 = 4$; then $a(1 + r + r^2) = 35$.
2. Solving in the real numbers gives $r = 2$ or $r = -2$.
3. Back-substitution gives $a = 5$ or $a = \frac{35}{3}$, respectively.

Correct option: C

Final answer

$$(a, r) = (5, 2) \text{ or } (a, r) = \left(\frac{35}{3}, -2\right)$$

Worked Solutions

Geometric Sequences

Q161 MCQ

Given $S_3 = \frac{19}{2}$ and the sum from the 3rd to the 5th terms is $\frac{171}{8}$, find all real pairs (a, r) :

Solution

1. The shifted block is $r^2 S_3 = \frac{171}{8}$, so $r^2 = \frac{9}{4}$; then $a(1 + r + r^2) = \frac{19}{2}$.
2. Solving in the real numbers gives $r = \frac{3}{2}$ or $r = -\frac{3}{2}$.
3. Back-substitution gives $a = 2$ or $a = \frac{38}{7}$, respectively.

Correct option: D**Final answer**

$$(a, r) = \left(2, \frac{3}{2}\right) \text{ or } (a, r) = \left(\frac{38}{7}, -\frac{3}{2}\right)$$

Worked Solutions

Geometric Sequences

Q162 Written

Find the first term and common ratio of a geometric sequence if $S_3 = 9$ and the sum from the 4th to the 6th term is -72 .

Solution

1. The 4th+6th block is $r^3 S_3 = -72$, so $r^3 = -8$ and $r = -2$.
2. Then $a(1 - 2 + 4) = 9$, giving $a = 3$.

Final answer

$$(a, r) = (3, -2)$$

Worked Solutions

Geometric Sequences

Q163 MCQ

Given $S_3 = 7$ and the sum from the 3rd to the 5th terms is $\frac{7}{4}$, find all real pairs (a, r) :

Solution

1. The shifted block is $r^2 S_3 = \frac{7}{4}$, so $r^2 = \frac{1}{4}$; then $a(1 + r + r^2) = 7$.
2. Solving in the real numbers gives $r = \frac{1}{2}$ or $r = -\frac{1}{2}$.
3. Back-substitution gives $a = 4$ or $a = \frac{28}{3}$, respectively.

Correct option: C

Final answer

$$(a, r) = \left(4, \frac{1}{2}\right) \text{ or } (a, r) = \left(\frac{28}{3}, -\frac{1}{2}\right)$$

Worked Solutions

Geometric Sequences

Q164 Written

Find the first term and common ratio of a geometric sequence if $S_3 = 35$ and the sum from the 3rd to the 5th term is 140.

Solution

1. Write $S_3 = a(1 + r + r^2) = 35$. The shifted block is $r^2 S_3 = 140$, so $r^2 = 4$ and $r = \pm 2$.
2. For $r = 2$, $7a = 35$, giving $a = 5$. For $r = -2$, $3a = 35$, giving $a = \frac{35}{3}$.

Final answer

$$(a, r) = (5, 2) \text{ or } (a, r) = \left(\frac{35}{3}, -2\right)$$

Q165 Written

Find the first term and common ratio of a geometric sequence if $S_3 = 21$ and the sum from the 3rd to the 5th term is 84.

Solution

1. The shifted block is $r^2 S_3 = 84$, so $r^2 = 4$ and $r = \pm 2$.
2. Using $a(1 + r + r^2) = 21$ gives $a = 3$ for $r = 2$ and $a = 7$ for $r = -2$.

Final answer

$$(a, r) = (3, 2) \text{ or } (a, r) = (7, -2)$$

Worked Solutions

Geometric Sequences

Q166 Written

Find the first term and common ratio of a geometric sequence if $S_3 = 9$ and the sum from the 3rd to the 5th term is 36.

Solution

1. The shifted block gives $r^2 S_3 = 36$, hence $r = \pm 2$.
2. Substitution into $a(1 + r + r^2) = 9$ gives $a = \frac{9}{7}$ for $r = 2$, or $a = 3$ for $r = -2$.

Final answer

$$(a, r) = \left(\frac{9}{7}, 2\right) \text{ or } (a, r) = (3, -2)$$

Q167 Written

Find the first term and common ratio of a geometric sequence if $S_3 = 78$ and the sum from the 3rd to the 5th term is 702.

Solution

1. The shifted block gives $r^2 S_3 = 702$, so $r^2 = 9$ and $r = \pm 3$.
2. Using $a(1 + r + r^2) = 78$ gives $a = 6$ for $r = 3$, or $a = \frac{78}{7}$ for $r = -3$.

Final answer

$$(a, r) = (6, 3) \text{ or } (a, r) = \left(\frac{78}{7}, -3\right)$$

Worked Solutions

Geometric Sequences

Q168 MCQ

Given $S_3 = 6$ and the sum from the 3rd to the 5th terms is 24, find all real pairs (a, r) :

Solution

1. The shifted block is $r^2 S_3 = 24$, so $r^2 = 4$; then $a(1 + r + r^2) = 6$.
2. Solving in the real numbers gives $r = -2$ or $r = 2$.
3. Back-substitution gives $a = 2$ or $a = \frac{6}{7}$, respectively.

Correct option: B

Final answer

$$(a, r) = (2, -2) \text{ or } (a, r) = \left(\frac{6}{7}, 2\right)$$

Worked Solutions

Geometric Sequences

Q169 MCQ

Given $S_3 = 9$ and the sum from the 3rd to the 5th terms is 36, find all real pairs (a, r) :

Solution

1. The shifted block is $r^2 S_3 = 36$, so $r^2 = 4$; then $a(1 + r + r^2) = 9$.
2. Solving in the real numbers gives $r = -2$ or $r = 2$.
3. Back-substitution gives $a = 3$ or $a = \frac{9}{7}$, respectively.

Correct option: A

Final answer

$$(a, r) = (3, -2) \text{ or } (a, r) = \left(\frac{9}{7}, 2\right)$$

Worked Solutions

Geometric Sequences

Q170 Written

Find the first term and common ratio of a geometric sequence if the 2nd term is 3 and $S_3 = 13$.

Solution

1. From $ar = 3$, write $a = 3/r$. Then $3/r + 3 + 3r = 13$, so $3r^2 - 10r + 3 = 0$.
2. Hence $r = 3$ or $r = \frac{1}{3}$, giving $a = 1$ or $a = 9$, respectively.

Final answer

$$(a, r) = (1, 3) \text{ or } (a, r) = \left(9, \frac{1}{3}\right)$$

Q171 Written

Find the first term and common ratio of a geometric sequence if the 2nd term is 6, and $S_3 = 21$.

Solution

1. From $ar = 6$, $a = 6/r$. Substitution into $S_3 = 21$ gives $6/r + 6 + 6r = 21$, hence $2r^2 - 5r + 2 = 0$.
2. Therefore $r = 2$ or $r = \frac{1}{2}$, giving $a = 3$ or $a = 12$, respectively.

Final answer

$$(a, r) = (3, 2) \text{ or } (a, r) = \left(12, \frac{1}{2}\right)$$

Worked Solutions

Geometric Sequences

Q172 MCQ

The second term is 6 and $S_3 = 21$. Find all real pairs (a, r) :**Solution**

1. Use $ar = 6$ and $a(1 + r + r^2) = 21$; after substitution, solve the quadratic in r .
2. Solving in the real numbers gives $r = 2$ or $r = \frac{1}{2}$.
3. Back-substitution gives $a = 3$ or $a = 12$, respectively.

Correct option: C**Final answer**

$$(a, r) = (3, 2) \text{ or } (a, r) = \left(12, \frac{1}{2}\right)$$

Worked Solutions

Geometric Sequences

Q173 MCQ

The second term is 6 and $S_3 = 26$. Find all real pairs (a, r) :

Solution

1. Use $ar = 6$ and $a(1 + r + r^2) = 26$; after substitution, solve the quadratic in r .
2. Solving in the real numbers gives $r = 3$ or $r = \frac{1}{3}$.
3. Back-substitution gives $a = 2$ or $a = 18$, respectively.

Correct option: D

Final answer

$$(a, r) = (2, 3) \text{ or } (a, r) = \left(18, \frac{1}{3}\right)$$

Worked Solutions

Geometric Sequences

Q174 MCQ

The second term is 8 and $S_3 = 28$. Find all real pairs (a, r) :**Solution**

1. Use $ar = 8$ and $a(1 + r + r^2) = 28$; after substitution, solve the quadratic in r .
2. Solving in the real numbers gives $r = 2$ or $r = \frac{1}{2}$.
3. Back-substitution gives $a = 4$ or $a = 16$, respectively.

Correct option: B**Final answer**

$$(a, r) = (4, 2) \text{ or } (a, r) = \left(16, \frac{1}{2}\right)$$

Worked Solutions

Geometric Sequences

Q175 MCQ

The second term is 3 and $S_3 = 13$. Find all real pairs (a, r) :

Solution

1. Use $ar = 3$ and $a(1 + r + r^2) = 13$; after substitution, solve the quadratic in r .
2. Solving in the real numbers gives $r = 3$ or $r = \frac{1}{3}$.
3. Back-substitution gives $a = 1$ or $a = 9$, respectively.

Correct option: C

Final answer

$$(a, r) = (1, 3) \text{ or } (a, r) = \left(9, \frac{1}{3}\right)$$

Worked Solutions

Geometric Sequences

Q176 MCQ

The second term is $\frac{5}{2}$ and $S_3 = \frac{35}{4}$. Find all real pairs (a, r) :

Solution

1. Use $ar = \frac{5}{2}$ and $a(1 + r + r^2) = \frac{35}{4}$; after substitution, solve the quadratic in r .
2. Solving in the real numbers gives $r = \frac{1}{2}$ or $r = 2$.
3. Back-substitution gives $a = 5$ or $a = \frac{5}{4}$, respectively.

Correct option: D**Final answer**

$$(a, r) = \left(5, \frac{1}{2}\right) \text{ or } (a, r) = \left(\frac{5}{4}, 2\right)$$

Worked Solutions

Geometric Sequences

Q177 Written

Find the first term and common ratio of a geometric sequence if the 3rd term is 4 and $S_3 = 7$.

Solution

1. The data give $ar^2 = 4$ and $a + ar + ar^2 = 7$. Substitute $a = 4/r^2$: $4 + 4r + 4r^2 = 7r^2$, so $3r^2 - 4r - 4 = 0$.
2. Thus $r = 2$ or $r = -\frac{2}{3}$. Back-substitution gives $a = 1$ or $a = 9$, respectively.

Final answer

$$(a, r) = (1, 2) \text{ or } (a, r) = \left(9, -\frac{2}{3}\right)$$

Q178 Written

Find the first term and common ratio of a geometric sequence if the 3rd term is 12, and $S_3 = 9$.

Solution

1. Use $ar^2 = 12$ and $a + ar + ar^2 = 9$. After multiplying by r^2 , $r^2 + 4r + 4 = 0$, so $r = -2$.
2. Then $a = 12/(-2)^2 = 3$.

Final answer

$$(a, r) = (3, -2)$$

Worked Solutions

Geometric Sequences

Q179 Written

Find the first term and common ratio of a geometric sequence if the 3rd term is 20, and $S_3 = 35$.

Solution

1. Use $ar^2 = 20$ and $a + ar + ar^2 = 35$. Multiplication by r^2 gives $3r^2 - 4r - 4 = 0$.
2. Thus $r = 2$ or $r = -\frac{2}{3}$, and $a = 5$ or $a = 45$, respectively.

Final answer

$$(a, r) = (5, 2) \text{ or } (a, r) = \left(45, -\frac{2}{3}\right)$$

Q180 Written

Find the first term and common ratio of a geometric sequence if $S_3 = 21$ and $S_6 = 189$.

Solution

1. Use $S_6 = S_3 + r^3 S_3$: $189 = 21(1 + r^3)$, hence $r^3 = 8$ and $r = 2$ in the real numbers.
2. Then $a(1 + 2 + 4) = 21$, so $a = 3$.

Final answer

$$(a, r) = (3, 2)$$

Worked Solutions

Geometric Sequences

Q181 Written

Find the first term and common ratio of a geometric sequence if $S_3 = 26$ and $S_6 = 728$.

Solution

1. Since $S_6 = S_3 + r^3 S_3$, $728 = 26(1 + r^3)$, so $r^3 = 27$ and $r = 3$.
2. Then $a(1 + 3 + 9) = 26$, giving $a = 2$.

Final answer

$$(a, r) = (2, 3)$$

Q182 Written

Find the first term and common ratio of a geometric sequence if $S_3 = 65$ and $S_6 = 1820$.

Solution

1. Use $S_6 = S_3(1 + r^3)$: $1820 = 65(1 + r^3)$, so $r^3 = 27$ and $r = 3$.
2. Then $13a = 65$, giving $a = 5$.

Final answer

$$(a, r) = (5, 3)$$

Worked Solutions

Geometric Sequences

Q183 Written

Find the first term and common ratio of a geometric sequence if the sum of the first and second terms is 3, and $S_4 = 51$.

Solution

1. Let $a + ar = 3$. Factor $S_4 = (a + ar)(1 + r^2)$, so $3(1 + r^2) = 51$ and $r = \pm 4$.
2. Then $a = 3/(1 + r)$, giving $a = \frac{3}{5}$ for $r = 4$, or $a = -1$ for $r = -4$.

Final answer

$$(a, r) = \left(\frac{3}{5}, 4\right) \text{ or } (a, r) = (-1, -4)$$

Q184 Written

Find the first term and common ratio of a geometric sequence if the sum of the first and second terms is -8 , and $S_4 = -80$.

Solution

1. Factor $S_4 = (a + ar)(1 + r^2)$. Thus $-8(1 + r^2) = -80$, so $r = \pm 3$.
2. Using $a(1 + r) = -8$ gives $a = -2$ for $r = 3$, or $a = 4$ for $r = -3$.

Final answer

$$(a, r) = (-2, 3) \text{ or } (a, r) = (4, -3)$$

Worked Solutions

Geometric Sequences

Q185 MCQ

A geometric sequence has $S_6 = 63$, $S_7 = 127$, and $S_8 = 255$. Find a_{10} :

Solution

1. $a_7 = S_7 - S_6 = 64$ and $a_8 = S_8 - S_7 = 128$.
2. Thus $r = a_8/a_7 = 2$, and $a_{10} = a_8 r^2 = 512$.

Correct option: D

Final answer

512

Worked Solutions

Geometric Sequences

Q186 Challenge

If S_n is the sum of the first n terms of a geometric sequence, $S_8 = 255$, $S_6 = 63$, and $a_8 - a_7 = 64$, find a_{10} .

Solution

1. $S_8 - S_6 = a_7 + a_8 = 255 - 63 = 192$. Together with $a_8 - a_7 = 64$, this gives $a_7 = 64$ and $a_8 = 128$.
2. Hence $r = a_8/a_7 = 2$, and $a_{10} = a_8 r^2 = 128(4) = 512$.

Final answer

$$a_7 = 64, a_8 = 128, r = 2, a_{10} = 512$$

Worked Solutions

Geometric Sequences

Quiz MCQ 1 [Concept Quiz · MCQ](#)

Select the correct option: For $a = 3, r = 2$, find a_6 :

Solution

1. $a_6 = 3(2)^5 = 96$

Correct option: B

Final answer

96

Quiz MCQ 2 [Concept Quiz · MCQ](#)

Select the correct option: If 2, x , 8 are consecutive geometric terms, find x :

Solution

1. $x^2 = 16 \Rightarrow x = \pm 4$

Correct option: C

Final answer

± 4

Worked Solutions

Geometric Sequences

Quiz MCQ 3 [Concept Quiz](#) · [MCQ](#)

For 1, 3, 9, ..., find S_n :

Solution

1. $a = 1, r = 3$, so $S_n = (3^n - 1)/2$.

Correct option: B

Final answer

$$\frac{3^n - 1}{2}$$

Quiz MCQ 4 [Concept Quiz](#) · [MCQ](#)

A geometric sequence has $S_6 = 63$, $S_7 = 127$, and $S_8 = 255$. Find a_{10} :

Solution

1. $a_7 = S_7 - S_6 = 64$, $a_8 = S_8 - S_7 = 128$, so $r = 2$.

2. Therefore $a_{10} = 128(2^2) = 512$.

Correct option: D

Final answer

512

Worked Solutions

Geometric Sequences

Quiz WRITTEN 5 Concept Quiz · Written

Find the general term of $-4, 8, -16, 32, \dots$

Solution

1. The ratio is -2 .
2. $a_n = -4(-2)^{n-1}$

Final answer

$$-4(-2)^{n-1}$$

Quiz WRITTEN 6 Concept Quiz · Written

If $1, x, 25$ are consecutive geometric terms, find x .

Solution

1. $x^2 = 25 \Rightarrow x = \pm 5$

Final answer

$$x = \pm 5$$

Worked Solutions

Geometric Sequences

Quiz WRITTEN 7 Concept Quiz · Written

A four-stage process has outputs 5, 15, 45, 135. Find the total output.

Solution

1. $S_4 = 5(1 + 3 + 9 + 27) = 200.$

Final answer

200

Quiz WRITTEN 8 Concept Quiz · Written

A geometric sequence has $S_6 = 63$, $S_7 = 127$, and $S_8 = 255$. Find a_{10} .

Solution

1. $a_7 = 64$, $a_8 = 128$, hence $r = 2$.

2. $a_{10} = a_8 r^2 = 512.$

Final answer

512

Answer key

Use the question number shown on each page.

Q001 $2 \times 3^{n-1}$

Q002 $3 \times 4^{n-1}$

Q003 $5 \times 2^{n-1}$

Q004 $2 \times 5^{n-1}$

Q005 $7 \times 4^{n-1}$

Q006 A

Q007 A

Q008 A

Q009 A

Q010 A

Q011 A

Q012 $5(-2)^{n-1}$

Q013 $2(-3)^{n-1}$

Q014 $3(-5)^{n-1}$

Q015 $4(-2)^{n-1}$

Q016 A

Q017 A

Q018 A

Q019 A

Q020 A

Q021 $8\left(-\frac{1}{3}\right)^{n-1}$

Q022 $4\left(\frac{1}{3}\right)^{n-1}$

Q023 $3\left(-\frac{1}{2}\right)^{n-1}$

Q024 $6\left(\frac{1}{2}\right)^{n-1}$

Q025 A

Q026 A

Q027 A

Q028 A

Q029 A

Q030 $5 \times 2^{n-1}$

Q031 $2 \times 3^{n-1}$

Q032 $3 \times 2^{n-1}$

Q033 A

Q034 A

Q035 A

Q036 A

Q037 A

Q038 $2(-3)^{n-1}$

Q039 $5(-2)^{n-1}$

Q040 A

Q041 A

Q042 A

Q043 A

Q044 $-4\left(-\frac{3}{2}\right)^{n-1}$

Q045 $-8\left(-\frac{3}{2}\right)^{n-1}$

Q046 $2\left(\frac{1}{3}\right)^{n-1}$

Q047 $48\left(\frac{1}{2}\right)^{n-1}$

Q048 $\frac{1}{6}\left(\frac{3}{2}\right)^{n-1}$

Q049 $8(2\sqrt{2})^{n-1}$

Q050 $3\left(-\frac{5}{2}\right)^{n-1}$

Q051 $9\left(-\frac{\sqrt{3}}{3}\right)^{n-1}$

Q052 $5\left(-\frac{2}{3}\right)^{n-1}$

Q053 A

Q054 A

Q055 A

Q056 A

Q057 A

Q058 A

Q059 A

Q060 $r = -\frac{1}{3}, A = 54, B = -2$

Q061 A

Q062 $r = \frac{1}{2}, A = 48, B = 6$

Q063 $r = -3, A = 1, B = -27$

Q064 $r = \frac{1}{3}, A = 1, B = \frac{1}{27}$

Q065 $r = -\frac{1}{2}, A = \frac{1}{2}, B = -\frac{1}{16}$

Q066 $a_7 = 648$

Q067 $a_n = 3(2)^{n-1}$ or $-3(-2)^{n-1}$

Q068 $a_n = 2(3)^{n-1}$ or $2(-3)^{n-1}$

Q069 $a_n = 2(3)^{n-1}$ or $-2(-3)^{n-1}$

Q070 $a_n = 3(3)^{n-1}$ or $3(-3)^{n-1}$

Q071 $a_n = 7(2)^{n-1}$

Q072 $a_n = 3(-2)^{n-1}$

Q073 $a_n = 3(-2)^{n-1}$

Q074 $a_n = 3(2)^{n-1}$ or $3(-2)^{n-1}$

Q075 $a_n = -7(2)^{n-1}$ or $-7(-2)^{n-1}$

Q076 $a_n = -2(-2)^{n-1}$

Q077 $a_n = -5(-2)^{n-1}$

Q078 $a_n = \frac{3}{4}(4)^{n-1}$ or $-\frac{3}{4}(-4)^{n-1}$

Q079 $a_n = 4\left(\frac{3}{2}\right)^{n-1}$ or $4\left(-\frac{3}{2}\right)^{n-1}$

Q080 $a_n = \frac{2}{243}(3)^{n-1}$ or $-\frac{2}{243}(-3)^{n-1}$

Q081 A

Q082 A

Q083 A

Q084 A

Q085 A

Q086 A

Q087 A

Q088 A

Q089 A

Q090 A

Q091 A

Q092 A

Q093 A

Q094 A

Q095 A	Q122 $\frac{189}{16}$	Q148 $\frac{1 - (-3)^n}{2}$	Q171 $(a, r) = (3, 2)$ or $(a, r) = \left(12, \frac{1}{2}\right)$
Q096 A	Q123 $\frac{13}{9}$	Q149 $\frac{16}{3}\{1 - (-\frac{1}{2})^n\}$	Q172 C
Q097 $x = \pm 4$	Q124 $\frac{255}{128}$	Q150 $\frac{1 - (-3)^n}{4}$	Q173 D
Q098 $x = \pm 5$	Q125 B	Q151 $\frac{32}{3}\{1 - (-\frac{1}{2})^n\}$	Q174 B
Q099 $x = \pm 2$	Q126 A	Q152 $\frac{1 - (-2)^n}{3}$	Q175 C
Q100 $x = \pm 2\sqrt{3}$	Q127 D	Q153 A	Q176 D
Q101 A	Q128 C	Q154 $\frac{40}{9}$	Q177 $(a, r) = (1, 2)$ or $(a, r) = \left(9, -\frac{2}{3}\right)$
Q102 A	Q129 B	Q155 D	Q178 $(a, r) = (3, -2)$
Q103 A	Q130 A	Q156 C	Q179 $(a, r) = (5, 2)$ or $(a, r) = \left(45, -\frac{2}{3}\right)$
Q104 A	Q131 -30	Q157 B	Q180 $(a, r) = (3, 2)$
Q105 A	Q132 -20	Q158 A	Q181 $(a, r) = (2, 3)$
Q106 A	Q133 -364	Q159 D	Q182 $(a, r) = (5, 3)$
Q107 $(a, b) = (9, 12)$ or $(1, -4)$	Q134 D	Q160 C	Q183 $(a, r) = \left(\frac{3}{5}, 4\right)$ or $(a, r) = (-1, -4)$
Q108 $(a, b) = (2, 2)$ or $(-4, -1)$	Q135 C	Q161 D	Q184 $(a, r) = (-2, 3)$ or $(a, r) = (4, -3)$
Q109 $(a, b) = (2, 6)$ or $(\frac{1}{2}, 3)$	Q136 B	Q162 $(a, r) = (3, -2)$	Q185 D
Q110 $(a, b) = (5, 15)$ or $(\frac{5}{4}, \frac{15}{2})$	Q137 A	Q163 C	Q186 $a_7 = 64, a_8 = 128, r = 2, a_{10} = 512$
Q111 484	Q138 D	Q164 $(a, r) = (5, 2)$ or $(a, r) = \left(\frac{35}{3}, -2\right)$	Quiz MCQ 1 B
Q112 93	Q139 C	Q165 $(a, r) = (3, 2)$ or $(a, r) = (7, -2)$	Quiz MCQ 2 C
Q113 200	Q140 $1 - (-2)^n$	Q166 $(a, r) = \left(\frac{9}{7}, 2\right)$ or $(a, r) = (3, -2)$	Quiz MCQ 3 B
Q114 3069	Q141 $7(2^n - 1)$	Q167 $(a, r) = (6, 3)$ or $(a, r) = \left(\frac{78}{7}, -3\right)$	Quiz MCQ 4 D
Q115 D	Q142 $3\{1 - (\frac{1}{3})^n\}$	Q168 B	Quiz WRITTEN 5 $-4(-2)^{n-1}$
Q116 C	Q143 $4(2^n - 1)$	Q169 A	Quiz WRITTEN 6 $x = \pm 5$
Q117 B	Q144 C	Q170 $(a, r) = (1, 3)$ or $(a, r) = \left(9, \frac{1}{3}\right)$	Quiz WRITTEN 7 200
Q118 A	Q145 D		Quiz WRITTEN 8 512
Q119 D	Q146 A		
Q120 C	Q147 $3^n - 1$		
Q121 $\frac{63}{4}$			

Concept 3 | Summation

English Student Edition • Focused formulas and guided practice

Formula opener

Linearity

$$\sum (au_k + bv_k) = a \sum u_k + b \sum v_k$$

Standard sums

$$\sum k = \frac{n(n+1)}{2}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Worked Solutions

Summation

Q001 Written

Write $\sum_{k=1}^5 2k$ as a sum of terms.

Solution

1. Substitute $k = 1, 2, 3, 4, 5$.
2. $\sum_{k=1}^5 2k = 2 + 4 + 6 + 8 + 10$

Final answer

$$2 + 4 + 6 + 8 + 10$$

Q002 Written

Write $\sum_{k=1}^5 (2k + 1)$ as a sum of terms.

Solution

1. Substitute $k = 1, 2, 3, 4, 5$.
2. $\sum_{k=1}^5 (2k + 1) = 3 + 5 + 7 + 9 + 11$

Final answer

$$3 + 5 + 7 + 9 + 11$$

Worked Solutions

Summation

Q003 Written

Write $\sum_{k=1}^5 3k$ as a sum of terms.

Solution

1. Substitute $k = 1, 2, 3, 4, 5$.
2. $\sum_{k=1}^5 3k = 3 + 6 + 9 + 12 + 15$

Final answer

$$3 + 6 + 9 + 12 + 15$$

Q004 Written

Write $\sum_{k=1}^4 (4k - 2)$ as a sum of terms.

Solution

1. Substitute $k = 1, 2, 3, 4$.
2. $\sum_{k=1}^4 (4k - 2) = 2 + 6 + 10 + 14$

Final answer

$$2 + 6 + 10 + 14$$

Worked Solutions

Summation

Q005 Written

Write $\sum_{k=1}^5 (2k - 3)$ as a sum of terms.

Solution

1. Substitute $k = 1, 2, 3, 4, 5$.
2. $\sum_{k=1}^5 (2k - 3) = -1 + 1 + 3 + 5 + 7$

Final answer

$$-1 + 1 + 3 + 5 + 7$$

Q006 MCQ

Select the correct option: Expand $\sum_{k=1}^3 4k$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$4 + 8 + 12$$

Worked Solutions

Summation

Q007 MCQ

Select the correct option: Expand $\sum_{k=1}^4 2^k$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$2 + 4 + 8 + 16$$

Q008 MCQ

Select the correct option: Expand $\sum_{k=1}^3 (5^k - 1)$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$4 + 24 + 124$$

Worked Solutions

Summation

Q009 MCQ

Select the correct option: Expand $\sum_{k=1}^5 (k + 4)$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$5 + 6 + 7 + 8 + 9$$

Q010 MCQ

Select the correct option: Expand $\sum_{k=2}^5 (2k + 3)$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$7 + 9 + 11 + 13$$

Worked Solutions

Summation

Q011 MCQ

Select the correct option: Expand $\sum_{k=1}^3 (2^k + 1)$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$3 + 5 + 9$$

Q012 MCQ

Select the correct option: Expand $\sum_{k=1}^3 (k^2 + 3k)$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$4 + 10 + 18$$

Worked Solutions

Summation

Q013 MCQ

Select the correct option: Expand $\sum_{k=2}^5 (k^2 + 1)$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$5 + 10 + 17 + 26$$

Q014 Written

Write $\sum_{k=1}^3 (k^2 - 3)$ as a sum of terms.

Solution

1. Substitute $k = 1, 2, 3$.
2. $\sum_{k=1}^3 (k^2 - 3) = -2 + 1 + 6$

Final answer

$$-2 + 1 + 6$$

Worked Solutions

Summation

Q015 Written

Write $\sum_{k=1}^3 (3k^2 + 2)$ as a sum of terms.

Solution

1. Substitute $k = 1, 2, 3$.
2. $\sum_{k=1}^3 (3k^2 + 2) = 5 + 14 + 29$

Final answer

$$5 + 14 + 29$$

Q016 Written

Write $\sum_{k=1}^4 k^2$ as a sum of terms.

Solution

1. Substitute $k = 1, 2, 3, 4$.
2. $\sum_{k=1}^4 k^2 = 1 + 4 + 9 + 16$

Final answer

$$1 + 4 + 9 + 16$$

Worked Solutions

Summation

Q017 Written

Write $\sum_{k=1}^3 (2k^2 + 3)$ as a sum of terms.

Solution

1. Substitute $k = 1, 2, 3$.
2. $\sum_{k=1}^3 (2k^2 + 3) = 5 + 11 + 21$

Final answer

$$5 + 11 + 21$$

Q018 Written

Write $\sum_{k=1}^4 (k^2 - 2k)$ as a sum of terms.

Solution

1. Substitute $k = 1, 2, 3, 4$.
2. $\sum_{k=1}^4 (k^2 - 2k) = -1 + 0 + 3 + 8$

Final answer

$$-1 + 0 + 3 + 8$$

Worked Solutions

Summation

Q019 MCQ

Select the correct option: Expand $\sum_{k=1}^4 (2k + 1)$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$3 + 5 + 7 + 9$$

Q020 MCQ

Select the correct option: Expand $\sum_{k=2}^5 (3k - 1)$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$5 + 8 + 11 + 14$$

Worked Solutions

Summation

Q021 MCQ

Select the correct option: Expand $\sum_{k=1}^4 (2k^2 - 3)$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$-1 + 5 + 15 + 29$$

Q022 MCQ

Select the correct option: Expand $\sum_{k=1}^4 (3k^2 + 1)$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$4 + 13 + 28 + 49$$

Worked Solutions

Summation

Q023 MCQ

Select the correct option: Expand $\sum_{k=1}^4 (k^2 - k)$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$0 + 2 + 6 + 12$

Q024 MCQ

Select the correct option: Expand $\sum_{k=2}^4 5k$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$10 + 15 + 20$

Worked Solutions

Summation

Q025 MCQ

Select the correct option: Expand $\sum_{k=1}^4 3$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$3 + 3 + 3 + 3$$

Q026 MCQ

Select the correct option: Expand $\sum_{k=1}^4 (5k + 1)$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$6 + 11 + 16 + 21$$

Worked Solutions

Summation

Q027 Written

Write $\sum_{k=1}^4 3^k$ as a sum of terms.

Solution

1. Substitute $k = 1, 2, 3, 4$.
2. $\sum_{k=1}^4 3^k = 3 + 9 + 27 + 81$

Final answer

$$3 + 9 + 27 + 81$$

Q028 Written

Write $\sum_{k=1}^3 2$ as a sum of terms.

Solution

1. The summand is constant.
2. $\sum_{k=1}^3 2 = 2 + 2 + 2$

Final answer

$$2 + 2 + 2$$

Worked Solutions

Summation

Q029 Written

Write $\sum_{k=1}^4 2^k$ as a sum of terms.

Solution

1. Substitute $k = 1, 2, 3, 4$.
2. $\sum_{k=1}^4 2^k = 2 + 4 + 8 + 16$

Final answer

$$2 + 4 + 8 + 16$$

Q030 Written

Write $\sum_{k=1}^6 3$ as a sum of terms.

Solution

1. The summand is constant.
2. $\sum_{k=1}^6 3 = 3 + 3 + 3 + 3 + 3 + 3$

Final answer

$$3 + 3 + 3 + 3 + 3 + 3$$

Worked Solutions

Summation

Q031 Written

Write $\sum_{k=1}^3 3^k$ as a sum of terms.

Solution

1. Substitute $k = 1, 2, 3$.
2. $\sum_{k=1}^3 3^k = 3 + 9 + 27$

Final answer

$$3 + 9 + 27$$

Q032 Written

Write $\sum_{k=1}^3 (5^k + 3)$ as a sum of terms.

Solution

1. Substitute $k = 1, 2, 3$.
2. $\sum_{k=1}^3 (5^k + 3) = 8 + 28 + 128$

Final answer

$$8 + 28 + 128$$

Worked Solutions

Summation

Q033 Written

Write $\sum_{k=1}^4 (2^k - 3)$ as a sum of terms.

Solution

1. Substitute $k = 1, 2, 3, 4$.
2. $\sum_{k=1}^4 (2^k - 3) = -1 + 1 + 5 + 13$

Final answer

$$-1 + 1 + 5 + 13$$

Q034 Written

Write $\sum_{k=1}^4 5$ as a sum of terms.

Solution

1. The summand is constant.
2. $\sum_{k=1}^4 5 = 5 + 5 + 5 + 5$

Final answer

$$5 + 5 + 5 + 5$$

Worked Solutions

Summation

Q035 Written

Write $\sum_{k=1}^3 3$ as a sum of terms.

Solution

1. The summand is constant.
2. $\sum_{k=1}^3 3 = 3 + 3 + 3$

Final answer

$$3 + 3 + 3$$

Q036 Written

Write $\sum_{k=1}^5 4$ as a sum of terms.

Solution

1. The summand is constant.
2. $\sum_{k=1}^5 4 = 4 + 4 + 4 + 4 + 4$

Final answer

$$4 + 4 + 4 + 4 + 4$$

Worked Solutions

Summation

Q037 MCQ

Select the correct option: Expand $\sum_{k=1}^3 (k^2 + 2)$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$3 + 6 + 11$$

Q038 MCQ

Select the correct option: Expand $\sum_{k=1}^5 7$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$7 + 7 + 7 + 7 + 7$$

Worked Solutions

Summation

Q039 MCQ

Select the correct option: Expand $\sum_{k=3}^6 k^2$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$9 + 16 + 25 + 36$$

Q040 MCQ

Select the correct option: Expand $\sum_{k=1}^3 6^k$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$6 + 36 + 216$$

Worked Solutions

Summation

Q041 MCQ

Select the correct option: Expand $\sum_{k=1}^5 (4k - 3)$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$1 + 5 + 9 + 13 + 17$$

Q042 MCQ

Select the correct option: Expand $\sum_{k=1}^4 (3k - 5)$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$-2 + 1 + 4 + 7$$

Worked Solutions

Summation

Q043 MCQ

Select the correct option: Expand $\sum_{k=1}^3 (4^k - 2)$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$2 + 14 + 62$$

Q044 MCQ

Select the correct option: Expand $\sum_{k=1}^3 (2k^2 - k)$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$1 + 6 + 15$$

Worked Solutions

Summation

Q045 Written

Express $3 + 7 + 11 + \cdots + (4n - 1)$ **using the summation symbol** Σ .**Solution**

1. The general term is $4k - 1$.
2. $\sum_{k=1}^n (4k - 1)$

Final answer

$$\sum_{k=1}^n (4k - 1)$$

Q046 Written

Express $5 + 8 + 11 + \cdots + (3n + 2)$ **using** Σ .**Solution**

1. The general term is $3k + 2$.
2. $\sum_{k=1}^n (3k + 2)$

Final answer

$$\sum_{k=1}^n (3k + 2)$$

Worked Solutions

Summation

Q047 Written

Write the sum of the first n terms of the sequence $a_n = 2n + 1$ using Σ .

Solution

1. Change the term index from n to k .
2. $\sum_{k=1}^n (2k + 1)$

Final answer

$$\sum_{k=1}^n (2k + 1)$$

Q048 Written

Express $1 + 3 + 5 + \dots + (2n - 1)$ using Σ .

Solution

1. Identify the general term and the endpoint indices.
2. $\sum_{k=1}^n (2k - 1)$

Final answer

$$\sum_{k=1}^n (2k - 1)$$

Worked Solutions

Summation

Q049 Written

Write the sum of the first n terms of $a_n = 7n - 1$ using Σ .**Solution**

1. Identify the general term and the endpoint indices.
2. $\sum_{k=1}^n (7k - 1)$

Final answer

$$\sum_{k=1}^n (7k - 1)$$

Q050 Written

Write the sum of the first n terms of $a_n = 4n + 5$ using Σ .**Solution**

1. Identify the general term and the endpoint indices.
2. $\sum_{k=1}^n (4k + 5)$

Final answer

$$\sum_{k=1}^n (4k + 5)$$

Worked Solutions

Summation

Q051 Written

Write the sum of the first n terms of $a_n = 3n - 2$ using Σ .

Solution

1. Identify the general term and the endpoint indices.
2. $\sum_{k=1}^n (3k - 2)$

Final answer

$$\sum_{k=1}^n (3k - 2)$$

Q052 MCQ

Select the correct option: Express $2 + 5 + 8 + \cdots + (3n - 1)$ using the summation symbol Σ :

Solution

1. Find the general term and the first/last index.

Correct option: A

Final answer

$$\sum_{k=1}^n (3k - 1)$$

Worked Solutions

Summation

Q053 MCQ

Select the correct option: Express $7 + 11 + 15 + \dots + 43$ using the summation symbol Σ :**Solution**

1. Find the general term and the first/last index.

Correct option: A**Final answer**

$$\sum_{k=1}^{10} (4k + 3)$$

Q054 MCQ

Select the correct option: Express $6 + 10 + 14 + \dots + (4n + 2)$ using the summation symbol Σ :**Solution**

1. Find the general term and the first/last index.

Correct option: A**Final answer**

$$\sum_{k=1}^n (4k + 2)$$

Worked Solutions

Summation

Q055 MCQ

Select the correct option: Express $8 + 13 + 18 + \dots + 103$ using the summation symbol Σ :**Solution**

1. Find the general term and the first/last index.

Correct option: A

Final answer

$$\sum_{k=1}^{20} (5k + 3)$$

Q056 MCQ

Select the correct option: Express $11^2 + 12^2 + \dots + 19^2$ using the summation symbol Σ :**Solution**

1. Find the general term and the first/last index.

Correct option: A

Final answer

$$\sum_{k=11}^{19} k^2$$

Worked Solutions

Summation

Q057 MCQ

Select the correct option: Express $2^2 + 3^2 + \dots + 10^2$ using the summation symbol Σ :**Solution**

1. Find the general term and the first/last index.

Correct option: A**Final answer**

$$\sum_{k=2}^{10} k^2$$

Q058 MCQ

Select the correct option: Express $1 + 3 + 9 + \dots + 3^{n-1}$ using the summation symbol Σ :**Solution**

1. Find the general term and the first/last index.

Correct option: A**Final answer**

$$\sum_{k=0}^{n-1} 3^k$$

Worked Solutions

Summation

Q059 MCQ

Select the correct option: Express $2 + 7 + 12 + \dots + (5n - 3)$ using the summation symbol Σ :

Solution

1. Find the general term and the first/last index.

Correct option: A

Final answer

$$\sum_{k=1}^n (5k - 3)$$

Q060 Written

Express $6^2 + 7^2 + 8^2 + \dots + 15^2$ using Σ .

Solution

1. The index starts at 6 and ends at 15.

2. $\sum_{k=6}^{15} k^2$

Final answer

$$\sum_{k=6}^{15} k^2$$

Worked Solutions

Summation

Q061 Written

Express $4^2 + 5^2 + 6^2 + \cdots + 13^2$ **using** Σ .**Solution**

1. The index starts at 4 and ends at 13.
2. $\sum_{k=4}^{13} k^2$

Final answer

$$\sum_{k=4}^{13} k^2$$

Q062 Written

Express $1 + 7 + 25 + 79 + \cdots + (3^n - 2)$ **using** Σ .**Solution**

1. Identify the general term and the endpoint indices.
2. $\sum_{k=1}^n (3^k - 2)$

Final answer

$$\sum_{k=1}^n (3^k - 2)$$

Worked Solutions

Summation

Q063 Written

Express $1 + 2 + 4 + 8 + \dots + 2^{n-1}$ using Σ .

Solution

1. Identify the general term and the endpoint indices.
2. $\sum_{k=1}^n 2^{k-1}$

Final answer

$$\sum_{k=1}^n 2^{k-1}$$

Q064 Written

Express $7^2 + 8^2 + \dots + 21^2$ using Σ .

Solution

1. Identify the general term and the endpoint indices.
2. $\sum_{k=7}^{21} k^2$

Final answer

$$\sum_{k=7}^{21} k^2$$

Worked Solutions

Summation

Q065 Written

Express $12^2 + 13^2 + \dots + 18^2$ **using** Σ .**Solution**

1. Identify the general term and the endpoint indices.
2. $\sum_{k=12}^{18} k^2$

Final answer

$$\sum_{k=12}^{18} k^2$$

Q066 Written

Express $10^2 + 11^2 + \dots + 20^2$ **using** Σ .**Solution**

1. Identify the general term and the endpoint indices.
2. $\sum_{k=10}^{20} k^2$

Final answer

$$\sum_{k=10}^{20} k^2$$

Worked Solutions

Summation

Q067 MCQ

Select the correct option: Express $4^2 + 5^2 + \dots + 12^2$ using the summation symbol Σ :

Solution

1. Find the general term and the first/last index.

Correct option: A

Final answer

$$\sum_{k=4}^{12} k^2$$

Q068 MCQ

Select the correct option: Express $3^1 + 3^2 + \dots + 3^n$ using the summation symbol Σ :

Solution

1. Find the general term and the first/last index.

Correct option: A

Final answer

$$\sum_{k=1}^n 3^k$$

Worked Solutions

Summation

Q069 MCQ

Select the correct option: Express $2 + 4 + 8 + \dots + 2^n$ using the summation symbol Σ :**Solution**

1. Find the general term and the first/last index.

Correct option: A**Final answer**

$$\sum_{k=1}^n 2^k$$

Q070 MCQ

Select the correct option: Express $1 + 2^1 + 2^2 + \dots + 2^{n-1}$ using the summation symbol Σ :**Solution**

1. Find the general term and the first/last index.

Correct option: A**Final answer**

$$\sum_{k=0}^{n-1} 2^k$$

Worked Solutions

Summation

Q071 MCQ

Select the correct option: Express $7 + 14 + 21 + \dots + 7n$ using the summation symbol Σ :**Solution**

1. Find the general term and the first/last index.

Correct option: A

Final answer

$$\sum_{k=1}^n 7k$$

Q072 MCQ

Select the correct option: Express $10 + 15 + 20 + \dots + 5n$ using the summation symbol Σ :**Solution**

1. Find the general term and the first/last index.

Correct option: A

Final answer

$$\sum_{k=2}^n 5k$$

Worked Solutions

Summation

Q073 MCQ

Select the correct option: Express $6^2 + 7^2 + \dots + 16^2$ using the summation symbol Σ :

Solution

1. Find the general term and the first/last index.

Correct option: A

Final answer

$$\sum_{k=6}^{16} k^2$$

Q074 Written

Express $2 + 5 + 8 + \dots + 29$ using Σ .

Solution

1. The general term is $3k - 1$, and $29 = 3(10) - 1$.
2. $\sum_{k=1}^{10} (3k - 1)$

Final answer

$$\sum_{k=1}^{10} (3k - 1)$$

Worked Solutions

Summation

Q075 **Written****Express** $8 + 16 + 24 + \cdots + 56$ **using** Σ .**Solution**

1. Identify the general term and the endpoint indices.
2. $\sum_{k=1}^7 8k$

Final answer

$$\sum_{k=1}^7 8k$$

Q076 **Written****Express** $7 + 13 + 19 + \cdots + 73$ **using** Σ .**Solution**

1. Identify the general term and the endpoint indices.
2. $\sum_{k=1}^{12} (6k + 1)$

Final answer

$$\sum_{k=1}^{12} (6k + 1)$$

Worked Solutions

Summation

Q077 Written

Express $4 + 9 + 14 + \dots + 119$ using Σ .**Solution**

1. Identify the general term and the endpoint indices.
2. $\sum_{k=1}^{24} (5k - 1)$

Final answer

$$\sum_{k=1}^{24} (5k - 1)$$

Worked Solutions

Summation

Q078 Challenge

A green-tech startup installs 1 solar panel in Week 1, 4 in Week 2, 7 in Week 3, and $3n - 2$ in Week n . Write one expression using Σ for the total panels installed from Week 1 through Week n .

Solution

1. The weekly pattern is arithmetic with general term $3k - 2$.
2. Total = $\sum_{k=1}^n (3k - 2)$

Final answer

$$\sum_{k=1}^n (3k - 2)$$

Worked Solutions

Summation

Q079 Written

Find $\sum_{k=1}^n (-3)$.

Solution

1. There are n copies of -3 .
2. $\sum_{k=1}^n (-3) = -3n$

Final answer

$$-3n$$

Q080 Written

Find $\sum_{k=1}^n 3$.

Solution

1. There are n copies of 3.
2. $\sum_{k=1}^n 3 = 3n$

Final answer

$$3n$$

Worked Solutions

Summation

Q081 Written

Find $\sum_{k=1}^n 6$.

Solution

1. $\sum_{k=1}^n 6 = 6n$

Final answer

$$6n$$

Q082 Written

Find $\sum_{k=1}^n (-4)$.

Solution

1. $\sum_{k=1}^n (-4) = -4n$

Final answer

$$-4n$$

Worked Solutions

Summation

Q083 Written

Find $\sum_{k=1}^n 1$.

Solution

1. $\sum_{k=1}^n 1 = n$

Final answer n

Q084 MCQ

Select the correct option: Find $\sum_{k=1}^n 9$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer $9n$

Worked Solutions

Summation

Q085 MCQ

Select the correct option: Find $\sum_{k=1}^n (-2)$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer $-2n$

Q086 MCQ

Select the correct option: Find $\sum_{k=1}^7 4$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

28

Worked Solutions

Summation

Q087 MCQ

Select the correct option: Find $\sum_{k=1}^{12} (-5)$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

-60

Q088 MCQ

Select the correct option: Find $\sum_{k=1}^n k$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer
$$\frac{n(n+1)}{2}$$

Worked Solutions

Summation

Q089 MCQ

Select the correct option: Find $\sum_{k=1}^n 4k$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

$$2n(n + 1)$$

Q090 MCQ

Select the correct option: Find $\sum_{k=1}^{10} k$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

$$55$$

Worked Solutions

Summation

Q091 MCQ

Select the correct option: Find $\sum_{k=1}^8 3k$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

108

Q092 MCQ

Select the correct option: Find $\sum_{k=1}^6 5k$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

105

Worked Solutions

Summation

Q093 MCQ

Select the correct option: Find $\sum_{k=1}^{20} k$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

210

Q094 MCQ

Select the correct option: Find $\sum_{k=1}^n k^2$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer
$$\frac{n(n+1)(2n+1)}{6}$$

Worked Solutions

Summation

Q095 MCQ

Select the correct option: Find $\sum_{k=1}^n 6k^2$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

$$n(n+1)(2n+1)$$

Q096 MCQ

Select the correct option: Find $\sum_{k=1}^5 k^2$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

$$55$$

Worked Solutions

Summation

Q097 MCQ

Select the correct option: Find $\sum_{k=1}^4 2k^2$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

60

Q098 MCQ

Select the correct option: Find $\sum_{k=1}^3 4k^2$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

56

Worked Solutions

Summation

Q099 MCQ

Select the correct option: Find $\sum_{k=1}^7 k^2$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

140

Q100 Written

Find $\sum_{k=1}^n (5k + 4)$.

Solution

1. $\sum (5k + 4) = 5 \sum k + \sum 4$

2. $= \frac{5n(n+1)}{2} + 4n = \frac{n(5n+13)}{2}$

Final answer

$$\frac{n(5n + 13)}{2}$$

Worked Solutions

Summation

Q101 Written

Find $\sum_{k=1}^n (2k - 3)$.

Solution

$$1. \ 2 \sum k - \sum 3 = n(n+1) - 3n$$

$$2. \ = n(n-2)$$

Final answer

$$n(n-2)$$

Q102 Written

Find $\sum_{k=1}^n (4k - 1)$.

Solution

$$1. \ 4 \sum k - \sum 1 = 2n(n+1) - n$$

$$2. \ = n(2n+1)$$

Final answer

$$n(2n+1)$$

Worked Solutions

Summation

Q103 Written

Find $\sum_{k=1}^n (3k + 2)$.

Solution

$$1. \quad 3 \sum k + \sum 2 = \frac{3n(n+1)}{2} + 2n$$

$$2. \quad = \frac{n(3n+7)}{2}$$

Final answer

$$\frac{n(3n+7)}{2}$$

Q104 Written

Find $\sum_{k=1}^n (6k - 3)$.

Solution

$$1. \quad 6 \sum k - \sum 3 = 3n(n+1) - 3n$$

$$2. \quad = 3n^2$$

Final answer

$$3n^2$$

Worked Solutions

Summation

Q105 MCQ

Select the correct option: Find $\sum_{k=1}^n (3k - 1)$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

$$\frac{n(3n + 1)}{2}$$

Q106 MCQ

Select the correct option: Find $\sum_{k=1}^n (2k + 5)$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

$$n(n + 6)$$

Worked Solutions

Summation

Q107 MCQ

Select the correct option: Find $\sum_{k=1}^n (8k - 4)$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

$$4n^2$$

Q108 MCQ

Select the correct option: Find $\sum_{k=1}^n (5k + 3)$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

$$\frac{n(5n + 11)}{2}$$

Worked Solutions

Summation

Q109 Written

Find $\sum_{k=1}^n (k-1)^2$.

Solution

$$1. \sum (k^2 - 2k + 1) = \sum k^2 - 2 \sum k + \sum 1$$

$$2. = \frac{n(n-1)(2n-1)}{6}$$

Final answer

$$\frac{n(n-1)(2n-1)}{6}$$

Q110 MCQ

Select the correct option: Find $\sum_{k=1}^n (k^2 + k)$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

$$\frac{n(n+1)(n+2)}{3}$$

Worked Solutions

Summation

Q111 MCQ

Select the correct option: Find $\sum_{k=1}^n (2k^2 - k)$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

$$\frac{n(n+1)(4n-1)}{6}$$

Q112 MCQ

Select the correct option: Find $\sum_{k=1}^n (k^2 - 1)$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

$$\frac{n(2n^2 + 3n - 5)}{6}$$

Worked Solutions

Summation

Q113 MCQ

Select the correct option: Find $\sum_{k=1}^n (3k^2 + 2k)$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

$$\frac{n(n+1)(2n+3)}{2}$$

Q114 MCQ

Select the correct option: Find $\sum_{k=1}^n (4k^2 - 3k + 2)$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

$$\frac{n(8n^2 + 3n + 7)}{6}$$

Worked Solutions

Summation

Q115 Written

Find $1 \cdot 3 + 2 \cdot 8 + 3 \cdot 13 + \cdots + n(5n - 2)$.

Solution

$$\begin{aligned} 1. \quad & \sum_{k=1}^n k(5k - 2) = 5 \sum k^2 - 2 \sum k \\ 2. \quad & = \frac{5n(n+1)(2n+1)}{6} - n(n+1) = \frac{n(n+1)(10n-1)}{6} \end{aligned}$$

Final answer

$$\frac{n(n+1)(10n-1)}{6}$$

Q116 Written

Find $\sum_{k=1}^n (k-1)(3k-1)$.

Solution

$$\begin{aligned} 1. \quad & (k-1)(3k-1) = 3k^2 - 4k + 1 \\ 2. \quad & 3 \sum k^2 - 4 \sum k + \sum 1 = \frac{n(n-1)(2n+1)}{2} \end{aligned}$$

Final answer

$$\frac{n(n-1)(2n+1)}{2}$$

Worked Solutions

Summation

Q117 Written

Find $1 \cdot 3 + 2 \cdot 5 + 3 \cdot 7 + \cdots + n(2n + 1)$.

Solution

$$1. \sum_{k=1}^n k(2k + 1) = 2 \sum k^2 + \sum k$$

$$2. = \frac{n(n+1)(4n+5)}{6}$$

Final answer

$$\frac{n(n+1)(4n+5)}{6}$$

Q118 Written

Find $\sum_{k=1}^n k(k - 2)$.

Solution

$$1. \sum (k^2 - 2k) = \sum k^2 - 2 \sum k$$

$$2. = \frac{n(n+1)(2n-5)}{6}$$

Final answer

$$\frac{n(n+1)(2n-5)}{6}$$

Worked Solutions

Summation

Q119 Written

Find $\sum_{k=1}^n k(2k - 2)$.

Solution

$$1. \ 2 \sum k^2 - 2 \sum k$$

$$2. \ = \frac{2n(n+1)(n-1)}{3}$$

Final answer

$$\frac{2n(n+1)(n-1)}{3}$$

Q120 Written

Find $1 \cdot 4 + 2 \cdot 5 + 3 \cdot 6 + \cdots + n(n+3)$.

Solution

$$1. \ \sum_{k=1}^n k(k+3) = \sum k^2 + 3 \sum k$$

$$2. \ = \frac{n(n+1)(n+5)}{3}$$

Final answer

$$\frac{n(n+1)(n+5)}{3}$$

Worked Solutions

Summation

Q121 Written

Find $1 \cdot 5 + 2 \cdot 8 + 3 \cdot 11 + \cdots + n(3n + 2)$.**Solution**

$$1. \sum_{k=1}^n k(3k + 2) = 3 \sum k^2 + 2 \sum k$$

$$2. = \frac{n(n+1)(2n+3)}{2}$$

Final answer

$$\frac{n(n+1)(2n+3)}{2}$$

Q122 Written

Find $2 \cdot 5 + 3 \cdot 6 + 4 \cdot 7 + \cdots + (n+1)(n+4)$.**Solution**

$$1. \sum_{k=1}^n (k+1)(k+4) = \sum (k^2 + 5k + 4)$$

$$2. = \frac{n(n+4)(n+5)}{3}$$

Final answer

$$\frac{n(n+4)(n+5)}{3}$$

Worked Solutions

Summation

Q123 Written

Find $\sum_{k=1}^4 (2k - 1)$.

Solution

1. $2 \sum_{k=1}^4 k - \sum_{k=1}^4 1 = 2(10) - 4$
2. $= 16$

Final answer

16

Q124 Written

Find $\sum_{k=1}^5 (3k - 2)$.

Solution

1. $3 \sum_{k=1}^5 k - 2 \sum_{k=1}^5 1 = 3(15) - 10$
2. $= 35$

Final answer

35

Worked Solutions

Summation

Q125 Written

Find $\sum_{k=1}^{20} (2k - 7)$.

Solution

1. $2 \sum_{k=1}^{20} k - 7 \sum_{k=1}^{20} 1 = 2(210) - 140$

2. $= 280$

Final answer

280

Q126 Written

Find $\sum_{k=1}^8 (4k + 5)$.

Solution

1. $4 \sum_{k=1}^8 k + 5 \sum_{k=1}^8 1 = 4(36) + 40$

2. $= 184$

Final answer

184

Worked Solutions

Summation

Q127 Written

Find $\sum_{k=1}^{10} (2k - 1)$.

Solution

$$\begin{aligned} 1. & 2 \sum_{k=1}^{10} k - \sum_{k=1}^{10} 1 = 2(55) - 10 \\ 2. & = 100 \end{aligned}$$

Final answer

100

Worked Solutions

Summation

Q128 Challenge

Find $\sum_{r=4}^8 (r^2 + r + 5)$.

Solution

$$\begin{aligned} 1. \sum_{r=4}^8 (r^2 + r + 5) &= \sum_{r=1}^8 (r^2 + r + 5) - \sum_{r=1}^3 (r^2 + r + 5) \\ 2. &= (204 + 36 + 40) - (14 + 6 + 15) = 245 \end{aligned}$$

Final answer

245

Worked Solutions

Summation

Q129 Written

Find the sum $\sum_{k=1}^n k^2(k+1)$.

Solution

1. Expand: $k^2(k+1) = k^3 + k^2$.
2. Use $\sum k^3 = \frac{n^2(n+1)^2}{4}$ and $\sum k^2 = \frac{n(n+1)(2n+1)}{6}$.
3. $\sum_{k=1}^n k^2(k+1) = \frac{1}{12}n(n+1)(n+2)(3n+1)$

Final answer

$$\frac{n(n+1)(n+2)(3n+1)}{12}$$

Q130 Written

Find the sum $\sum_{k=1}^n k^2(k-2)$.

Solution

1. Expand: $k^2(k-2) = k^3 - 2k^2$.
2. Substitute the formulae for $\sum k^3$ and $\sum k^2$, then factor.
3. $\sum_{k=1}^n k^2(k-2) = \frac{n(n+1)(3n^2-5n-4)}{12}$

Final answer

$$\frac{n(n+1)(3n^2-5n-4)}{12}$$

Worked Solutions

Summation

Q131 Written

Find the sum $\sum_{k=1}^n k^2(k+2)$.

Solution

1. Expand: $k^2(k+2) = k^3 + 2k^2$.
2. Substitute the standard sums and factor the common $n(n+1)$.
3. $\sum_{k=1}^n k^2(k+2) = \frac{n(n+1)(3n^2 + 11n + 4)}{12}$

Final answer

$$\frac{n(n+1)(3n^2 + 11n + 4)}{12}$$

Q132 Written

Find the sum $\sum_{k=1}^n k^2(k-3)$.

Solution

1. Expand: $k^2(k-3) = k^3 - 3k^2$.
2. Substitute the standard sums and simplify the common factor.
3. $\sum_{k=1}^n k^2(k-3) = \frac{n(n+1)(n^2-3n-2)}{4}$

Final answer

$$\frac{n(n+1)(n^2-3n-2)}{4}$$

Worked Solutions

Summation

Q133 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k+1)$:**Solution**

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A**Final answer**

$$\frac{n(n+1)(n+2)(3n+1)}{12}$$

Q134 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k-2)$:**Solution**

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A**Final answer**

$$\frac{n(n+1)(3n^2-5n-4)}{12}$$

Worked Solutions

Summation

Q135 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k+2)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(3n^2+11n+4)}{12}$$

Q136 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k-3)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(n^2-3n-2)}{4}$$

Worked Solutions

Summation

Q137 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k+3)$:**Solution**

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A**Final answer**

$$\frac{n(n+1)(n^2+5n+2)}{4}$$

Q138 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k-4)$:**Solution**

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A**Final answer**

$$\frac{n(n+1)(3n^2-13n-8)}{12}$$

Worked Solutions

Summation

Q139 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k+4)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(3n^2+19n+8)}{12}$$

Q140 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k-1)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(3n^2-5n-4)}{12}$$

Worked Solutions

Summation

Q141 Written

Find the sum $\sum_{k=1}^n k(k^2 - 1)$.

Solution

1. Expand: $k(k^2 - 1) = k^3 - k$.
2. Use $\sum k^3$ and $\sum k$, then factor $n^2 + n - 2 = (n - 1)(n + 2)$.
3. $\sum_{k=1}^n k(k^2 - 1) = \frac{n(n + 1)(n - 1)(n + 2)}{4}$

Final answer

$$\frac{n(n + 1)(n - 1)(n + 2)}{4}$$

Q142 Written

Find the sum $\sum_{k=1}^n k(k^2 + 2)$.

Solution

1. Expand: $k(k^2 + 2) = k^3 + 2k$.
2. Use $\sum k^3$ and $\sum k$, then factor.
3. $\sum_{k=1}^n k(k^2 + 2) = \frac{n(n+1)(n^2 + n + 4)}{4}$

Final answer

$$\frac{n(n+1)(n^2 + n + 4)}{4}$$

Worked Solutions

Summation

Q143 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2 - 1)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(n-1)(n+2)}{4}$$

Q144 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2 + 2)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(n^2 + n + 4)}{4}$$

Worked Solutions

Summation

Q145 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2 + 3)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(n^2+n+6)}{4}$$

Q146 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2 - 2)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(n^2+n-4)}{4}$$

Worked Solutions

Summation

Q147 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2 + 4)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(n^2+n+8)}{4}$$

Q148 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2 - 3)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(n^2+n-6)}{4}$$

Worked Solutions

Summation

Q149 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k+5)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(3n^2+23n+10)}{12}$$

Q150 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k-5)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(3n^2-17n-10)}{12}$$

Worked Solutions

Summation

Q151 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2 + 5)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(n^2+n+10)}{4}$$

Q152 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2 - 5)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(n^2+n-10)}{4}$$

Worked Solutions

Summation

Q153 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k+6)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(3n^2+27n+12)}{12}$$

Q154 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2+6)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(n^2+n+12)}{4}$$

Worked Solutions

Summation

Q155 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k - 6)$:**Solution**

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A**Final answer**

$$\frac{n(n+1)(3n^2 - 25n - 12)}{12}$$

Q156 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2 - 6)$:**Solution**

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A**Final answer**

$$\frac{n(n+1)(n^2 + n - 12)}{4}$$

Worked Solutions

Summation

Q157 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k+7)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(3n^2+31n+14)}{12}$$

Q158 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k+8)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(3n^2+35n+16)}{12}$$

Worked Solutions

Summation

Q159 Written

Find the sum $\sum_{k=1}^n 6 \times 3^{k-1}$.

Solution

1. Use $a = 6, r = 3$.
2. $S_n = \frac{6(3^n - 1)}{3 - 1} = 3(3^n - 1)$

Final answer

$$3(3^n - 1)$$

Q160 Written

Find the sum $\sum_{k=1}^n 4 \times 2^{k-1}$.

Solution

1. Use $a = 4, r = 2$.
2. $S_n = \frac{4(2^n - 1)}{2 - 1} = 4(2^n - 1)$

Final answer

$$4(2^n - 1)$$

Worked Solutions

Summation

Q161 Written

Find the sum $\sum_{k=1}^n 2^k$.

Solution

1. Rewrite as $(2 \times 2^{k-1})$.
2. $S_n = \frac{2(2^n - 1)}{2 - 1} = 2(2^n - 1)$

Final answer

$$2(2^n - 1)$$

Q162 Written

Find the sum $\sum_{k=1}^n 2 \times (-3)^{k-1}$.

Solution

1. Here $a = 2$ and $r = -3$.
2. $S_n = \frac{2(1 - (-3)^n)}{1 - (-3)} = \frac{1 - (-3)^n}{2}$

Final answer

$$\frac{1 - (-3)^n}{2}$$

Worked Solutions

Summation

Q163 Written

Find the sum $\sum_{k=1}^n 3 \times (-2)^{k-1}$.

Solution

1. Use $a = 3, r = -2$.
2. $S_n = \frac{3(1-(-2)^n)}{1-(-2)} = 1 - (-2)^n$

Final answer

$$1 - (-2)^n$$

Q164 Written

Find the sum $\sum_{k=1}^n (-5)^k$.

Solution

1. Rewrite as $(-5) \times (-5)^{k-1}$, so $a = -5, r = -5$.
2. $S_n = \frac{-5(1-(-5)^n)}{1-(-5)} = \frac{5((-5)^n - 1)}{6}$

Final answer

$$\frac{5((-5)^n - 1)}{6}$$

Worked Solutions

Summation

Q165 Written

Find the sum $\sum_{k=1}^n 2 \times (-1)^{k-1}$.

Solution

1. Use $a = 2, r = -1$.
2. $S_n = \frac{2(1-(-1)^n)}{2} = 1 - (-1)^n$

Final answer

$$1 - (-1)^n$$

Q166 Written

Find the sum $\sum_{k=1}^n 8 \times (-3)^{k-1}$.

Solution

1. Use $a = 8, r = -3$.
2. $S_n = \frac{8(1-(-3)^n)}{4} = 2(1 - (-3)^n)$

Final answer

$$2(1 - (-3)^n)$$

Worked Solutions

Summation

Q167 Written

Find the sum $\sum_{k=1}^n (-2)^{k-1}$.

Solution

1. Use $(a = 1, r = -2)$.
2. $S_n = \frac{1 - (-2)^n}{1 - (-2)} = \frac{1 - (-2)^n}{3}$

Final answer

$$\frac{1 - (-2)^n}{3}$$

Q168 Written

Find the sum $\sum_{k=1}^n (-3)^k$.

Solution

1. Rewrite as $((-3)(-3)^{k-1})$.
2. $S_n = \frac{-3(1 - (-3)^n)}{4} = \frac{3((-3)^n - 1)}{4}$

Final answer

$$\frac{3((-3)^n - 1)}{4}$$

Worked Solutions

Summation

Q169 Written

Find the sum $\sum_{k=1}^n 4^k$.

Solution

1. Rewrite $4^k = 4 \times 4^{k-1}$, so $a = 4, r = 4$.

2. $S_n = \frac{4(4^n - 1)}{4 - 1} = \frac{4(4^n - 1)}{3}$

Final answer

$$\frac{4(4^n - 1)}{3}$$

Q170 Written

Find the sum $\sum_{k=1}^n 4^{k-1}$.

Solution

1. The first term is 1 and the common ratio is 4.

2. $S_n = \frac{4^n - 1}{4 - 1} = \frac{4^n - 1}{3}$

Final answer

$$\frac{4^n - 1}{3}$$

Worked Solutions

Summation

Q171 Written

Find the sum $\sum_{k=1}^n 3^{k-1}$.

Solution

1. The first term is 1 and the ratio is 3.
2. $S_n = \frac{3^n - 1}{3 - 1} = \frac{3^n - 1}{2}$

Final answer

$$\frac{3^n - 1}{2}$$

Q172 Written

Find the sum $\sum_{k=1}^4 (-2)^{k-1}$.

Solution

1. This is a finite geometric sum with $(a = 1, r = -2, n = 4)$.
2. $S_4 = 1 - 2 + 4 - 8 = -5$

Final answer

$$-5$$

Worked Solutions

Summation

Q173 Written

Find the sum $\sum_{k=1}^3 3^k$.

Solution

1. Use $(a = 3, r = 3, n = 3)$.
2. $S_3 = 3 + 9 + 27 = 39$

Final answer

39

Q174 Written

Find the sum $\sum_{k=1}^5 2^k$.

Solution

1. Use $(a = 2, r = 2, n = 5)$.
2. $S_5 = 2 + 4 + 8 + 16 + 32 = 62$

Final answer

62

Worked Solutions

Summation

Q175 Written

Find the sum $\sum_{k=1}^4 (-3)^k$.

Solution

1. Use $(a = -3, r = -3, n = 4)$.
2. $S_4 = -3 + 9 - 27 + 81 = 60$

Final answer

60

Worked Solutions

Summation

Q176 Challenge

Find the finite sum $\sum_{r=5}^9 (2r + 7)$.

Solution

1. Substitute ($r = 5, 6, 7, 8, 9$): ($17+19+21+23+25$).
2. $17 + 19 + 21 + 23 + 25 = 105$

Final answer

105

Worked Solutions

Summation

Q177 Written

Find $\frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \frac{1}{4 \times 5} + \cdots + \frac{1}{(n+1)(n+2)}.$

Solution

1. Split each term as $\frac{1}{j} - \frac{1}{j+1}.$
2. $\frac{1}{2} - \frac{1}{n+2} = \frac{n}{2(n+2)}$

Final answer

$$\frac{n}{2(n+2)}$$

Q178 Written

Find $\frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \cdots + \frac{1}{n(n+1)}.$

Solution

1. Use $\frac{1}{j(j+1)} = \frac{1}{j} - \frac{1}{j+1}.$
2. $1 - \frac{1}{n+1} = \frac{n}{n+1}$

Final answer

$$\frac{n}{n+1}$$

Worked Solutions

Summation

Q179 Written

Find $\frac{1}{3 \times 4} + \frac{1}{4 \times 5} + \frac{1}{5 \times 6} + \cdots + \frac{1}{(n+2)(n+3)}$.

Solution

1. The gap is 1 and the first term starts at 3.

$$2. \frac{1}{3} - \frac{1}{n+3} = \frac{n}{3(n+3)}$$

Final answer

$$\frac{n}{3(n+3)}$$

Q180 MCQ

Select the correct option: Find the sum $\frac{1}{1 \times 2} + \cdots + \frac{1}{n(n+1)}$:

Solution

1. Split into two fractions, cancel the chain, and retain the first and last terms.

Correct option: A

Final answer

$$\frac{n}{n+1}$$

Worked Solutions

Summation

Q181 MCQ

Select the correct option: Find the sum $\frac{1}{2 \times 3} + \cdots + \frac{1}{(n+1)(n+2)}$:

Solution

1. Split into two fractions, cancel the chain, and retain the first and last terms.

Correct option: A

Final answer

$$\frac{n}{2(n+2)}$$

Q182 MCQ

Select the correct option: Find the sum $\frac{1}{3 \times 4} + \cdots + \frac{1}{(n+2)(n+3)}$:

Solution

1. Split into two fractions, cancel the chain, and retain the first and last terms.

Correct option: A

Final answer

$$\frac{n}{3(n+3)}$$

Worked Solutions

Summation

Q183 MCQ

Select the correct option: Find the sum $\frac{1}{4 \times 5} + \cdots + \frac{1}{(n+3)(n+4)}$:

Solution

1. Split into two fractions, cancel the chain, and retain the first and last terms.

Correct option: A

Final answer

$$\frac{n}{4(n+4)}$$

Q184 MCQ

Select the correct option: Find the sum $\frac{1}{6 \times 7} + \cdots + \frac{1}{(n+5)(n+6)}$:

Solution

1. Split into two fractions, cancel the chain, and retain the first and last terms.

Correct option: A

Final answer

$$\frac{n}{6(n+6)}$$

Worked Solutions

Summation

Q185 MCQ

Select the correct option: Find the sum $\frac{1}{7 \times 8} + \cdots + \frac{1}{(n+6)(n+7)}$:

Solution

1. Split into two fractions, cancel the chain, and retain the first and last terms.

Correct option: A

Final answer

$$\frac{n}{7(n+7)}$$

Q186 MCQ

Select the correct option: Find the sum $\frac{1}{9 \times 10} + \cdots + \frac{1}{(n+8)(n+9)}$:

Solution

1. Split into two fractions, cancel the chain, and retain the first and last terms.

Correct option: A

Final answer

$$\frac{n}{9(n+9)}$$

Worked Solutions

Summation

Q187 MCQ

Select the correct option: Find the sum $\frac{1}{10 \times 11} + \cdots + \frac{1}{(n+9)(n+10)}$:

Solution

1. Split into two fractions, cancel the chain, and retain the first and last terms.

Correct option: A

Final answer

$$\frac{n}{10(n+10)}$$

Q188 MCQ

Select the correct option: Find the sum $\frac{1}{11 \times 12} + \cdots + \frac{1}{(n+10)(n+11)}$:

Solution

1. Split into two fractions, cancel the chain, and retain the first and last terms.

Correct option: A

Final answer

$$\frac{n}{11(n+11)}$$

Worked Solutions

Summation

Q189 Written

Find the following sum: $\frac{1}{2 \times 5} + \frac{1}{5 \times 8} + \frac{1}{8 \times 11} + \cdots + \frac{1}{(3n-1)(3n+2)}$.

Solution

1. Use $\frac{1}{ab} = \frac{1}{b-a} \left(\frac{1}{a} - \frac{1}{b} \right)$ with gap 3.
2. $\frac{1}{3} \left(\frac{1}{2} - \frac{1}{3n+2} \right) = \frac{n}{2(3n+2)}$

Final answer

$$\frac{n}{2(3n+2)}$$

Q190 Written

Find $\frac{1}{1 \times 3} + \frac{1}{3 \times 5} + \frac{1}{5 \times 7} + \cdots + \frac{1}{(2n-1)(2n+1)}$.

Solution

1. The gap is 2, so factor out $\frac{1}{2}$.
2. $\frac{1}{2} \left(1 - \frac{1}{2n+1} \right) = \frac{n}{2n+1}$

Final answer

$$\frac{n}{2n+1}$$

Worked Solutions

Summation

Q191 Written

Find $\frac{1}{1 \times 3} + \frac{1}{2 \times 4} + \frac{1}{3 \times 5} + \cdots + \frac{1}{n(n+2)}.$

Solution

1. Use the constant gap 2 decomposition; two end terms remain.
2. $\frac{1}{2}(1 + \frac{1}{2} - \frac{1}{n+1} - \frac{1}{n+2})$

Final answer

$$\frac{1}{2}(1 + \frac{1}{2} - \frac{1}{n+1} - \frac{1}{n+2})$$

Q192 Written

Find $\frac{1}{1 \times 4} + \frac{1}{4 \times 7} + \frac{1}{7 \times 10} + \cdots + \frac{1}{(3n-2)(3n+1)}.$

Solution

1. Use gap 3 and the first denominator 1.
2. $\frac{1}{3}(1 - \frac{1}{3n+1}) = \frac{n}{3n+1}$

Final answer

$$\frac{n}{3n+1}$$

Worked Solutions

Summation

Q193 Written

Find $\frac{1}{1 \times 7} + \frac{1}{4 \times 10} + \frac{1}{7 \times 13} + \cdots + \frac{1}{(3n-2)(3n+4)}$.

Solution

1. The two factors differ by 6, while the first factors advance by 3; two terms remain at each end.
2. $\frac{1}{6}(1 + \frac{1}{4} - \frac{1}{3n+1} - \frac{1}{3n+4})$

Final answer

$$\frac{1}{6}(1 + \frac{1}{4} - \frac{1}{3n+1} - \frac{1}{3n+4})$$

Q194 MCQ

Select the correct option: Find the sum $\frac{1}{1 \times 3} + \cdots + \frac{1}{(2n-1)(2n+1)}$:

Solution

1. Split into two fractions, cancel the chain, and retain the first and last terms.

Correct option: A

Final answer

$$\frac{n}{2n+1}$$

Worked Solutions

Summation

Q195 MCQ

Select the correct option: Find the sum $\frac{1}{1 \times 4} + \cdots + \frac{1}{(3n-2)(3n+1)}$:

Solution

1. Split into two fractions, cancel the chain, and retain the first and last terms.

Correct option: A

Final answer

$$\frac{n}{3n+1}$$

Q196 MCQ

Select the correct option: Find the sum $\frac{1}{1 \times 7} + \cdots + \frac{1}{(3n-2)(3n+4)}$:

Solution

1. Split into two fractions, cancel the chain, and retain the first and last terms.

Correct option: A

Final answer

$$\frac{1}{6} \left(1 + \frac{1}{4} - \frac{1}{3n+1} - \frac{1}{3n+4} \right)$$

Worked Solutions

Summation

Q197 MCQ

Select the correct option: Find the sum $\frac{1}{2 \times 6} + \cdots + \frac{1}{(4n-2)(4n+2)}$:

Solution

1. Split into two fractions, cancel the chain, and retain the first and last terms.

Correct option: A

Final answer

$$\frac{n}{2(4n+2)}$$

Q198 MCQ

Select the correct option: Find the sum $\frac{1}{3 \times 8} + \cdots + \frac{1}{(5n-2)(5n+3)}$:

Solution

1. Split into two fractions, cancel the chain, and retain the first and last terms.

Correct option: A

Final answer

$$\frac{n}{3(5n+3)}$$

Worked Solutions

Summation

Q199 MCQ

Select the correct option: Find the sum $\frac{1}{1 \times 5} + \cdots + \frac{1}{(4n-3)(4n+1)}$:

Solution

1. Split into two fractions, cancel the chain, and retain the first and last terms.

Correct option: A

Final answer

$$\frac{n}{4n+1}$$

Q200 MCQ

Select the correct option: Find the sum $\frac{1}{2 \times 7} + \cdots + \frac{1}{(5n-3)(5n+2)}$:

Solution

1. Split into two fractions, cancel the chain, and retain the first and last terms.

Correct option: A

Final answer

$$\frac{n}{2(5n+2)}$$

Worked Solutions

Summation

Q201 MCQ

Select the correct option: Find the sum $\frac{1}{4 \times 10} + \cdots + \frac{1}{(6n-2)(6n+4)}$:

Solution

1. Split into two fractions, cancel the chain, and retain the first and last terms.

Correct option: A

Final answer

$$\frac{n}{4(6n+4)}$$

Q202 Written

Find $\frac{1}{4 \times 6} + \frac{1}{5 \times 7} + \frac{1}{6 \times 8} + \cdots + \frac{1}{(n+3)(n+5)}$.

Solution

1. Write the general term as $\frac{1}{(j+3)(j+5)}$, gap 2.

2. $\frac{1}{2} \left(\frac{1}{4} + \frac{1}{5} - \frac{1}{n+4} - \frac{1}{n+5} \right)$

Final answer

$$\frac{1}{2} \left(\frac{1}{4} + \frac{1}{5} - \frac{1}{n+4} - \frac{1}{n+5} \right)$$

Worked Solutions

Summation

Q203 Written

Find $\frac{1}{2 \times 4} + \frac{1}{4 \times 6} + \frac{1}{6 \times 8} + \cdots + \frac{1}{2n(2n+2)}$.

Solution

1. Factor each denominator as $4j(j+1)$.
2. $\frac{1}{4}\left(1 - \frac{1}{n+1}\right) = \frac{n}{4(n+1)}$

Final answer

$$\frac{n}{4(n+1)}$$

Q204 Written

Find the following sum: $\sum_{k=1}^n \frac{1}{\sqrt{k+2} + \sqrt{k+1}}$.

Solution

1. Multiply by the conjugate.
2. $\frac{\sqrt{k+2}-\sqrt{k+1}}{(k+2)-(k+1)} = \sqrt{k+2} - \sqrt{k+1}$
3. $\sum_{k=1}^n (\sqrt{k+2} - \sqrt{k+1}) = \sqrt{n+2} - \sqrt{2}$

Final answer

$$\sqrt{n+2} - \sqrt{2}$$

Worked Solutions

Summation

Q205 Written

Find $\sum_{k=1}^n \frac{1}{\sqrt{k+1} + \sqrt{k}}.$

Solution

1. Rationalise with $\sqrt{k+1} - \sqrt{k}.$
2. $\sum_{k=1}^n (\sqrt{k+1} - \sqrt{k}) = \sqrt{n+1} - 1$

Final answer

$$\sqrt{n+1} - 1$$

Q206 Written

Find $\sum_{k=1}^n \frac{1}{\sqrt{k} + \sqrt{k-1}}.$

Solution

1. Rationalise: the denominator difference is 1.
2. $\sum_{k=1}^n (\sqrt{k} - \sqrt{k-1}) = \sqrt{n}$

Final answer

$$\sqrt{n}$$

Worked Solutions

Summation

Q207 Written

Find $\frac{1}{\sqrt{1} + \sqrt{2}} + \frac{1}{\sqrt{2} + \sqrt{3}} + \dots + \frac{1}{\sqrt{n} + \sqrt{n+1}}.$

Solution

1. Use the conjugate to obtain $\sqrt{j+1} - \sqrt{j}.$
2. $\sum_{j=1}^n (\sqrt{j+1} - \sqrt{j}) = \sqrt{n+1} - 1$

Final answer

$$\sqrt{n+1} - 1$$

Q208 Written

Find $\sum_{k=1}^n \frac{1}{\sqrt{2k-1} + \sqrt{2k+1}}.$

Solution

1. Rationalise; the square-difference gap is 2.
2. $\frac{1}{2} \sum_{k=1}^n (\sqrt{2k+1} - \sqrt{2k-1}) = \frac{\sqrt{2n+1}-1}{2}$

Final answer

$$\frac{\sqrt{2n+1} - 1}{2}$$

Worked Solutions

Summation

Q209 Written

Find $\frac{1}{\sqrt{2} + \sqrt{3}} + \frac{1}{\sqrt{3} + \sqrt{4}} + \cdots + \frac{1}{\sqrt{n+1} + \sqrt{n+2}}.$

Solution

1. Multiply by each conjugate.
2. $\sum_{j=1}^n (\sqrt{j+2} - \sqrt{j+1}) = \sqrt{n+2} - \sqrt{2}$

Final answer

$$\sqrt{n+2} - \sqrt{2}$$

Worked Solutions

Summation

Q210 Challenge

Find the sum, showing your steps: $\sum_{r=1}^{30} \frac{1}{r(r+2)}$.

Solution

1. Use $\frac{1}{r(r+2)} = \frac{1}{2} \left(\frac{1}{r} - \frac{1}{r+2} \right)$.
2. $S = \frac{1}{2} \left(1 + \frac{1}{2} - \frac{1}{31} - \frac{1}{32} \right)$
3. $S = \frac{1425}{1984}$

Final answer

$$\frac{1425}{1984}$$

Worked Solutions

Summation

Q211 Written

Find the following sum $S = 3 \times 1 + 5 \times 3 + 7 \times 3^2 + 9 \times 3^3 + \dots + (2n + 1) \times 3^{n-1}$.

Solution

1. The arithmetic coefficient is $2k + 1$ and the geometric ratio is $r = 3$.
2. Write S and $3S$, subtract with aligned powers, and simplify: $-2S = -2n \times 3^n$.
3. $S = n \times 3^n$

Final answer

$$n \times 3^n$$

Q212 Written

Find the sum $\sum_{k=1}^n (2k + 1) \times (3)^{k-1}$.

Solution

1. Write the first sum and multiply by the common ratio.
2. Subtract and keep the final shifted term with its correct sign.
3. $3^n n$

Final answer

$$3^n n$$

Worked Solutions

Summation

Q213 Written

Find the sum $\sum_{k=1}^4 (2k + 1) \times (3)^{k-1}$.

Solution

1. Identify the arithmetic coefficient and geometric ratio.
2. Apply (S-rS) over the finite limit ($n = 4$).
3. 324

Final answer

324

Q214 Written

Find the sum $\sum_{k=1}^n (2k - 1) \times (4)^{k-1}$.

Solution

1. Write the first sum and multiply by the common ratio.
2. Subtract and keep the final shifted term with its correct sign.
3. $\frac{6 \cdot 2^{2n}n - 5 \cdot 2^{2n} + 5}{9}$

Final answer

$$\frac{6 \cdot 2^{2n}n - 5 \cdot 2^{2n} + 5}{9}$$

Worked Solutions

Summation

Q215 Written

Find the sum $\sum_{k=1}^6 (2k - 1) \times (4)^{k-1}$.

Solution

1. Identify the arithmetic coefficient and geometric ratio.
2. Apply (S-rS) over the finite limit ($n = 6$).
3. 14109

Final answer

14109

Q216 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n (2k + 1) \times (3)^{k-1}$:

Solution

1. Identify the arithmetic coefficient and the ratio, then apply S-rS.

Correct option: A

Final answer $3^n n$

Worked Solutions

Summation

Q217 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n (2k - 1) \times (4)^{k-1}$:

Solution

1. Identify the arithmetic coefficient and the ratio, then apply S-rS.

Correct option: A

Final answer

$$\frac{6 \cdot 2^{2n}n - 5 \cdot 2^{2n} + 5}{9}$$

Q218 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n (k + 4) \times (2)^{k-1}$:

Solution

1. Identify the arithmetic coefficient and the ratio, then apply S-rS.

Correct option: A

Final answer

$$2^n n + 3 \cdot 2^n - 3$$

Worked Solutions

Summation

Q219 Written

Find the following sum $S = 2 \times 1 + 3 \times 3 + 4 \times 3^2 + 5 \times 3^3 + \dots + (n + 1) \times 3^{n-1}$.

Solution

1. Here the coefficient is $k + 1$ and $r = 3$.
2. Subtract $3S$ from S ; the interior coefficients reduce to a geometric sum of 1s.
3. $-2S = \frac{1 - (2n+1) \times 3^n}{2}$
4. $S = \frac{(2n+1) \times 3^n - 1}{4}$

Final answer

$$\frac{(2n + 1) \times 3^n - 1}{4}$$

Q220 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n (k + 1) \times (3)^{k-1}$:

Solution

1. Identify the arithmetic coefficient and the ratio, then apply S-rS.

Correct option: A

Final answer

$$\frac{2 \cdot 3^n n + 3^n - 1}{4}$$

Worked Solutions

Summation

Q221 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n (k-2) \times (3)^{k-1}$:

Solution

1. Identify the arithmetic coefficient and the ratio, then apply S-rS.

Correct option: A

Final answer

$$\frac{2 \cdot 3^n n - 5 \cdot 3^n + 5}{4}$$

Q222 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n (2k-3) \times (3)^{k-1}$:

Solution

1. Identify the arithmetic coefficient and the ratio, then apply S-rS.

Correct option: A

Final answer

$$3^n n - 2 \cdot 3^n + 2$$

Worked Solutions

Summation

Q223 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n (2k + 1) \times (-2)^{k-1}$:

Solution

1. Identify the arithmetic coefficient and the ratio, then apply S-rS.

Correct option: A

Final answer

$$-\frac{6(-2)^n n + 5(-2)^n - 5}{9}$$

Worked Solutions

Summation

Q224 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n (k+2) \times (-3)^{k-1}$:

Solution

1. Identify the arithmetic coefficient and the ratio, then apply S-rS.

Correct option: A

Final answer

$$-\frac{4(-3)^n n + 9(-3)^n - 9}{16}$$

Worked Solutions

Summation

Q225 Written

Find the following sum $S = 1 \times 1 + 2 \times 2 + 3 \times 2^2 + 4 \times 2^3 + \cdots + n \times 2^{n-1}$.

Solution

1. Use $a_k = k$ and $r = 2$.
2. Applying $S - rS$ gives $-S = 1 - (n - 1) \times 2^n$.
3. $S = (n - 1) \times 2^n + 1$

Final answer

$$(n - 1) \times 2^n + 1$$

Q226 Written

Find the following sum $S = 1 \times 3 + 3 \times 3^2 + 5 \times 3^3 + 7 \times 3^4 + \cdots + (2n - 1) \times 3^n$.

Solution

1. Factor out the leading 3, leaving $\sum_{k=1}^n (2k - 1)3^{k-1}$.
2. The $S - rS$ subtraction gives $\sum_{k=1}^n (2k - 1)3^{k-1} = (n - 1)3^n + 1$.
3. $S = (n - 1) \times 3^{n+1} + 3$

Final answer

$$(n - 1) \times 3^{n+1} + 3$$

Worked Solutions

Summation

Q227 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n (3k) \times (3)^{k-1}$:

Solution

1. Identify the arithmetic coefficient and the ratio, then apply S-rS.

Correct option: A

Final answer

$$\frac{3(2 \cdot 3^n n - 3^n + 1)}{4}$$

Worked Solutions

Summation

Q228 Challenge

Find the sum of the partial sums $5 + (5 + 9) + (5 + 9 + 13) + \cdots + [5 + 9 + \cdots + (4k + 1)]$.

Solution

1. The j -th inner arithmetic sum is $5 + 9 + \cdots + (4j + 1) = j(2j + 3)$.
2. Sum the partial sums: $\sum_{j=1}^k (2j^2 + 3j)$.
3. $\frac{k(k+1)(4k+11)}{6}$

Final answer

$$\frac{k(k+1)(4k+11)}{6}$$

Worked Solutions

Summation

Q229 Written

For the sequence 3, 3 + 5, 3 + 5 + 7, 3 + 5 + 7 + 9, ..., find the k -th term.

Solution

1. The inner terms form an arithmetic sequence with first term 3 and common difference 2.
2. $u_i = 2i + 1$, $a_k = \sum_{i=1}^k u_i$
3. $a_k = \frac{k}{2} [2(3) + (k - 1)(2)] = k(k + 2)$

Final answer

$$a_k = k(k + 2)$$

Q230 Written

For the sequence 1, 1 + 3, 1 + 3 + 5, 1 + 3 + 5 + 7, ..., find the k -th term.

Solution

1. The inner terms form an arithmetic sequence with first term 1 and common difference 2.
2. $u_i = 2i - 1$, $a_k = \sum_{i=1}^k u_i$
3. $a_k = \frac{k}{2} [2(1) + (k - 1)(2)] = k^2$

Final answer

$$a_k = k^2$$

Worked Solutions

Summation

Q231 Written

For the sequence $2, 2 + 4, 2 + 4 + 6, 2 + 4 + 6 + 8, \dots$, **find the k -th term.**

Solution

1. The inner terms form an arithmetic sequence with first term 2 and common difference 2.
2. $u_i = 2i, \quad a_k = \sum_{i=1}^k u_i$
3. $a_k = \frac{k}{2} [2(2) + (k-1)(2)] = k(k+1)$

Final answer

$$a_k = k(k+1)$$

Q232 Written

For the sequence $1, 1 + 5, 1 + 5 + 9, 1 + 5 + 9 + 13, \dots$, **find the k -th term.**

Solution

1. The inner terms form an arithmetic sequence with first term 1 and common difference 4.
2. $u_i = 4i - 3, \quad a_k = \sum_{i=1}^k u_i$
3. $a_k = \frac{k}{2} [2(1) + (k-1)(4)] = k(2k-1)$

Final answer

$$a_k = k(2k-1)$$

Worked Solutions

Summation

Q233 MCQ

Select the correct option: For the cumulative sequence $2, 2 + 4, 2 + 4 + 6, 2 + 4 + 6 + 8, \dots$, what is a_k :

Solution

1. The inner terms form an arithmetic sequence with first term 2 and common difference 2.

2. $u_i = 2i, \quad a_k = \sum_{i=1}^k u_i$

3. $a_k = \frac{k}{2} [2(2) + (k-1)(2)] = k(k+1)$

Correct option: A

Final answer

$$k(k+1)$$

Q234 MCQ

Select the correct option: For the cumulative sequence $3, 3 + 6, 3 + 6 + 9, 3 + 6 + 9 + 12, \dots$, what is a_k :

Solution

1. The inner terms form an arithmetic sequence with first term 3 and common difference 3.

2. $u_i = 3i, \quad a_k = \sum_{i=1}^k u_i$

3. $a_k = \frac{k}{2} [2(3) + (k-1)(3)] = \frac{3k(k+1)}{2}$

Correct option: A

Final answer

$$\frac{3k(k+1)}{2}$$

Worked Solutions

Summation

Q235 MCQ

Select the correct option: For the cumulative sequence 4, 4 + 6, 4 + 6 + 8, 4 + 6 + 8 + 10, ..., what is a_k :

Solution

1. The inner terms form an arithmetic sequence with first term 4 and common difference 2.

2. $u_i = 2i + 2, \quad a_k = \sum_{i=1}^k u_i$

3. $a_k = \frac{k}{2} [2(4) + (k-1)(2)] = k(k+3)$

Correct option: A

Final answer

$$k(k+3)$$

Q236 MCQ

Select the correct option: For the cumulative sequence 2, 2 + 6, 2 + 6 + 10, 2 + 6 + 10 + 14, ..., what is a_k :

Solution

1. The inner terms form an arithmetic sequence with first term 2 and common difference 4.

2. $u_i = 4i - 2, \quad a_k = \sum_{i=1}^k u_i$

3. $a_k = \frac{k}{2} [2(2) + (k-1)(4)] = 2k^2$

Correct option: A

Final answer

$$2k^2$$

Worked Solutions

Summation

Q237 MCQ

Select the correct option: For the cumulative sequence 5, 5 + 8, 5 + 8 + 11, 5 + 8 + 11 + 14, ..., what is a_k :

Solution

1. The inner terms form an arithmetic sequence with first term 5 and common difference 3.

$$2. u_i = 3i + 2, \quad a_k = \sum_{i=1}^k u_i$$

$$3. a_k = \frac{k}{2} [2(5) + (k-1)(3)] = \frac{k(3k+7)}{2}$$

Correct option: A

Final answer

$$\frac{k(3k+7)}{2}$$

Q238 MCQ

Select the correct option: For the cumulative sequence 3, 3 + 8, 3 + 8 + 13, 3 + 8 + 13 + 18, ..., what is a_k :

Solution

1. The inner terms form an arithmetic sequence with first term 3 and common difference 5.

$$2. u_i = 5i - 2, \quad a_k = \sum_{i=1}^k u_i$$

$$3. a_k = \frac{k}{2} [2(3) + (k-1)(5)] = \frac{k(5k+1)}{2}$$

Correct option: A

Final answer

$$\frac{k(5k+1)}{2}$$

Worked Solutions

Summation

Q239 MCQ

Select the correct option: For the cumulative sequence 6, 6 + 8, 6 + 8 + 10, 6 + 8 + 10 + 12, ..., what is a_k :

Solution

1. The inner terms form an arithmetic sequence with first term 6 and common difference 2.

2. $u_i = 2i + 4, \quad a_k = \sum_{i=1}^k u_i$

3. $a_k = \frac{k}{2} [2(6) + (k-1)(2)] = k(k+5)$

Correct option: A

Final answer

$$k(k+5)$$

Q240 MCQ

Select the correct option: For the cumulative sequence 4, 4 + 8, 4 + 8 + 12, 4 + 8 + 12 + 16, ..., what is a_k :

Solution

1. The inner terms form an arithmetic sequence with first term 4 and common difference 4.

2. $u_i = 4i, \quad a_k = \sum_{i=1}^k u_i$

3. $a_k = \frac{k}{2} [2(4) + (k-1)(4)] = 2k(k+1)$

Correct option: A

Final answer

$$2k(k+1)$$

Worked Solutions

Summation

Q241 Written

For the sequence 3, 3 + 5, 3 + 5 + 7, 3 + 5 + 7 + 9, ..., find the sum from the first term to the n -th term.

Solution

1. First collapse the inner arithmetic sum: $a_k = k(k + 2)$.
2. $S_n = \sum_{k=1}^n a_k$; use $\sum k = \frac{n(n+1)}{2}$ and $\sum k^2 = \frac{n(n+1)(2n+1)}{6}$.
3. $S_n = \frac{n(n+1)(2n+7)}{6}$

Final answer

$$S_n = \frac{n(n+1)(2n+7)}{6}$$

Q242 Written

For the sequence 1, 1 + 3, 1 + 3 + 5, 1 + 3 + 5 + 7, ..., find the sum from the first term to the n -th term.

Solution

1. First collapse the inner arithmetic sum: $a_k = k^2$.
2. $S_n = \sum_{k=1}^n a_k$; use $\sum k = \frac{n(n+1)}{2}$ and $\sum k^2 = \frac{n(n+1)(2n+1)}{6}$.
3. $S_n = \frac{n(n+1)(2n+1)}{6}$

Final answer

$$S_n = \frac{n(n+1)(2n+1)}{6}$$

Worked Solutions

Summation

Q243 Written

For the sequence $1, 1 + 5, 1 + 5 + 9, 1 + 5 + 9 + 13, \dots$, **find the sum from the first term to the n -th term.**

Solution

1. First collapse the inner arithmetic sum: $a_k = k(2k - 1)$.
2. $S_n = \sum_{k=1}^n a_k$; use $\sum k = \frac{n(n+1)}{2}$ and $\sum k^2 = \frac{n(n+1)(2n+1)}{6}$.
3. $S_n = \frac{n(n+1)(4n-1)}{6}$

Final answer

$$S_n = \frac{n(n+1)(4n-1)}{6}$$

Q244 Written

For the sequence $2, 2 + 4, 2 + 4 + 6, 2 + 4 + 6 + 8, \dots$, **find the sum from the first term to the n -th term.**

Solution

1. First collapse the inner arithmetic sum: $a_k = k(k + 1)$.
2. $S_n = \sum_{k=1}^n a_k$; use $\sum k = \frac{n(n+1)}{2}$ and $\sum k^2 = \frac{n(n+1)(2n+1)}{6}$.
3. $S_n = \frac{n(n+1)(n+2)}{3}$

Final answer

$$S_n = \frac{n(n+1)(n+2)}{3}$$

Worked Solutions

Summation

Q245 MCQ

Select the correct option: For the cumulative sequence $2, 2 + 5, 2 + 5 + 8, 2 + 5 + 8 + 11, \dots$, find a_5 :

Solution

1. The inner terms form an arithmetic sequence with first term 2 and common difference 3.

2. $u_i = 3i - 1, \quad a_k = \sum_{i=1}^k u_i$

3. $a_k = \frac{k}{2} [2(2) + (k-1)(3)] = \frac{k(3k+1)}{2}$

4. $a_5 = 40$

Correct option: A

Final answer

40

Worked Solutions

Summation

Q246 Challenge

The sum of the first n terms of $\{a_n\}$ is $S_n = 2n^2 + 5n$. Find the general term.

Solution

1. $a_1 = S_1 = 7$
2. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
3. $a_n = (2n^2 + 5n) - [2(n-1)^2 + 5(n-1)] = 4n + 3$
4. The formula also gives $a_1 = 7$, so it is valid for every positive integer n .

Final answer

$$a_n = 4n + 3$$

Worked Solutions

Summation

Q247 **Written**

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = n(n - 3)$.

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = (n - 4)(n - 1)$
3. $a_n = 2(n - 2)$
4. $a_1 = S_1 = -2$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .

Final answer

$$a_n = 2(n - 2)$$

Worked Solutions

Summation

Q248 **Written**

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = n(2n + 5)$.

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = (n-1)(2n+3)$
3. $a_n = 4n + 3$
4. $a_1 = S_1 = 7$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .

Final answer

$$a_n = 4n + 3$$

Worked Solutions

Summation

Q249 Written

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = n^2$.

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = (n-1)^2$
3. $a_n = 2n - 1$
4. $a_1 = S_1 = 1$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .

Final answer

$$a_n = 2n - 1$$

Worked Solutions

Summation

Q250 Written

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = n(n - 4)$.

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = (n - 5)(n - 1)$
3. $a_n = 2n - 5$
4. $a_1 = S_1 = -3$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .

Final answer

$$a_n = 2n - 5$$

Worked Solutions

Summation

Q251 MCQ

Select the correct option: If $S_n = n(n + 2)$, the general term a_n is:

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = (n - 1)(n + 1)$
3. $a_n = 2n + 1$
4. $a_1 = S_1 = 3$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .

Correct option: D

Final answer

$$a_n = 2n + 1$$

Worked Solutions

Summation

Q252 MCQ

Select the correct option: If $S_n = 2n(n + 2)$, the general term a_n is:

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = 2(n-1)(n+1)$
3. $a_n = 2(2n+1)$
4. $a_1 = S_1 = 6$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .

Correct option: C

Final answer

$$a_n = 2(2n + 1)$$

Worked Solutions

Summation

Q253 MCQ

Select the correct option: If $S_n = 3n(n - 1)$, the general term a_n is:

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = 3(n - 2)(n - 1)$
3. $a_n = 6(n - 1)$
4. $a_1 = S_1 = 0$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .

Correct option: B

Final answer

$$a_n = 6(n - 1)$$

Worked Solutions

Summation

Q254 MCQ

Select the correct option: If $S_n = 2n(2n - 1)$, the general term a_n is:

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = 2(n-1)(2n-3)$
3. $a_n = 2(4n-3)$
4. $a_1 = S_1 = 2$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .

Correct option: A

Final answer

$$a_n = 2(4n - 3)$$

Worked Solutions

Summation

Q255 MCQ

Select the correct option: If $S_n = 5n(n + 1)$, the general term a_n is:

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = 5n(n - 1)$
3. $a_n = 10n$
4. $a_1 = S_1 = 10$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .

Correct option: D

Final answer

$$a_n = 10n$$

Worked Solutions

Summation

Q256 MCQ

Select the correct option: If $S_n = n(2n + 3)$, the general term a_n is:

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = (n-1)(2n+1)$
3. $a_n = 4n + 1$
4. $a_1 = S_1 = 5$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .

Correct option: C

Final answer

$$a_n = 4n + 1$$

Worked Solutions

Summation

Q257 MCQ

Select the correct option: If $S_n = n(3n + 1)$, the general term a_n is:

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = (n-1)(3n-2)$
3. $a_n = 2(3n-1)$
4. $a_1 = S_1 = 4$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .

Correct option: C

Final answer

$$a_n = 2(3n - 1)$$

Worked Solutions

Summation

Q258 MCQ

Select the correct option: If $S_n = n(n^2 + 2n + 3)$, the general term a_n is:

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = (n-1)(n^2 + 2)$
3. $a_n = 3n^2 + n + 2$
4. $a_1 = S_1 = 6$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .

Correct option: C

Final answer

$$a_n = 3n^2 + n + 2$$

Worked Solutions

Summation

Q259 MCQ

Select the correct option: If $S_n = n(n+1)(2n-1)$, the general term a_n is:

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = n(n-1)(2n-3)$
3. $a_n = 2n(3n-2)$
4. $a_1 = S_1 = 2$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .

Correct option: B

Final answer

$$a_n = 2n(3n-2)$$

Worked Solutions

Summation

Q260 MCQ

Select the correct option: If $S_n = n(3n^2 - 2n + 4)$, the general term a_n is:

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = (n-1)(3n^2 - 8n + 9)$
3. $a_n = 9n^2 - 13n + 9$
4. $a_1 = S_1 = 5$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .

Correct option: A

Final answer

$$a_n = 9n^2 - 13n + 9$$

Worked Solutions

Summation

Q261 MCQ

Select the correct option: If $S_n = n(n^2 - n + 2)$, the general term a_n is:

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = (n-1)(n^2 - 3n + 4)$
3. $a_n = 3n^2 - 5n + 4$
4. $a_1 = S_1 = 2$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .

Correct option: D

Final answer

$$a_n = 3n^2 - 5n + 4$$

Worked Solutions

Summation

Q262 MCQ

Select the correct option: If $S_n = n(n+1)(2n+1)$, the general term a_n is:

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = n(n-1)(2n-1)$
3. $a_n = 6n^2$
4. $a_1 = S_1 = 6$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .

Correct option: D

Final answer

$$a_n = 6n^2$$

Worked Solutions

Summation

Q263 MCQ

Select the correct option: If $S_n = n(n+1)(n+2)$, the general term a_n is:

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = n(n-1)(n+1)$
3. $a_n = 3n(n+1)$
4. $a_1 = S_1 = 6$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .

Correct option: B

Final answer

$$a_n = 3n(n+1)$$

Worked Solutions

Summation

Q264 **Written**

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = n^3 + 2n + 6$.

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = n^3 - 3n^2 + 5n + 3$
3. $a_n = 3(n^2 - n + 1)$
4. $a_1 = S_1 = 9$
5. The difference formula does not match a_1 at $n = 1$; keep the first term separately.

Final answer

$$a_1 = 9, \quad a_n = 3(n^2 - n + 1) \quad (n \geq 2)$$

Worked Solutions

Summation

Q265 Written

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = n^2 + 2n - 1$.

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = n^2 - 2$
3. $a_n = 2n + 1$
4. $a_1 = S_1 = 2$
5. The difference formula does not match a_1 at $n = 1$; keep the first term separately.

Final answer

$$a_1 = 2, \quad a_n = 2n + 1 \quad (n \geq 2)$$

Worked Solutions

Summation

Q266 Written

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = (n-1)(n^2 + n + 1)$.

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = (n-2)(n^2 - n + 1)$
3. $a_n = 3n^2 - 3n + 1$
4. $a_1 = S_1 = 0$
5. The difference formula does not match a_1 at $n = 1$; keep the first term separately.

Final answer

$$a_1 = 0, \quad a_n = 3n^2 - 3n + 1 \quad (n \geq 2)$$

Worked Solutions

Summation

Q267 **Written**

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = 5^n - 1$.

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = \frac{5^n - 5}{5}$
3. $a_n = 4 \cdot 5^{n-1}$
4. $a_1 = S_1 = 4$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .

Final answer

$$a_n = 4 \cdot 5^{n-1}$$

Worked Solutions

Summation

Q268 **Written**

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = 3^n - 1$.

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = \frac{3^n - 3}{3}$
3. $a_n = 2 \cdot 3^{n-1}$
4. $a_1 = S_1 = 2$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .

Final answer

$$a_n = 2 \cdot 3^{n-1}$$

Worked Solutions

Summation

Q269 **Written**

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = 2^n - 1$.

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = \frac{2^n - 2}{2}$
3. $a_n = 2^{n-1}$
4. $a_1 = S_1 = 1$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .

Final answer

$$a_n = 2^{n-1}$$

Worked Solutions

Summation

Q270 Written

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = 7^n - 1$.

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = \frac{7^n - 7}{7}$
3. $a_n = 6 \cdot 7^{n-1}$
4. $a_1 = S_1 = 6$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .

Final answer

$$a_n = 6 \cdot 7^{n-1}$$

Worked Solutions

Summation

Q271 **Written**

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = 5^n - 2$.

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = \frac{5^n - 10}{5}$
3. $a_n = 4 \cdot 5^{n-1}$
4. $a_1 = S_1 = 3$
5. The difference formula does not match a_1 at $n = 1$; keep the first term separately.

Final answer

$$a_1 = 3, \quad a_n = 4 \cdot 5^{n-1} \quad (n \geq 2)$$

Worked Solutions

Summation

Q272 Written

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = 4 \cdot 2^n - 3$.

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = 4 \cdot 2^{n-1} - 3$
3. $a_n = 2^{n+1}$
4. $a_1 = S_1 = 5$
5. The difference formula does not match a_1 at $n = 1$; keep the first term separately.

Final answer

$$a_1 = 5, \quad a_n = 2^{n+1} \quad (n \geq 2)$$

Worked Solutions

Summation

Q273 MCQ

Select the correct option: If $S_n = n(n + 4)$, find a_5 :**Solution**

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = (n - 1)(n + 3)$
3. $a_n = 2n + 3$
4. $a_1 = S_1 = 5$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .
6. $a_5 = 13$

Correct option: B**Final answer**

13

Worked Solutions

Summation

Q274 Challenge

Find the general term of the sequence $\{a_n\}$ where S_n is given by $S_n = 3 \cdot 3^n - 2n + 1$.

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = 3^n - 2n + 3$
3. $a_n = 2(3^n - 1)$
4. $a_1 = S_1 = 8$
5. The difference formula does not match a_1 at $n = 1$; keep the first term separately.

Final answer

$$a_1 = 8, \quad a_n = 2(3^n - 1) \quad (n \geq 2)$$

Worked Solutions

Summation

Quiz MCQ 1 **Concept Quiz · MCQ**

Select the correct option: Expand $\sum_{k=1}^4 (3k - 1)$ into a sum of terms:

Solution

1. Substitute $k = 1, 2, 3, 4$.

Correct option: A

Final answer

$$2 + 5 + 8 + 11$$

Quiz MCQ 2 **Concept Quiz · MCQ**

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2 - 1)$:

Solution

1. Expand $k(k^2 - 1) = k^3 - k$.

Correct option: A

Final answer

$$\frac{n(n+1)(n-1)(n+2)}{4}$$

Worked Solutions

Summation

Quiz MCQ 3 **Concept Quiz · MCQ**

Select the correct option: Find the sum $\sum_{k=1}^n k \times 2^{k-1}$:

Solution

1. Write S and rS, subtract, and isolate S.

Correct option: A

Final answer

$$(n - 1) \times 2^n + 1$$

Worked Solutions

Summation

Quiz MCQ 4 Concept Quiz · MCQ

Select the correct option: If $S_n = 2^n - 3$, find a_6 :

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = \frac{2^n - 6}{2}$
3. $a_n = 2^{n-1}$
4. $a_1 = S_1 = -1$
5. The difference formula does not match a_1 at $n = 1$; keep the first term separately.
6. $a_6 = 32$

Correct option: C

Final answer

32

Worked Solutions

Summation

Quiz WRITTEN 5 Concept Quiz · Written

Expand $\sum_{k=1}^5 (k^2 - 2)$ into a sum of terms.

Solution

1. Substitute $k = 1, 2, 3, 4, 5$.

Final answer

$$-1 + 2 + 7 + 14 + 23$$

Quiz WRITTEN 6 Concept Quiz · Written

Find the sum $\sum_{k=1}^n k^2(k + 5)$.

Solution

1. Expand into $(k^3 + 5k^2)$.
2. Use the standard power sums and factor.

Final answer

$$\frac{n(n+1)(3n^2 + 23n + 10)}{12}$$

Worked Solutions

Summation

Quiz WRITTEN 7 Concept Quiz · Written

Find the sum $\sum_{k=1}^n (4k - 1) \times 3^k$.

Solution

1. Show the S-rS alignment (or the inner arithmetic sum for the challenge).
2. $\frac{3(4 \cdot 3^n n - 3 \cdot 3^n + 3)}{2}$

Final answer

$$\frac{3(4 \cdot 3^n n - 3 \cdot 3^n + 3)}{2}$$

Worked Solutions

Summation

Quiz WRITTEN 8 Concept Quiz · Written

Find the general term of the sequence $\{a_n\}$ when S_n is $2 \cdot 2^n - 2n + 1$.

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = 2^n - 2n + 3$
3. $a_n = 2^n - 2$
4. $a_1 = S_1 = 3$
5. The difference formula does not match a_1 at $n = 1$; keep the first term separately.

Final answer

$$a_1 = 3, \quad a_n = 2^n - 2 \quad (n \geq 2)$$

Answer key

Use the question number shown on each page.

Q001 $2 + 4 + 6 + 8 + 10$

Q002 $3 + 5 + 7 + 9 + 11$

Q003 $3 + 6 + 9 + 12 + 15$

Q004 $2 + 6 + 10 + 14$

Q005 $-1 + 1 + 3 + 5 + 7$

Q006 A

Q007 A

Q008 A

Q009 A

Q010 A

Q011 A

Q012 A

Q013 A

Q014 $-2 + 1 + 6$

Q015 $5 + 14 + 29$

Q016 $1 + 4 + 9 + 16$

Q017 $5 + 11 + 21$

Q018 $-1 + 0 + 3 + 8$

Q019 A

Q020 A

Q021 A

Q022 A

Q023 A

Q024 A

Q025 A

Q026 A

Q027 $3 + 9 + 27 + 81$

Q028 $2 + 2 + 2$

Q029 $2 + 4 + 8 + 16$

Q030 $3 + 3 + 3 + 3 + 3 + 3$

Q031 $3 + 9 + 27$

Q032 $8 + 28 + 128$

Q033 $-1 + 1 + 5 + 13$

Q034 $5 + 5 + 5 + 5$

Q035 $3 + 3 + 3$

Q036 $4 + 4 + 4 + 4 + 4$

Q037 A

Q038 A

Q039 A

Q040 A

Q041 A

Q042 A

Q043 A

Q044 A

Q045 $\sum_{k=1}^n (4k - 1)$

Q046 $\sum_{k=1}^n (3k + 2)$

Q047 $\sum_{k=1}^n (2k + 1)$

Q048 $\sum_{k=1}^n (2k - 1)$

Q049 $\sum_{k=1}^n (7k - 1)$

Q050 $\sum_{k=1}^n (4k + 5)$

Q051 $\sum_{k=1}^n (3k - 2)$

Q052 A

Q053 A

Q054 A

Q055 A

Q056 A

Q057 A

Q058 A

Q059 A

Q060 $\sum_{k=6}^{15} k^2$

Q061 $\sum_{k=4}^{13} k^2$

Q062 $\sum_{k=1}^n (3^k - 2)$

Q063 $\sum_{k=1}^n 2^{k-1}$

Q064 $\sum_{k=7}^{21} k^2$

Q065 $\sum_{k=12}^{18} k^2$

Q066 $\sum_{k=10}^{20} k^2$

Q067 A

Q068 A

Q069 A

Q070 A

Q071 A

Q072 A

Q073 A

Q074 $\sum_{k=1}^{10} (3k - 1)$

Q075 $\sum_{k=1}^7 8k$

Q076 $\sum_{k=1}^{12} (6k + 1)$

Q077 $\sum_{k=1}^{24} (5k - 1)$

Q078 $\sum_{k=1}^n (3k - 2)$

Q079 $-3n$

Q080 $3n$

Q081 $6n$

Q082 $-4n$

Q083 n

Q084 A

Q085 A

Q086 A

Q087 A

Q088 A

Q089 A	Q118 $\frac{n(n+1)(2n-5)}{6}$	Q145 A	Q174 62
Q090 A	Q119 $\frac{2n(n+1)(n-1)}{3}$	Q146 A	Q175 60
Q091 A	Q120 $\frac{n(n+1)(n+5)}{3}$	Q147 A	Q176 105
Q092 A	Q121 $\frac{n(n+1)(2n+3)}{2}$	Q148 A	Q177 $\frac{n}{2(n+2)}$
Q093 A	Q122 $\frac{n(n+4)(n+5)}{3}$	Q149 A	Q178 $\frac{n}{n+1}$
Q094 A	Q123 16	Q150 A	Q179 $\frac{n}{3(n+3)}$
Q095 A	Q124 35	Q151 A	Q180 A
Q096 A	Q125 280	Q152 A	Q181 A
Q097 A	Q126 184	Q153 A	Q182 A
Q098 A	Q127 100	Q154 A	Q183 A
Q099 A	Q128 245	Q155 A	Q184 A
Q100 $\frac{n(5n+13)}{2}$	Q129 $\frac{n(n+1)(n+2)(3n+1)}{12}$	Q156 A	Q185 A
Q101 $n(n-2)$	Q130 $\frac{n(n+1)(3n^2-5n-4)}{12}$	Q157 A	Q186 A
Q102 $n(2n+1)$	Q131 $\frac{n(n+1)(3n^2+11n+4)}{12}$	Q158 A	Q187 A
Q103 $\frac{n(3n+7)}{2}$	Q132 $\frac{n(n+1)(n^2-3n-2)}{4}$	Q159 $3(3^n-1)$	Q188 A
Q104 $3n^2$	Q133 A	Q160 $4(2^n-1)$	Q189 $\frac{n}{2(3n+2)}$
Q105 A	Q134 A	Q161 $2(2^n-1)$	Q190 $\frac{n}{2n+1}$
Q106 A	Q135 A	Q162 $\frac{1-(-3)^n}{2}$	Q191 $\frac{1}{2}(1+\frac{1}{2}-\frac{1}{n+1}-\frac{1}{n+2})$
Q107 A	Q136 A	Q163 $1-(-2)^n$	Q192 $\frac{n}{3n+1}$
Q108 A	Q137 A	Q164 $\frac{5((-5)^n-1)}{6}$	Q193 $\frac{1}{6}(1+\frac{1}{4}-\frac{1}{3n+1}-\frac{1}{3n+4})$
Q109 $\frac{n(n-1)(2n-1)}{6}$	Q138 A	Q165 $1-(-1)^n$	Q194 A
Q110 A	Q139 A	Q166 $2(1-(-3)^n)$	Q195 A
Q111 A	Q140 A	Q167 $\frac{1-(-2)^n}{3}$	Q196 A
Q112 A	Q141 $\frac{n(n+1)(n-1)(n+2)}{4}$	Q168 $\frac{3((-3)^n-1)}{4}$	Q197 A
Q113 A	Q142 $\frac{n(n+1)(n^2+n+4)}{4}$	Q169 $\frac{4(4^n-1)}{3}$	Q198 A
Q114 A	Q143 A	Q170 $\frac{4^n-1}{3}$	Q199 A
Q115 $\frac{n(n+1)(10n-1)}{6}$	Q144 A	Q171 $\frac{3^n-1}{2}$	Q200 A
Q116 $\frac{n(n-1)(2n+1)}{2}$		Q172 -5	Q201 A
Q117 $\frac{n(n+1)(4n+5)}{6}$		Q173 39	Q202 $\frac{1}{2}(\frac{1}{4}+\frac{1}{5}-\frac{1}{n+4}-\frac{1}{n+5})$

$$\text{Q203 } \frac{n}{4(n+1)}$$

$$\text{Q204 } \sqrt{n+2} - \sqrt{2}$$

$$\text{Q205 } \sqrt{n+1} - 1$$

$$\text{Q206 } \sqrt{n}$$

$$\text{Q207 } \sqrt{n+1} - 1$$

$$\text{Q208 } \frac{\sqrt{2n+1} - 1}{2}$$

$$\text{Q209 } \sqrt{n+2} - \sqrt{2}$$

$$\text{Q210 } \frac{1425}{1984}$$

$$\text{Q211 } n \times 3^n$$

$$\text{Q212 } 3^n n$$

$$\text{Q213 } 324$$

$$\text{Q214 } \frac{6 \cdot 2^{2n} n - 5 \cdot 2^{2n} + 5}{9}$$

$$\text{Q215 } 14109$$

$$\text{Q216 } A$$

$$\text{Q217 } A$$

$$\text{Q218 } A$$

$$\text{Q219 } \frac{(2n+1) \times 3^n - 1}{4}$$

$$\text{Q220 } A$$

$$\text{Q221 } A$$

$$\text{Q222 } A$$

$$\text{Q223 } A$$

$$\text{Q224 } A$$

$$\text{Q225 } (n-1) \times 2^n + 1$$

$$\text{Q226 } (n-1) \times 3^{n+1} + 3$$

$$\text{Q227 } A$$

$$\text{Q228 } \frac{k(k+1)(4k+11)}{6}$$

$$\text{Q229 } a_k = k(k+2)$$

$$\text{Q230 } a_k = k^2$$

$$\text{Q231 } a_k = k(k+1)$$

$$\text{Q232 } a_k = k(2k-1)$$

$$\text{Q233 } A$$

$$\text{Q234 } A$$

$$\text{Q235 } A$$

$$\text{Q236 } A$$

$$\text{Q237 } A$$

$$\text{Q238 } A$$

$$\text{Q239 } A$$

$$\text{Q240 } A$$

$$\text{Q241 } S_n = \frac{n(n+1)(2n+7)}{6}$$

$$\text{Q242 } S_n = \frac{n(n+1)(2n+1)}{6}$$

$$\text{Q243 } S_n = \frac{n(n+1)(4n-1)}{6}$$

$$\text{Q244 } S_n = \frac{n(n+1)(n+2)}{3}$$

$$\text{Q245 } A$$

$$\text{Q246 } a_n = 4n + 3$$

$$\text{Q247 } a_n = 2(n-2)$$

$$\text{Q248 } a_n = 4n + 3$$

$$\text{Q249 } a_n = 2n - 1$$

$$\text{Q250 } a_n = 2n - 5$$

$$\text{Q251 } D$$

$$\text{Q252 } C$$

$$\text{Q253 } B$$

$$\text{Q254 } A$$

$$\text{Q255 } D$$

$$\text{Q256 } C$$

$$\text{Q257 } C$$

$$\text{Q258 } C$$

$$\text{Q259 } B$$

$$\text{Q260 } A$$

$$\text{Q261 } D$$

$$\text{Q262 } D$$

$$\text{Q263 } B$$

$$\text{Q264 } a_1 = 9, \quad a_n = 3(n^2 - n + 1) \quad (n \geq 2)$$

$$\text{Q265 } a_1 = 2, \quad a_n = 2n + 1 \quad (n \geq 2)$$

$$\text{Q266 } a_1 = 0, \quad a_n = 3n^2 - 3n + 1 \quad (n \geq 2)$$

$$\text{Q267 } a_n = 4 \cdot 5^{n-1}$$

$$\text{Q268 } a_n = 2 \cdot 3^{n-1}$$

$$\text{Q269 } a_n = 2^{n-1}$$

$$\text{Q270 } a_n = 6 \cdot 7^{n-1}$$

$$\text{Q271 } a_1 = 3, \quad a_n = 4 \cdot 5^{n-1} \quad (n \geq 2)$$

$$\text{Q272 } a_1 = 5, \quad a_n = 2^{n+1} \quad (n \geq 2)$$

$$\text{Q273 } B$$

$$\text{Q274 } a_1 = 8, \quad a_n = 2(3^n - 1) \quad (n \geq 2)$$

$$\text{Quiz MCQ 1 } A$$

$$\text{Quiz MCQ 2 } A$$

$$\text{Quiz MCQ 3 } A$$

$$\text{Quiz MCQ 4 } C$$

$$\text{Quiz WRITTEN 5 } -1 + 2 + 7 + 14 + 23$$

$$\text{Quiz WRITTEN 6 } \frac{n(n+1)(3n^2 + 23n + 10)}{12}$$

$$\text{Quiz WRITTEN 7 } \frac{3(4 \cdot 3^n n - 3 \cdot 3^n + 3)}{2}$$

$$\text{Quiz WRITTEN 8 } a_1 = 3, \quad a_n = 2^n - 2 \quad (n \geq 2)$$

Concept 4 | Advanced Sequences

English Student Edition • Focused formulas and guided practice

Formula opener

Differences

$$d_n = a_{n+1} - a_n$$

Rebuild

$$a_n = a_1 + \sum_{k=1}^{n-1} d_k$$

Grouping

odd/even index $\rightarrow$ separate rules

Worked Solutions

Advanced Sequences

Q001 Written

Find the general term of the sequence $\{2, 5, 12, 23, 38, 57, \dots\}$.

Solution

1. The first differences are 3, 7, 11, 15, 19,
2. Hence $b_k = 4k - 1$.
3. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
4. $a_n = 2 + \sum_{k=1}^{n-1} (4k - 1) = 2n^2 - 3n + 3$.
5. Substituting $n = 1$ gives $a_1 = 2$, matching the given first term.

Final answer

$$a_n = 2n^2 - 3n + 3$$

Worked Solutions

Advanced Sequences

Q002 MCQ

Select the correct option: If $a_1 = 1$ and $b_k = 2k + 1$, then a_n is:

Solution

1. The first differences are represented by $b_k = 2k + 1$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
3. $a_n = 1 + \sum_{k=1}^{n-1} (2k + 1) = n^2$.
4. Substituting $n = 1$ gives $a_1 = 1$, matching the given first term.

Correct option: C

Final answer

$$a_n = n^2$$

Q003 MCQ

Select the correct option: If $a_1 = 2$ and $b_k = 3k - 2$, then a_n is:

Solution

1. The first differences are represented by $b_k = 3k - 2$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
3. $a_n = 2 + \sum_{k=1}^{n-1} (3k - 2) = \frac{3n^2 - 7n + 8}{2}$.
4. Substituting $n = 1$ gives $a_1 = 2$, matching the given first term.

Correct option: B

Final answer

$$a_n = \frac{3n^2 - 7n + 8}{2}$$

Worked Solutions

Advanced Sequences

Q004 MCQ

Select the correct option: If $a_1 = 7$ and $b_k = 6k - 5$, then a_n is:

Solution

1. The first differences are represented by $b_k = 6k - 5$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
3. $a_n = 7 + \sum_{k=1}^{n-1} (6k - 5) = 3n^2 - 8n + 12$.
4. Substituting $n = 1$ gives $a_1 = 7$, matching the given first term.

Correct option: A

Final answer

$$a_n = 3n^2 - 8n + 12$$

Worked Solutions

Advanced Sequences

Q005 MCQ

Select the correct option: If $a_1 = 1$ and $b_k = 2k + 1$, find a_5 :**Solution**

1. The first differences are represented by $b_k = 2k + 1$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
3. $a_n = 1 + \sum_{k=1}^{n-1} (2k + 1) = n^2$.
4. Substituting $n = 1$ gives $a_1 = 1$, matching the given first term.
5. $a_5 = 25$

Correct option: B**Final answer**

25

Worked Solutions

Advanced Sequences

Q006 MCQ

Select the correct option: If $a_1 = 2$ and $b_k = 3k - 2$, find a_6 :

Solution

1. The first differences are represented by $b_k = 3k - 2$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
3. $a_n = 2 + \sum_{k=1}^{n-1} (3k - 2) = \frac{3n^2 - 7n + 8}{2}$.
4. Substituting $n = 1$ gives $a_1 = \frac{2}{2}$, matching the given first term.
5. $a_6 = 37$

Correct option: A

Final answer

37

Worked Solutions

Advanced Sequences

Q007 MCQ

Select the correct option: If $a_1 = 7$ and $b_k = 6k - 5$, find a_6 :**Solution**

1. The first differences are represented by $b_k = 6k - 5$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
3. $a_n = 7 + \sum_{k=1}^{n-1} (6k - 5) = 3n^2 - 8n + 12$.
4. Substituting $n = 1$ gives $a_1 = 7$, matching the given first term.
5. $a_6 = 72$

Correct option: D**Final answer**

72

Worked Solutions

Advanced Sequences

Q008 Written

Find the general term of the sequence $\{2, 10, 24, 44, 70, 102, \dots\}$.

Solution

1. The first differences are 8, 14, 20, 26, 32,
2. Hence $b_k = 2(3k + 1)$.
3. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
4. $a_n = 2 + \sum_{k=1}^{n-1} (2(3k + 1)) = n(3n - 1)$.
5. Substituting $n = 1$ gives $a_1 = 2$, matching the given first term.

Final answer

$$a_n = n(3n - 1)$$

Worked Solutions

Advanced Sequences

Q009 Written

Find the general term of the sequence $\{1, 6, 15, 28, 45, 66, \dots\}$.

Solution

1. The first differences are 5, 9, 13, 17, 21,
2. Hence $b_k = 4k + 1$.
3. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
4. $a_n = 1 + \sum_{k=1}^{n-1} (4k + 1) = n(2n - 1)$.
5. Substituting $n = 1$ gives $a_1 = 1$, matching the given first term.

Final answer

$$a_n = n(2n - 1)$$

Worked Solutions

Advanced Sequences

Q010 MCQ

Select the correct option: If $a_1 = 3$ and $b_k = 5k + 4$, then a_n is:

Solution

1. The first differences are represented by $b_k = 5k + 4$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
3. $a_n = 3 + \sum_{k=1}^{n-1} (5k + 4) = \frac{(n+1)(5n-2)}{2}$.
4. Substituting $n = 1$ gives $a_1 = 3$, matching the given first term.

Correct option: A

Final answer

$$a_n = \frac{(n+1)(5n-2)}{2}$$

Q011 MCQ

Select the correct option: If $a_1 = 4$ and $b_k = 7k - 3$, then a_n is:

Solution

1. The first differences are represented by $b_k = 7k - 3$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
3. $a_n = 4 + \sum_{k=1}^{n-1} (7k - 3) = \frac{7n^2 - 13n + 14}{2}$.
4. Substituting $n = 1$ gives $a_1 = 4$, matching the given first term.

Correct option: D

Final answer

$$a_n = \frac{7n^2 - 13n + 14}{2}$$

Worked Solutions

Advanced Sequences

Q012 MCQ

Select the correct option: If $a_1 = 6$ and $b_k = 2(2k + 3)$, then a_n is:**Solution**

1. The first differences are represented by $b_k = 2(2k + 3)$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
3. $a_n = 6 + \sum_{k=1}^{n-1} (2(2k + 3)) = 2n(n + 2)$.
4. Substituting $n = 1$ gives $a_1 = 6$, matching the given first term.

Correct option: B**Final answer**

$$a_n = 2n(n + 2)$$

Worked Solutions

Advanced Sequences

Q013 Written

Find the general term of the sequence $\{5, 8, 9, 8, 5, 0, \dots\}$.

Solution

1. The first differences are $3, 1, -1, -3, -5, \dots$
2. Hence $b_k = 5 - 2k$.
3. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
4. $a_n = 5 + \sum_{k=1}^{n-1} (5 - 2k) = -n(n - 6)$.
5. Substituting $n = 1$ gives $a_1 = 5$, matching the given first term.

Final answer

$$a_n = -n(n - 6)$$

Worked Solutions

Advanced Sequences

Q014 Written

Find the general term of the sequence $\{2, 3, 6, 15, 42, 123, \dots\}$.

Solution

1. The first differences are 1, 3, 9, 27, 81,
2. Hence $b_k = 3^{k-1}$.
3. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
4. $a_n = 2 + \sum_{k=1}^{n-1} (3^{k-1}) = \frac{3^n + 9}{6}$.
5. Substituting $n = 1$ gives $a_1 = 2$, matching the given first term.

Final answer

$$a_n = \frac{3^n + 9}{6}$$

Worked Solutions

Advanced Sequences

Q015 Written

Find the general term of the sequence $\{1, 2, 5, 14, 41, 122, \dots\}$.

Solution

1. The first differences are $1, 3, 9, 27, 81, \dots$
2. Hence $b_k = 3^{k-1}$.
3. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
4. $a_n = 1 + \sum_{k=1}^{n-1} (3^{k-1}) = \frac{3^n + 3}{6}$.
5. Substituting $n = 1$ gives $a_1 = 1$, matching the given first term.

Final answer

$$a_n = \frac{3^n + 3}{6}$$

Worked Solutions

Advanced Sequences

Q016 Written

Find the general term of the sequence $\{5, 7, 11, 19, 35, \dots\}$.

Solution

1. The first differences are 2, 4, 8, 16,
2. Hence $b_k = 2^k$.
3. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
4. $a_n = 5 + \sum_{k=1}^{n-1} (2^k) = 2^n + 3$.
5. Substituting $n = 1$ gives $a_1 = 5$, matching the given first term.

Final answer

$$a_n = 2^n + 3$$

Worked Solutions

Advanced Sequences

Q017 Written

Find the general term of the sequence $\{2, 3, 1, 5, -3, 13, \dots\}$.

Solution

1. The first differences are $1, -2, 4, -8, 16, \dots$
2. Hence $b_k = (-2)^{k-1}$.
3. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
4. $a_n = 2 + \sum_{k=1}^{n-1} ((-2)^{k-1}) = \frac{(-2)^n + 14}{6}$.
5. Substituting $n = 1$ gives $a_1 = 2$, matching the given first term.

Final answer

$$a_n = \frac{(-2)^n + 14}{6}$$

Worked Solutions

Advanced Sequences

Q018 Written

Find the general term of the sequence $\{5, 4, 7, -2, 25, -56, \dots\}$.

Solution

1. The first differences are $-1, 3, -9, 27, -81, \dots$
2. Hence $b_k = \frac{(-3)^k}{3}$.
3. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
4. $a_n = 5 + \sum_{k=1}^{n-1} \left(\frac{(-3)^k}{3}\right) = -\frac{(-3)^n - 57}{12}$.
5. Substituting $n = 1$ gives $a_1 = 5$, matching the given first term.

Final answer

$$a_n = -\frac{(-3)^n - 57}{12}$$

Worked Solutions

Advanced Sequences

Q019 MCQ

Select the correct option: If $a_1 = 1$ and $b_k = k^2$, then a_n is:

Solution

1. The first differences are represented by $b_k = k^2$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
3. $a_n = 1 + \sum_{k=1}^{n-1} (k^2) = \frac{(n+1)(2n^2-5n+6)}{6}$.
4. Substituting $n = 1$ gives $a_1 = 1$, matching the given first term.

Correct option: A

Final answer

$$a_n = \frac{(n+1)(2n^2-5n+6)}{6}$$

Worked Solutions

Advanced Sequences

Q020 Challenge

An oncology team records tissue volumes 1, 3, 7, 13, 21, 31, ... in mm^3 . Find the general formula a_n and calculate a_{12} to decide whether the 130 mm^3 threshold is exceeded.

Solution

1. The first differences are represented by $b_k = 2k$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
3. $a_n = 1 + \sum_{k=1}^{n-1} (2k) = n^2 - n + 1$.
4. Substituting $n = 1$ gives $a_1 = 1$, matching the given first term.
5. $a_{12} = 133$; this is greater than 130, so immediate surgery is required.

Final answer

$$a_n = n^2 - n + 1, \quad a_{12} = 133; \text{ surgery required}$$

Worked Solutions

Advanced Sequences

Q021 Written

For groups with $a_j = 2 + (j - 1)2$ and $c_j = 2n - 1$ terms in group j , find the first term of group 10.

Solution

1. $P_9 = 81$ terms precede the group.
2. First position = $P_9 + 1 = 82$
3. First value = 164

Final answer

164

Q022 Written

For groups with $a_j = 1 + (j - 1)2$ and $c_j = 2n - 1$ terms in group j , find the first term of group 10.

Solution

1. $P_9 = 81$ terms precede the group.
2. First position = $P_9 + 1 = 82$
3. First value = 163

Final answer

163

Worked Solutions

Advanced Sequences

Q023 Written

For groups with $a_j = 1 + (j - 1)1$ and $c_j = n$ terms in group j , find the first term of group 10.

Solution

1. $P_9 = 45$ terms precede the group.
2. First position $= P_9 + 1 = 46$
3. First value $= 46$

Final answer

46

Q024 MCQ

Select the correct option: find the first term of group 3:

Solution

1. $P_2 = 3$
2. Move to the next sequence position, then substitute in the ungrouped arithmetic sequence.
3. The first value is 4.

Correct option: B

Final answer

4

Worked Solutions

Advanced Sequences

Q025 MCQ

Select the correct option: find the first term of group 6:**Solution**

1. $P_5 = 25$
2. Move to the next sequence position, then substitute in the ungrouped arithmetic sequence.
3. The first value is 78.

Correct option: C**Final answer**

78

Q026 MCQ

Select the correct option: find the first term of group 9:**Solution**

1. $P_8 = 36$
2. Move to the next sequence position, then substitute in the ungrouped arithmetic sequence.
3. The first value is 37.

Correct option: D**Final answer**

37

Worked Solutions

Advanced Sequences

Q027 MCQ

Select the correct option: find the first term of group 4:**Solution**

1. $P_3 = 9$
2. Move to the next sequence position, then substitute in the ungrouped arithmetic sequence.
3. The first value is 30.

Correct option: B**Final answer**

30

Q028 MCQ

Select the correct option: find the first term of group 7:**Solution**

1. $P_6 = 21$
2. Move to the next sequence position, then substitute in the ungrouped arithmetic sequence.
3. The first value is 22.

Correct option: C**Final answer**

22

Worked Solutions

Advanced Sequences

Q029 MCQ

Select the correct option: find the first term of group 10:**Solution**

1. $P_9 = 81$
2. Move to the next sequence position, then substitute in the ungrouped arithmetic sequence.
3. The first value is 246.

Correct option: D**Final answer**

246

Q030 MCQ

Select the correct option: find the first term of group 4:**Solution**

1. $P_3 = 9$
2. Move to the next sequence position, then substitute in the ungrouped arithmetic sequence.
3. The first value is 20.

Correct option: A**Final answer**

20

Worked Solutions

Advanced Sequences

Q031 MCQ

Select the correct option: find the first term of group 7:**Solution**

1. $P_6 = 27$
2. Move to the next sequence position, then substitute in the ungrouped arithmetic sequence.
3. The first value is 110.

Correct option: B**Final answer**

110

Q032 MCQ

Select the correct option: find the first term of group 10:**Solution**

1. $P_9 = 81$
2. Move to the next sequence position, then substitute in the ungrouped arithmetic sequence.
3. The first value is 164.

Correct option: C**Final answer**

164

Worked Solutions

Advanced Sequences

Q033 MCQ

Select the correct option: find the first term of group 5:**Solution**

1. $P_4 = 14$
2. Move to the next sequence position, then substitute in the ungrouped arithmetic sequence.
3. The first value is 58.

Correct option: A**Final answer**

58

Q034 MCQ

Select the correct option: find the first term of group 8:**Solution**

1. $P_7 = 49$
2. Move to the next sequence position, then substitute in the ungrouped arithmetic sequence.
3. The first value is 100.

Correct option: B**Final answer**

100

Worked Solutions

Advanced Sequences

Q035 **Written**

For groups with $a_j = 1 + (j - 1)3$ and $c_j = 3n - 1$ terms in group j , find the first term of group 10.

Solution

1. $P_9 = 126$ terms precede the group.
2. First position = $P_9 + 1 = 127$
3. First value = 379

Final answer

379

Q036 **MCQ**

Select the correct option: find the first term of group 5:

Solution

1. $P_4 = 14$
2. Move to the next sequence position, then substitute in the ungrouped arithmetic sequence.
3. The first value is 29.

Correct option: D**Final answer**

29

Worked Solutions

Advanced Sequences

Q037 MCQ

Select the correct option: find the first term of group 8:**Solution**

1. $P_7 = 28$
2. Move to the next sequence position, then substitute in the ungrouped arithmetic sequence.
3. The first value is 89.

Correct option: A**Final answer**

89

Q038 Written

For groups with $a_j = 2 + (j - 1)2$ and $c_j = 2n - 1$ terms in group j , derive the sum of the terms in group n .**Solution**

1. First value $= 2(n^2 - 2n + 2)$
2. Last value $= 2n^2$
3. Number of terms $= 2n - 1$
4. $S_n = \frac{c_n}{2}(\text{first} + \text{last}) = 2(2n - 1)(n^2 - n + 1)$

Final answer $2(2n - 1)(n^2 - n + 1)$

Worked Solutions

Advanced Sequences

Q039 Written

For the same grouped sequence, find the sum of the terms in group 10.

Solution

1. First value = 163
2. Last value = 199
3. Number of terms = 19
4. $S_{10} = 3439$

Final answer

3439

Q040 Written

For the same grouped sequence, find the sum of the terms in group 10.

Solution

1. First value = 46
2. Last value = 55
3. Number of terms = 10
4. $S_{10} = 505$

Final answer

505

Worked Solutions

Advanced Sequences

Q041 **Written****For the same grouped sequence, find the sum of the terms in group 10.****Solution**

1. First value = 379
2. Last value = 463
3. Number of terms = 29
4. $S_{10} = 12209$

Final answer

12209

Q042 **Written****In the grouped arithmetic sequence, locate the value 88: state its group and its position within that group.****Solution**

1. Sequence position = $(88 - 2)/2 + 1 = 44$
2. $P_6 = 36 < 44 \leq P_7 = 49$
3. Therefore the value is in Group 7, position 8.

Final answer

Group 7, position 8

Worked Solutions

Advanced Sequences

Q043 Written

In the grouped arithmetic sequence, locate the value 101: state its group and its position within that group.

Solution

1. Sequence position $= (101 - 1)/2 + 1 = 51$
2. $P_7 = 49 < 51 \leq P_8 = 64$
3. Therefore the value is in Group 8, position 2.

Final answer

Group 8, position 2

Q044 Written

In the grouped arithmetic sequence, locate the value 124: state its group and its position within that group.

Solution

1. Sequence position $= (124 - 1)/1 + 1 = 124$
2. $P_{15} = 120 < 124 \leq P_{16} = 136$
3. Therefore the value is in Group 16, position 4.

Final answer

Group 16, position 4

Worked Solutions

Advanced Sequences

Q045 **Written**

In the grouped arithmetic sequence, locate the value 100: state its group and its position within that group.

Solution

1. Sequence position $= (100 - 1)/3 + 1 = 34$
2. $P_4 = 26 < 34 \leq P_5 = 40$
3. Therefore the value is in Group 5, position 8.

Final answer

Group 5, position 8

Worked Solutions

Advanced Sequences

Q046 Challenge

For the grouped sequence $2, 6 \mid 10, 14, 18 \mid 22, 26, 30, 34 \mid \dots$, solve: (a) the first term of Group 12; (b) the sum in Group n ; (c) the group and position of 202; (d) the total through Group 12.

Solution

1. $c_n = n + 1, \quad P_n = \frac{n(n+3)}{2}$
2. (a) first term = 310
3. (b) group sum = $2(n+1)(n^2 + 2n - 1)$
4. (c) 202 is in Group 9, position 7.
5. (d) total = 16200

Final answer

(a) 310; (b) $2(n+1)(n^2 + 2n - 1)$; (c) Group 9, position 7; (d) 16200

Worked Solutions

Advanced Sequences

Quiz MCQ 1 Concept Quiz · MCQ

Select the correct option: If $a_1 = 2$ and $b_k = 3k + 1$, then a_n is:

Solution

1. The first differences are represented by $b_k = 3k + 1$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
3. $a_n = 2 + \sum_{k=1}^{n-1} (3k + 1) = \frac{3n^2 - n + 2}{2}$.
4. Substituting $n = 1$ gives $a_1 = \frac{2}{2}$, matching the given first term.

Correct option: A

Final answer

$$a_n = \frac{3n^2 - n + 2}{2}$$

Quiz MCQ 2 Concept Quiz · MCQ

Select the correct option: If $a_1 = 3$ and $b_k = (-2)^{k-1}$, then a_n is:

Solution

1. The first differences are represented by $b_k = (-2)^{k-1}$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
3. $a_n = 3 + \sum_{k=1}^{n-1} ((-2)^{k-1}) = \frac{(-2)^n + 20}{6}$.
4. Substituting $n = 1$ gives $a_1 = 3$, matching the given first term.

Correct option: C

Final answer

$$a_n = \frac{(-2)^n + 20}{6}$$

Worked Solutions

Advanced Sequences

Quiz MCQ 3 **Concept Quiz · MCQ**

Select the correct option: find the first term of group 9:

Solution

1. $P_8 = 64$
2. Move to the next sequence position, then substitute in the ungrouped arithmetic sequence.
3. The first value is 195.

Correct option: A

Final answer

195

Quiz MCQ 4 **Concept Quiz · MCQ**

Select the correct option: the value 29 is in:

Solution

1. The value is sequence position 15.
2. It lies in Group 5, position 1.

Correct option: B

Final answer

Group 5, position 1

Worked Solutions

Advanced Sequences

Quiz WRITTEN 5 Concept Quiz · Written

A sequence has $a_1 = 1$ and difference sequence $b_k = 4k - 1$. Find the required general term a_n .

Solution

1. The first differences are represented by $b_k = 4k - 1$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
3. $a_n = 1 + \sum_{k=1}^{n-1} (4k - 1) = 2n^2 - 3n + 2$.
4. Substituting $n = 1$ gives $a_1 = 1$, matching the given first term.

Final answer

$$a_n = 2n^2 - 3n + 2$$

Quiz WRITTEN 6 Concept Quiz · Written

A sequence has $a_1 = 2$ and difference sequence $b_k = (-2)^{k-1}$. Find the required general term a_n .

Solution

1. The first differences are represented by $b_k = (-2)^{k-1}$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
3. $a_n = 2 + \sum_{k=1}^{n-1} ((-2)^{k-1}) = \frac{(-2)^n + 14}{6}$.
4. Substituting $n = 1$ gives $a_1 = 2$, matching the given first term.

Final answer

$$a_n = \frac{(-2)^n + 14}{6}$$

Worked Solutions

Advanced Sequences

Quiz WRITTEN 7 Concept Quiz · Written

For groups with $a_j = 5 + (j - 1)3$ and $c_j = n$ terms in group j , find the first term of group 12.

Solution

1. $P_{11} = 66$ terms precede the group.
2. First position $= P_{11} + 1 = 67$
3. First value $= 203$

Final answer

203

Quiz WRITTEN 8 Concept Quiz · Written

In the grouped arithmetic sequence, locate the value 66: state its group and its position within that group.

Solution

1. Sequence position $= (66 - 2)/4 + 1 = 17$
2. $P_4 = 14 < 17 \leq P_5 = 20$
3. Therefore the value is in Group 5, position 3.

Final answer

Group 5, position 3

Answer key

Use the question number shown on each page.

Q001 $a_n = 2n^2 - 3n + 3$

Q002 C

Q003 B

Q004 A

Q005 B

Q006 A

Q007 D

Q008 $a_n = n(3n - 1)$

Q009 $a_n = n(2n - 1)$

Q010 A

Q011 D

Q012 B

Q013 $a_n = -n(n - 6)$

Q014 $a_n = \frac{3^n + 9}{6}$

Q015 $a_n = \frac{3^n + 3}{6}$

Q016 $a_n = 2^n + 3$

Q017 $a_n = \frac{(-2)^n + 14}{6}$

Q018 $a_n = -\frac{(-3)^n - 57}{12}$

Q019 A

Q020 $a_n = n^2 - n + 1$, $a_{12} = 133$; surgery required

Q021 164

Q022 163

Q023 46

Q024 B

Q025 C

Q026 D

Q027 B

Q028 C

Q029 D

Q030 A

Q031 B

Q032 C

Q033 A

Q034 B

Q035 379

Q036 D

Q037 A

Q038 $2(2n - 1)(n^2 - n + 1)$

Q039 3439

Q040 505

Q041 12209

Q042 Group 7, position 8

Q043 Group 8, position 2

Q044 Group 16, position 4

Q045 Group 5, position 8

Q046 (a) 310; (b) $2(n + 1)(n^2 + 2n - 1)$; (c) Group 9, position 7; (d) 16200

Quiz MCQ 1 A

Quiz MCQ 2 C

Quiz MCQ 3 A

Quiz MCQ 4 B

Quiz WRITTEN 5 $a_n = 2n^2 - 3n + 2$

Quiz WRITTEN 6 $a_n = \frac{(-2)^n + 14}{6}$

Quiz WRITTEN 7 203

Quiz WRITTEN 8 Group 5, position 3

Concept 5 | Recurrence and Induction

English Student Edition • Focused formulas and guided practice

Formula opener

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1), P(k) \Rightarrow P(k + 1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k + 1) = mq'$$

Worked Solutions

Recurrence and Induction

Q001 Written

For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 3a_n + (2)$. Find the 2nd through the 4th terms.

Solution

1. $a_2 = 8$.
2. $a_3 = 26$.
3. $a_4 = 80$.

Final answer

$$a_2 = 8, a_3 = 26, a_4 = 80$$

Q002 Written

For the sequence $\{a_n\}$ with $a_1 = -6$ and recurrence $a_{n+1} = 2a_n + (3)$. Find the 2nd through the 4th terms.

Solution

1. $a_2 = -9$.
2. $a_3 = -15$.
3. $a_4 = -27$.

Final answer

$$a_2 = -9, a_3 = -15, a_4 = -27$$

Worked Solutions

Recurrence and Induction

Q003 Written

For the sequence $\{a_n\}$ with $a_1 = -6$ and recurrence $a_{n+1} = 1a_n + (n(n-2))$. Find the 2nd through the 4th terms.

Solution

1. $a_2 = -7$.
2. $a_3 = -7$.
3. $a_4 = -4$.

Final answer

$$a_2 = -7, a_3 = -7, a_4 = -4$$

Q004 Written

For the sequence $\{a_n\}$ with $a_1 = -2$ and recurrence $a_{n+1} = 1a_n + (-n)$. Find the 2nd through the 4th terms.

Solution

1. $a_2 = -3$.
2. $a_3 = -5$.
3. $a_4 = -8$.

Final answer

$$a_2 = -3, a_3 = -5, a_4 = -8$$

Worked Solutions

Recurrence and Induction

Q005 Written

For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 1a_n + (2n)$. Find the 2nd through the 4th terms.

Solution

1. $a_2 = 4$.
2. $a_3 = 8$.
3. $a_4 = 14$.

Final answer

$$a_2 = 4, a_3 = 8, a_4 = 14$$

Q006 Written

For the sequence $\{a_n\}$ with $a_1 = -2$ and recurrence $a_{n+1} = 1a_n + (3^n)$. Find the 2nd through the 4th terms.

Solution

1. $a_2 = 1$.
2. $a_3 = 10$.
3. $a_4 = 37$.

Final answer

$$a_2 = 1, a_3 = 10, a_4 = 37$$

Worked Solutions

Recurrence and Induction

Q007 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 3$ and recurrence $a_{n+1} = 3a_n + (0)$. The fourth term is:

Solution

1. $a_2 = 9$.
2. $a_3 = 27$.
3. $a_4 = 81$.

Correct option: B

Final answer

81

Q008 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 4a_n + (0)$. The fourth term is:

Solution

1. $a_2 = 8$.
2. $a_3 = 32$.
3. $a_4 = 128$.

Correct option: D

Final answer

128

Worked Solutions

Recurrence and Induction

Q009 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 2a_n + (n + 1)$. The fourth term is:

Solution

1. $a_2 = 6$.
2. $a_3 = 15$.
3. $a_4 = 34$.

Correct option: A

Final answer

34

Q010 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 5$ and recurrence $a_{n+1} = 3a_n + (2n)$. The fourth term is:

Solution

1. $a_2 = 17$.
2. $a_3 = 55$.
3. $a_4 = 171$.

Correct option: B

Final answer

171

Worked Solutions

Recurrence and Induction

Q011 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 1$ and recurrence $a_{n+1} = 1a_n + (4)$. The fourth term is:

Solution

1. $a_2 = 5$.
2. $a_3 = 9$.
3. $a_4 = 13$.

Correct option: D

Final answer

13

Q012 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 4$ and recurrence $a_{n+1} = -2a_n + (5)$. The fourth term is:

Solution

1. $a_2 = -3$.
2. $a_3 = 11$.
3. $a_4 = -17$.

Correct option: A

Final answer

-17

Worked Solutions

Recurrence and Induction

Q013 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 1$ and recurrence $a_{n+1} = -3a_n + (2)$. The fourth term is:

Solution

1. $a_2 = -1$.
2. $a_3 = 5$.
3. $a_4 = -13$.

Correct option: C

Final answer

-13

Q014 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 4$ and recurrence $a_{n+1} = 1a_n + (n^2)$. The fourth term is:

Solution

1. $a_2 = 5$.
2. $a_3 = 9$.
3. $a_4 = 18$.

Correct option: D

Final answer

18

Worked Solutions

Recurrence and Induction

Q015 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 2a_n + (1)$. The fourth term is:

Solution

1. $a_2 = 5$.
2. $a_3 = 11$.
3. $a_4 = 23$.

Correct option: C

Final answer

23

Q016 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 5$ and recurrence $a_{n+1} = 1a_n + (-3)$. The fourth term is:

Solution

1. $a_2 = 2$.
2. $a_3 = -1$.
3. $a_4 = -4$.

Correct option: D

Final answer

-4

Worked Solutions

Recurrence and Induction

Q017 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 3$ and recurrence $a_{n+1} = 1a_n + (n)$. The fourth term is:

Solution

1. $a_2 = 4$.
2. $a_3 = 6$.
3. $a_4 = 9$.

Correct option: B

Final answer

9

Q018 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 1$ and recurrence $a_{n+1} = 2a_n + (-n)$. The fourth term is:

Solution

1. $a_2 = 1$.
2. $a_3 = 0$.
3. $a_4 = -3$.

Correct option: C

Final answer

-3

Worked Solutions

Recurrence and Induction

Q019 Written

For the sequence $\{a_n\}$ with $a_1 = -1$ and recurrence $a_{n+1} = 1a_n + (5)$. Find the general term.

Solution

1. The rule adds the difference term, so it is arithmetic or a finite sum.
2. $a_n = 5n - 6$.

Final answer

$$a_n = 5n - 6$$

Q020 Written

For the sequence $\{a_n\}$ with $a_1 = 1$ and recurrence $a_{n+1} = 1a_n + (3)$. Find the general term.

Solution

1. The rule adds the difference term, so it is arithmetic or a finite sum.
2. $a_n = 3n - 2$.

Final answer

$$a_n = 3n - 2$$

Worked Solutions

Recurrence and Induction

Q021 Written

For the sequence $\{a_n\}$ with $a_1 = 4$ and recurrence $a_{n+1} = 1a_n + (-2)$. Find the general term.

Solution

1. The rule adds the difference term, so it is arithmetic or a finite sum.
2. $a_n = -2(n - 3)$.

Final answer

$$a_n = -2(n - 3)$$

Q022 Written

For the sequence $\{a_n\}$ with $a_1 = 1$ and recurrence $a_{n+1} = 1a_n + (6)$. Find the general term.

Solution

1. The rule adds the difference term, so it is arithmetic or a finite sum.
2. $a_n = 6n - 5$.

Final answer

$$a_n = 6n - 5$$

Worked Solutions

Recurrence and Induction

Q023 Written

For the sequence $\{a_n\}$ with $a_1 = 3$ and recurrence $a_{n+1} = 1a_n + (-5)$. Find the general term.

Solution

1. The rule adds the difference term, so it is arithmetic or a finite sum.
2. $a_n = 8 - 5n$.

Final answer

$$a_n = 8 - 5n$$

Q024 Written

For the sequence $\{a_n\}$ with $a_1 = -2$ and recurrence $a_{n+1} = 1a_n + (3)$. Find the general term.

Solution

1. The rule adds the difference term, so it is arithmetic or a finite sum.
2. $a_n = 3n - 5$.

Final answer

$$a_n = 3n - 5$$

Worked Solutions

Recurrence and Induction

Q025 Written

For the sequence $\{a_n\}$ with $a_1 = 4$ and recurrence $a_{n+1} = 1a_n + (-4)$. Find the general term.

Solution

1. The rule adds the difference term, so it is arithmetic or a finite sum.
2. $a_n = -4(n - 2)$.

Final answer

$$a_n = -4(n - 2)$$

Q026 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 3$ and recurrence $a_{n+1} = 3a_n + (0)$. The general term is:

Solution

1. The rule multiplies by a constant ratio, so the sequence is geometric.
2. $a_n = 3^n$.

Correct option: D

Final answer

$$a_n = 3^n$$

Worked Solutions

Recurrence and Induction

Q027 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 4a_n + (0)$. The general term is:

Solution

1. The rule multiplies by a constant ratio, so the sequence is geometric.
2. $a_n = 2^{2n-1}$.

Correct option: A

Final answer

$$a_n = 2^{2n-1}$$

Q028 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 4$ and recurrence $a_{n+1} = 5a_n + (0)$. The general term is:

Solution

1. The rule multiplies by a constant ratio, so the sequence is geometric.
2. $a_n = 4 \cdot 5^{n-1}$.

Correct option: A

Final answer

$$a_n = 4 \cdot 5^{n-1}$$

Worked Solutions

Recurrence and Induction

Q029 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 6$ and recurrence $a_{n+1} = 2a_n + (0)$. The general term is:

Solution

1. The rule multiplies by a constant ratio, so the sequence is geometric.
2. $a_n = 3 \cdot 2^n$.

Correct option: A

Final answer

$$a_n = 3 \cdot 2^n$$

Q030 Written

For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 3a_n + (0)$. Find the general term.

Solution

1. The rule multiplies by a constant ratio, so the sequence is geometric.
2. $a_n = 2 \cdot 3^{n-1}$.

Final answer

$$a_n = 2 \cdot 3^{n-1}$$

Worked Solutions

Recurrence and Induction

Q031 Written

For the sequence $\{a_n\}$ with $a_1 = 3$ and recurrence $a_{n+1} = 5a_n + (0)$. Find the general term.

Solution

1. The rule multiplies by a constant ratio, so the sequence is geometric.
2. $a_n = 3 \cdot 5^{n-1}$.

Final answer

$$a_n = 3 \cdot 5^{n-1}$$

Q032 Written

For the sequence $\{a_n\}$ with $a_1 = 4$ and recurrence $a_{n+1} = 3a_n + (0)$. Find the general term.

Solution

1. The rule multiplies by a constant ratio, so the sequence is geometric.
2. $a_n = 4 \cdot 3^{n-1}$.

Final answer

$$a_n = 4 \cdot 3^{n-1}$$

Worked Solutions

Recurrence and Induction

Q033 Written

For the sequence $\{a_n\}$ with $a_1 = 1$ and recurrence $a_{n+1} = 4a_n + (0)$. Find the general term.

Solution

1. The rule multiplies by a constant ratio, so the sequence is geometric.
2. $a_n = 2^{2n-2}$.

Final answer

$$a_n = 2^{2n-2}$$

Q034 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 1$ and recurrence $a_{n+1} = 1a_n + (4)$. The general term is:

Solution

1. The rule adds the difference term, so it is arithmetic or a finite sum.
2. $a_n = 4n - 3$.

Correct option: B

Final answer

$$a_n = 4n - 3$$

Worked Solutions

Recurrence and Induction

Q035 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 4$ and recurrence $a_{n+1} = 1a_n + (n^2)$. The general term is:

Solution

1. The rule adds the difference term, so it is arithmetic or a finite sum.

2. $a_n = \frac{2n^3 - 3n^2 + n + 24}{6}$.

Correct option: B

Final answer

$$a_n = \frac{2n^3 - 3n^2 + n + 24}{6}$$

Q036 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 1$ and recurrence $a_{n+1} = -4a_n + (0)$. The general term is:

Solution

1. The rule multiplies by a constant ratio, so the sequence is geometric.

2. $a_n = (-4)^{n-1}$.

Correct option: D

Final answer

$$a_n = (-4)^{n-1}$$

Worked Solutions

Recurrence and Induction

Q037 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 5$ and recurrence $a_{n+1} = 1a_n + (3^n)$. The general term is:

Solution

1. The rule adds the difference term, so it is arithmetic or a finite sum.
2. $a_n = \frac{3^{n+7}}{2}$.

Correct option: A

Final answer

$$a_n = \frac{3^n + 7}{2}$$

Q038 Written

For the sequence $\{a_n\}$ with $a_1 = -3$ and recurrence $a_{n+1} = -2a_n + (0)$. Find the general term.

Solution

1. The rule multiplies by a constant ratio, so the sequence is geometric.
2. $a_n = \frac{3(-2)^n}{2}$.

Final answer

$$a_n = \frac{3(-2)^n}{2}$$

Worked Solutions

Recurrence and Induction

Q039 Written

For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = -5a_n + (0)$. Find the general term.

Solution

1. The rule multiplies by a constant ratio, so the sequence is geometric.
2. $a_n = 2(-5)^{n-1}$.

Final answer

$$a_n = 2(-5)^{n-1}$$

Q040 Written

For the sequence $\{a_n\}$ with $a_1 = -2$ and recurrence $a_{n+1} = -4a_n + (0)$. Find the general term.

Solution

1. The rule multiplies by a constant ratio, so the sequence is geometric.
2. $a_n = \frac{(-4)^n}{2}$.

Final answer

$$a_n = \frac{(-4)^n}{2}$$

Worked Solutions

Recurrence and Induction

Q041 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 5$ and recurrence $a_{n+1} = 1a_n + (-3)$. The general term is:

Solution

1. The rule adds the difference term, so it is arithmetic or a finite sum.
2. $a_n = 8 - 3n$.

Correct option: B

Final answer

$$a_n = 8 - 3n$$

Q042 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 3$ and recurrence $a_{n+1} = 1a_n + (n)$. The general term is:

Solution

1. The rule adds the difference term, so it is arithmetic or a finite sum.
2. $a_n = \frac{n^2 - n + 6}{2}$.

Correct option: C

Final answer

$$a_n = \frac{n^2 - n + 6}{2}$$

Worked Solutions

Recurrence and Induction

Q043 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 3$ and recurrence $a_{n+1} = 1a_n + (-5)$. The general term is:

Solution

1. The rule adds the difference term, so it is arithmetic or a finite sum.
2. $a_n = 8 - 5n$.

Correct option: B**Final answer**

$$a_n = 8 - 5n$$

Q044 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 1a_n + (6)$. The general term is:

Solution

1. The rule adds the difference term, so it is arithmetic or a finite sum.
2. $a_n = 2(3n - 2)$.

Correct option: C**Final answer**

$$a_n = 2(3n - 2)$$

Worked Solutions

Recurrence and Induction

Q045 Written

For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 3a_n + (-n)$. Find the 2nd through the 4th terms.

Solution

1. $a_2 = 5$.
2. $a_3 = 13$.
3. $a_4 = 36$.

Final answer

$$a_2 = 5, a_3 = 13, a_4 = 36$$

Q046 Written

For the sequence $\{a_n\}$ with $a_1 = 1$ and recurrence $a_{n+1} = -2a_n + (5)$. Find the 2nd through the 4th terms.

Solution

1. $a_2 = 3$.
2. $a_3 = -1$.
3. $a_4 = 7$.

Final answer

$$a_2 = 3, a_3 = -1, a_4 = 7$$

Worked Solutions

Recurrence and Induction

Q047 **Written**

For the sequence $\{a_n\}$ with $a_1 = 3$ and recurrence $a_{n+1} = 3a_n + (-n)$. Find the 2nd through the 4th terms.

Solution

1. $a_2 = 8$.
2. $a_3 = 22$.
3. $a_4 = 63$.

Final answer

$$a_2 = 8, a_3 = 22, a_4 = 63$$

Worked Solutions

Recurrence and Induction

Q048 Challenge

Solve the third-order recurrence $a_n = 6a_{n-1} - 11a_{n-2} + 6a_{n-3}$ for $n \geq 3$ with $a_0 = 0, a_1 = 1, a_2 = 6$.

Solution

1. $r^3 - 6r^2 + 11r - 6 = (r - 1)(r - 2)(r - 3)$

2. $a_n = A + B2^n + C3^n$

3. $A = \frac{1}{2}, B = -2, C = \frac{3}{2}$

4. $a_n = -\frac{4 \cdot 2^n - 3 \cdot 3^n - 1}{2}$

Final answer

$$a_n = -\frac{4 \cdot 2^n - 3 \cdot 3^n - 1}{2}$$

Worked Solutions

Recurrence and Induction

Q049 Written

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 1$ and $a_{n+1} = a_n + (2n - 3)$.

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = 2k - 3$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 1 + \sum_{k=1}^{n-1} (2k - 3) = (n - 2)^2$.
4. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = (n - 2)^2$$

Q050 Written

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 1$ and $a_{n+1} = a_n + (3n)$.

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = 3k$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 1 + \sum_{k=1}^{n-1} (3k) = \frac{3n^2 - 3n + 2}{2}$.
4. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = \frac{3n^2 - 3n + 2}{2}$$

Worked Solutions

Recurrence and Induction

Q051 Written

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 2$ and $a_{n+1} = a_n + (2n)$.

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = 2k$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 2 + \sum_{k=1}^{n-1} (2k) = n^2 - n + 2$.
4. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = n^2 - n + 2$$

Q052 Written

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 2$ and $a_{n+1} = a_n + (3n^2)$.

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = 3k^2$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 2 + \sum_{k=1}^{n-1} (3k^2) = \frac{2n^3 - 3n^2 + n + 4}{2}$.
4. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = \frac{2n^3 - 3n^2 + n + 4}{2}$$

Worked Solutions

Recurrence and Induction

Q053 Written

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 2$ and $a_{n+1} = a_n + (3n + 1)$.

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = 3k + 1$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 2 + \sum_{k=1}^{n-1} (3k + 1) = \frac{3n^2 - n + 2}{2}$.
4. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = \frac{3n^2 - n + 2}{2}$$

Q054 Written

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 2$ and $a_{n+1} = a_n + (n(2n + 1))$.

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = k(2k + 1)$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 2 + \sum_{k=1}^{n-1} (k(2k + 1)) = \frac{4n^3 - 3n^2 - n + 12}{6}$.
4. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = \frac{4n^3 - 3n^2 - n + 12}{6}$$

Worked Solutions

Recurrence and Induction

Q055 MCQ

Select the correct option: If $a_1 = 3$ and $a_{n+1} = a_n + (n^2)$, then a_n is:

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = k^2$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 3 + \sum_{k=1}^{n-1} (k^2) = \frac{2n^3 - 3n^2 + n + 18}{6}$.
4. Substitute $n = 1$ to verify the given first term.

Correct option: C

Final answer

$$a_n = \frac{2n^3 - 3n^2 + n + 18}{6}$$

Q056 MCQ

Select the correct option: If $a_1 = 2$ and $a_{n+1} = a_n + (3n^2)$, then a_n is:

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = 3k^2$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 2 + \sum_{k=1}^{n-1} (3k^2) = \frac{2n^3 - 3n^2 + n + 4}{2}$.
4. Substitute $n = 1$ to verify the given first term.

Correct option: A

Final answer

$$a_n = \frac{2n^3 - 3n^2 + n + 4}{2}$$

Worked Solutions

Recurrence and Induction

Q057 MCQ

Select the correct option: If $a_1 = 2$ and $a_{n+1} = a_n + (n^2 + 1)$, then a_n is:

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = k^2 + 1$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 2 + \sum_{k=1}^{n-1} (k^2 + 1) = \frac{2n^3 - 3n^2 + 7n + 6}{6}$.
4. Substitute $n = 1$ to verify the given first term.

Correct option: C

Final answer

$$a_n = \frac{2n^3 - 3n^2 + 7n + 6}{6}$$

Q058 MCQ

Select the correct option: If $a_1 = 5$ and $a_{n+1} = a_n + (n(4n - 1))$, then a_n is:

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = k(4k - 1)$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 5 + \sum_{k=1}^{n-1} (k(4k - 1)) = \frac{(n+1)(8n^2 - 23n + 30)}{6}$.
4. Substitute $n = 1$ to verify the given first term.

Correct option: D

Final answer

$$a_n = \frac{(n+1)(8n^2 - 23n + 30)}{6}$$

Worked Solutions

Recurrence and Induction

Q059 MCQ

Select the correct option: If $a_1 = 2$ and $a_{n+1} = a_n + (8n)$, then a_n is:

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = 8k$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 2 + \sum_{k=1}^{n-1} (8k) = 2(2n^2 - 2n + 1)$.
4. Substitute $n = 1$ to verify the given first term.

Correct option: C

Final answer

$$a_n = 2(2n^2 - 2n + 1)$$

Q060 MCQ

Select the correct option: find a_4 when $a_1 = 3$ and $a_{n+1} = a_n + (n^2)$:

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = k^2$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 3 + \sum_{k=1}^{n-1} (k^2) = \frac{2n^3 - 3n^2 + n + 18}{6}$.
4. Substitute $n = 1$ to verify the given first term.

Correct option: A

Final answer

17

Worked Solutions

Recurrence and Induction

Q061 MCQ

Select the correct option: find a_5 when $a_1 = 2$ and $a_{n+1} = a_n + (3n^2)$:

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = 3k^2$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 2 + \sum_{k=1}^{n-1} (3k^2) = \frac{2n^3 - 3n^2 + n + 4}{2}$.
4. Substitute $n = 1$ to verify the given first term.

Correct option: D

Final answer

92

Q062 MCQ

Select the correct option: find a_3 when $a_1 = 2$ and $a_{n+1} = a_n + (n^2 + 1)$:

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = k^2 + 1$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 2 + \sum_{k=1}^{n-1} (k^2 + 1) = \frac{2n^3 - 3n^2 + 7n + 6}{6}$.
4. Substitute $n = 1$ to verify the given first term.

Correct option: A

Final answer

9

Worked Solutions

Recurrence and Induction

Q063 MCQ

Select the correct option: find a_6 when $a_1 = 5$ and $a_{n+1} = a_n + (n(4n - 1))$:

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = k(4k - 1)$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 5 + \sum_{k=1}^{n-1} (k(4k - 1)) = \frac{(n+1)(8n^2 - 23n + 30)}{6}$.
4. Substitute $n = 1$ to verify the given first term.

Correct option: B

Final answer

210

Q064 MCQ

Select the correct option: find a_2 when $a_1 = 2$ and $a_{n+1} = a_n + (8n)$:

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = 8k$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 2 + \sum_{k=1}^{n-1} (8k) = 2(2n^2 - 2n + 1)$.
4. Substitute $n = 1$ to verify the given first term.

Correct option: A

Final answer

10

Worked Solutions

Recurrence and Induction

Q065 Written

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 2$ and $a_{n+1} = a_n + (n^2)$.

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = k^2$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 2 + \sum_{k=1}^{n-1} (k^2) = \frac{2n^3 - 3n^2 + n + 12}{6}$.
4. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = \frac{2n^3 - 3n^2 + n + 12}{6}$$

Q066 MCQ

Select the correct option: If $a_1 = 1$ and $a_{n+1} = a_n + (2n + 1)$, then a_n is:

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = 2k + 1$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 1 + \sum_{k=1}^{n-1} (2k + 1) = n^2$.
4. Substitute $n = 1$ to verify the given first term.

Correct option: A

Final answer

$$a_n = n^2$$

Worked Solutions

Recurrence and Induction

Q067 Written

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 1$ and $a_{n+1} = a_n + (2^n)$.

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = 2^k$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 1 + \sum_{k=1}^{n-1} (2^k) = 2^n - 1$.
4. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = 2^n - 1$$

Q068 Written

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 2$ and $a_{n+1} = a_n + (-3^n)$.

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = -3^k$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 2 + \sum_{k=1}^{n-1} (-3^k) = -\frac{3^n - 7}{2}$.
4. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = -\frac{3^n - 7}{2}$$

Worked Solutions

Recurrence and Induction

Q069 Written

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 4$ and $a_{n+1} = a_n + (-4^n)$.

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = -4^k$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 4 + \sum_{k=1}^{n-1} (-4^k) = -\frac{4^n - 16}{3}$.
4. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = -\frac{4^n - 16}{3}$$

Worked Solutions

Recurrence and Induction

Q070 Challenge

A technology startup has a reserve of 400,000 LE, represented by $a_1 = 4$, and burns $0.04n$ hundred-thousand LE each quarter: $a_{n+1} = a_n - 0.04n$. Find an explicit formula for a_n .

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = -\frac{k}{25}$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 4 + \sum_{k=1}^{n-1} \left(-\frac{k}{25}\right) = -\frac{n^2 - n - 200}{50}$.
4. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = -\frac{n^2 - n - 200}{50}$$

Worked Solutions

Recurrence and Induction

Q071 Written

Find the general term of $\{a_n\}$ given $a_1 = 1$ and $a_{n+1} = 3a_n + (4)$.**Solution**

1. $c = -2$ from $c = 3c + (4)$.
2. $a_{n+1} - c = 3(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = 3^n - 2$.
5. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = 3^n - 2$$

Worked Solutions

Recurrence and Induction

Q072 **Written****Find the general term of $\{a_n\}$ given $a_1 = 1$ and $a_{n+1} = 5a_n + (8)$.****Solution**

1. $c = -2$ from $c = 5c + (8)$.
2. $a_{n+1} - c = 5(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = \frac{3 \cdot 5^n - 10}{5}$.
5. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = \frac{3 \cdot 5^n - 10}{5}$$

Worked Solutions

Recurrence and Induction

Q073 Written

Find the general term of $\{a_n\}$ given $a_1 = 2$ and $a_{n+1} = 2a_n + (-1)$.

Solution

1. $c = 1$ from $c = 2c + (-1)$.
2. $a_{n+1} - c = 2(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = \frac{2^n + 2}{2}$.
5. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = \frac{2^n + 2}{2}$$

Worked Solutions

Recurrence and Induction

Q074 **Written****Find the general term of $\{a_n\}$ given $a_1 = -3$ and $a_{n+1} = 4a_n + (3)$.****Solution**

1. $c = -1$ from $c = 4c + (3)$.
2. $a_{n+1} - c = 4(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = -\frac{4^n + 2}{2}$.
5. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = -\frac{4^n + 2}{2}$$

Worked Solutions

Recurrence and Induction

Q075 Written

Find the general term of $\{a_n\}$ given $a_1 = 3$ and $a_{n+1} = 3a_n + (-2)$.

Solution

1. $c = 1$ from $c = 3c + (-2)$.
2. $a_{n+1} - c = 3(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = \frac{2 \cdot 3^n + 3}{3}$.
5. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = \frac{2 \cdot 3^n + 3}{3}$$

Worked Solutions

Recurrence and Induction

Q076 Written

Find the general term of $\{a_n\}$ given $a_1 = -2$ and $a_{n+1} = -3a_n + (4)$.

Solution

1. $c = 1$ from $c = -3c + (4)$.
2. $a_{n+1} - c = -3(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = (-3)^n + 1$.
5. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = (-3)^n + 1$$

Worked Solutions

Recurrence and Induction

Q077 Written

Find the general term of $\{a_n\}$ given $a_1 = 1$ and $a_{n+1} = \frac{1}{3}a_n + (2)$.

Solution

1. $c = 3$ from $c = \frac{1}{3}c + (2)$.
2. $a_{n+1} - c = \frac{1}{3}(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = 3 \cdot 3^{-n} (3^n - 2)$.
5. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = 3 \cdot 3^{-n} (3^n - 2)$$

Worked Solutions

Recurrence and Induction

Q078 **Written**

Find the general term of $\{a_n\}$ given $a_1 = 2$ and $a_{n+1} = \frac{2}{3}a_n + (-\frac{1}{2})$.

Solution

1. $c = -\frac{3}{2}$ from $c = \frac{2}{3}c + (-\frac{1}{2})$.
2. $a_{n+1} - c = \frac{2}{3}(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = \frac{3 \cdot 3^{-n}(7 \cdot 2^n - 2 \cdot 3^n)}{4}$.
5. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = \frac{3 \cdot 3^{-n}(7 \cdot 2^n - 2 \cdot 3^n)}{4}$$

Worked Solutions

Recurrence and Induction

Q079 Written

Find the general term of $\{a_n\}$ given $a_1 = 4$ and $a_{n+1} = -2a_n + (3)$.

Solution

1. $c = 1$ from $c = -2c + (3)$.
2. $a_{n+1} - c = -2(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = -\frac{3(-2)^n - 2}{2}$.
5. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = -\frac{3(-2)^n - 2}{2}$$

Worked Solutions

Recurrence and Induction

Q080 MCQ

Select the correct option: If $a_1 = 3$ and $a_{n+1} = -2a_n + (5)$, the general term is:

Solution

1. $c = \frac{5}{3}$ from $c = -2c + (5)$.
2. $a_{n+1} - c = -2(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = -\frac{2(-2)^n - 5}{3}$.
5. Substitute $n = 1$ to verify the given first term.

Correct option: B

Final answer

$$a_n = -\frac{2(-2)^n - 5}{3}$$

Worked Solutions

Recurrence and Induction

Q081 MCQ

Select the correct option: If $a_1 = 1$ and $a_{n+1} = \frac{2}{3}a_n + (-2)$, the general term is:

Solution

1. $c = -6$ from $c = \frac{2}{3}c + (-2)$.
2. $a_{n+1} - c = \frac{2}{3}(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = \frac{3 \cdot 3^{-n}(7 \cdot 2^n - 4 \cdot 3^n)}{2}$.
5. Substitute $n = 1$ to verify the given first term.

Correct option: C

Final answer

$$a_n = \frac{3 \cdot 3^{-n}(7 \cdot 2^n - 4 \cdot 3^n)}{2}$$

Worked Solutions

Recurrence and Induction

Q082 MCQ

Select the correct option: If $a_1 = 2$ and $a_{n+1} = 7a_n + (-4)$, the general term is:

Solution

1. $c = \frac{2}{3}$ from $c = 7c + (-4)$.
2. $a_{n+1} - c = 7(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = \frac{2(2 \cdot 7^n + 7)}{21}$.
5. Substitute $n = 1$ to verify the given first term.

Correct option: D

Final answer

$$a_n = \frac{2(2 \cdot 7^n + 7)}{21}$$

Worked Solutions

Recurrence and Induction

Q083 MCQ

Select the correct option: If $a_1 = 6$ and $a_{n+1} = 2a_n + (-5)$, the general term is:

Solution

1. $c = 5$ from $c = 2c + (-5)$.
2. $a_{n+1} - c = 2(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = \frac{2^n + 10}{2}$.
5. Substitute $n = 1$ to verify the given first term.

Correct option: B

Final answer

$$a_n = \frac{2^n + 10}{2}$$

Worked Solutions

Recurrence and Induction

Q084 MCQ

Select the correct option: If $a_1 = 5$ and $a_{n+1} = -2a_n + (7)$, the general term is:

Solution

1. $c = \frac{7}{3}$ from $c = -2c + (7)$.
2. $a_{n+1} - c = -2(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = -\frac{4(-2)^n - 7}{3}$.
5. Substitute $n = 1$ to verify the given first term.

Correct option: C

Final answer

$$a_n = -\frac{4(-2)^n - 7}{3}$$

Worked Solutions

Recurrence and Induction

Q085 MCQ

Select the correct option: If $a_1 = 4$ and $a_{n+1} = -5a_n + (-2)$, the general term is:

Solution

1. $c = -\frac{1}{3}$ from $c = -5c + (-2)$.
2. $a_{n+1} - c = -5(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = -\frac{13(-5)^n + 5}{15}$.
5. Substitute $n = 1$ to verify the given first term.

Correct option: D

Final answer

$$a_n = -\frac{13(-5)^n + 5}{15}$$

Worked Solutions

Recurrence and Induction

Q086 MCQ

Select the correct option: If $a_1 = 2$ and $a_{n+1} = -3a_n + (5)$, the general term is:

Solution

1. $c = \frac{5}{4}$ from $c = -3c + (5)$.
2. $a_{n+1} - c = -3(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = -\frac{(-3)^n - 5}{4}$.
5. Substitute $n = 1$ to verify the given first term.

Correct option: B

Final answer

$$a_n = -\frac{(-3)^n - 5}{4}$$

Worked Solutions

Recurrence and Induction

Q087 MCQ

Select the correct option: If $a_1 = 1$ and $a_{n+1} = \frac{1}{2}a_n + (7)$, the general term is:

Solution

1. $c = 14$ from $c = \frac{1}{2}c + (7)$.
2. $a_{n+1} - c = \frac{1}{2}(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = 2 \cdot 2^{-n} (7 \cdot 2^n - 13)$.
5. Substitute $n = 1$ to verify the given first term.

Correct option: C

Final answer

$$a_n = 2 \cdot 2^{-n} (7 \cdot 2^n - 13)$$

Worked Solutions

Recurrence and Induction

Q088 MCQ

Select the correct option: If $a_1 = 1$ and $a_{n+1} = 2a_n + (-1)$, the general term is:

Solution

1. $c = 1$ from $c = 2c + (-1)$.
2. $a_{n+1} - c = 2(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = 1$.
5. Substitute $n = 1$ to verify the given first term.

Correct option: D

Final answer

$$a_n = 1$$

Worked Solutions

Recurrence and Induction

Q089 MCQ

Select the correct option: If $a_1 = 3$ and $a_{n+1} = -2a_n + (5)$, the general term is:

Solution

1. $c = \frac{5}{3}$ from $c = -2c + (5)$.
2. $a_{n+1} - c = -2(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = -\frac{2(-2)^{n-5}}{3}$.
5. Substitute $n = 1$ to verify the given first term.

Correct option: D

Final answer

$$a_n = -\frac{2(-2)^n - 5}{3}$$

Worked Solutions

Recurrence and Induction

Q090 Challenge

A sequence satisfies $a_{n+1} = 4a_n - 9$ with $a_1 = 2$. Find the equilibrium value, the explicit formula, and a_5 .

Solution

1. $c = 3$ from $c = 4c + (-9)$.
2. $a_{n+1} - c = 4(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = -\frac{2^{2n}-12}{4}$.
5. Substitute $n = 1$ to verify the given first term.
6. $a_5 = -253$

Final answer

$$c = 3, \quad a_n = -\frac{2^{2n} - 12}{4}, \quad a_5 = -253$$

Worked Solutions

Recurrence and Induction

Q091 Written

Use mathematical induction to prove $\sum_{i=1}^n i = \frac{n(n+1)}{2}$.

Solution

1. [1] Base case $n = 1$: both sides equal 1.
2. [2] Assume the statement holds for $n = k$: $\sum_{i=1}^k i = \frac{k(k+1)}{2}$.
3. For $n = k + 1$, add the next term n to the left side and use the hypothesis.
4. $\frac{k(k+1)}{2} + n = \frac{(k+1)(k+2)}{2}$, which is the right side at $k + 1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Final answer

$$\frac{n(n+1)}{2}$$

Worked Solutions

Recurrence and Induction

Q092 Written

Use mathematical induction to prove $\sum_{i=1}^n (2i - 1)^2 = \frac{n(2n-1)(2n+1)}{3}$.

Solution

1. [1] Base case $n = 1$: both sides equal 1.
2. [2] Assume the statement holds for $n = k$: $\sum_{i=1}^k (2i - 1)^2 = \frac{k(2k-1)(2k+1)}{3}$.
3. For $n = k + 1$, add the next term $(2k + 1)^2$ to the left side and use the hypothesis.
4. $\frac{k(2k-1)(2k+1)}{3} + (2k + 1)^2 = \frac{(k+1)(2k+1)(2k+3)}{3}$, which is the right side at $k + 1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Final answer

$$\frac{n(2n-1)(2n+1)}{3}$$

Worked Solutions

Recurrence and Induction

Q093 Written

Use mathematical induction to prove $\sum_{i=1}^n i(i+1) = \frac{n(n+1)(n+2)}{3}$.

Solution

1. [1] Base case $n = 1$: both sides equal 2.
2. [2] Assume the statement holds for $n = k$: $\sum_{i=1}^k i(i+1) = \frac{k(k+1)(k+2)}{3}$.
3. For $n = k + 1$, add the next term $(k+1)(k+2)$ to the left side and use the hypothesis.
4. $\frac{k(k+1)(k+2)}{3} + (k+1)(k+2) = \frac{(k+1)(k+2)(k+3)}{3}$, which is the right side at $k + 1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Final answer

$$\frac{n(n+1)(n+2)}{3}$$

Worked Solutions

Recurrence and Induction

Q094 MCQ

Select the correct option: the right-hand side equivalent to $\sum_{i=1}^n i(i+1)$ is:

Solution

1. [1] Base case $n = 1$: both sides equal 2.
2. [2] Assume the statement holds for $n = k$: $\sum_{i=1}^k i(i+1) = \frac{k(k+1)(k+2)}{3}$.
3. For $n = k + 1$, add the next term $(k+1)(k+2)$ to the left side and use the hypothesis.
4. $\frac{k(k+1)(k+2)}{3} + (k+1)(k+2) = \frac{(k+1)(k+2)(k+3)}{3}$, which is the right side at $k + 1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Correct option: A

Final answer

$$\frac{n(n+1)(n+2)}{3}$$

Worked Solutions

Recurrence and Induction

Q095 MCQ

Select the correct option: the right-hand side equivalent to $\sum_{i=1}^n 3 \cdot 4^{i-1}$ is:

Solution

1. [1] Base case $n = 1$: both sides equal 3.
2. [2] Assume the statement holds for $n = k$: $\sum_{i=1}^k 3 \cdot 4^{i-1} = 4^k - 1$.
3. For $n = k + 1$, add the next term $3 \cdot 4^k$ to the left side and use the hypothesis.
4. $4^k - 1 + 3 \cdot 4^k = (2 \cdot 2^k - 1)(2 \cdot 2^k + 1)$, which is the right side at $k + 1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Correct option: B

Final answer

$$4^n - 1$$

Worked Solutions

Recurrence and Induction

Q096 MCQ

Select the correct option: the right-hand side equivalent to $\sum_{i=1}^n 2i + 3$ is:

Solution

1. [1] Base case $n = 1$: both sides equal 5.
2. [2] Assume the statement holds for $n = k$: $\sum_{i=1}^k 2i + 3 = k(k + 4)$.
3. For $n = k + 1$, add the next term $2k + 5$ to the left side and use the hypothesis.
4. $k(k + 4) + 2k + 5 = (k + 1)(k + 5)$, which is the right side at $k + 1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Correct option: C

Final answer

$$n(n + 4)$$

Worked Solutions

Recurrence and Induction

Q097 MCQ

Select the correct option: the right-hand side equivalent to $\sum_{i=1}^n 2i^2 + 3i$ is:

Solution

1. [1] Base case $n = 1$: both sides equal 5.
2. [2] Assume the statement holds for $n = k$: $\sum_{i=1}^k 2i^2 + 3i = \frac{k(k+1)(4k+11)}{6}$.
3. For $n = k + 1$, add the next term $(k + 1)(2k + 5)$ to the left side and use the hypothesis.
4. $\frac{k(k+1)(4k+11)}{6} + (k + 1)(2k + 5) = \frac{(k+1)(k+2)(4k+15)}{6}$, which is the right side at $k + 1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Correct option: B

Final answer

$$\frac{n(n+1)(4n+11)}{6}$$

Worked Solutions

Recurrence and Induction

Q098 MCQ

Select the correct option: the right-hand side equivalent to $\sum_{i=1}^n 3i^2 + 2i$ is:

Solution

1. [1] Base case $n = 1$: both sides equal 5.
2. [2] Assume the statement holds for $n = k$: $\sum_{i=1}^k 3i^2 + 2i = \frac{k(k+1)(2k+3)}{2}$.
3. For $n = k + 1$, add the next term $(k + 1)(3k + 5)$ to the left side and use the hypothesis.
4. $\frac{k(k+1)(2k+3)}{2} + (k + 1)(3k + 5) = \frac{(k+1)(k+2)(2k+5)}{2}$, which is the right side at $k + 1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Correct option: C

Final answer

$$\frac{n(n+1)(2n+3)}{2}$$

Worked Solutions

Recurrence and Induction

Q099 MCQ

Select the correct option: the right-hand side equivalent to $\sum_{i=1}^n 7 \cdot 2^{i-1}$ is:

Solution

1. [1] Base case $n = 1$: both sides equal 7.
2. [2] Assume the statement holds for $n = k$: $\sum_{i=1}^k 7 \cdot 2^{i-1} = 7(2^k - 1)$.
3. For $n = k + 1$, add the next term $7 \cdot 2^k$ to the left side and use the hypothesis.
4. $7(2^k - 1) + 7 \cdot 2^k = 7(2 \cdot 2^k - 1)$, which is the right side at $k + 1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Correct option: A

Final answer

$$7(2^n - 1)$$

Worked Solutions

Recurrence and Induction

Q100 MCQ

Select the correct option: the right-hand side equivalent to $\sum_{i=1}^n i(i+1)$ is:

Solution

1. [1] Base case $n = 1$: both sides equal 2.
2. [2] Assume the statement holds for $n = k$: $\sum_{i=1}^k i(i+1) = \frac{k(k+1)(k+2)}{3}$.
3. For $n = k + 1$, add the next term $(k+1)(k+2)$ to the left side and use the hypothesis.
4. $\frac{k(k+1)(k+2)}{3} + (k+1)(k+2) = \frac{(k+1)(k+2)(k+3)}{3}$, which is the right side at $k + 1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Correct option: C

Final answer

$$\frac{n(n+1)(n+2)}{3}$$

Worked Solutions

Recurrence and Induction

Q101 MCQ

Select the correct option: the right-hand side equivalent to $\sum_{i=1}^n 3 \cdot 4^{i-1}$ is:

Solution

1. [1] Base case $n = 1$: both sides equal 3.
2. [2] Assume the statement holds for $n = k$: $\sum_{i=1}^k 3 \cdot 4^{i-1} = 4^k - 1$.
3. For $n = k + 1$, add the next term $3 \cdot 4^k$ to the left side and use the hypothesis.
4. $4^k - 1 + 3 \cdot 4^k = (2 \cdot 2^k - 1)(2 \cdot 2^k + 1)$, which is the right side at $k + 1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Correct option: A

Final answer

$$4^n - 1$$

Worked Solutions

Recurrence and Induction

Q102 MCQ

Select the correct option: the right-hand side equivalent to $\sum_{i=1}^n 2i + 3$ is:

Solution

1. [1] Base case $n = 1$: both sides equal 5.
2. [2] Assume the statement holds for $n = k$: $\sum_{i=1}^k 2i + 3 = k(k + 4)$.
3. For $n = k + 1$, add the next term $2k + 5$ to the left side and use the hypothesis.
4. $k(k + 4) + 2k + 5 = (k + 1)(k + 5)$, which is the right side at $k + 1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Correct option: B

Final answer

$$n(n + 4)$$

Worked Solutions

Recurrence and Induction

Q103 Written

Use mathematical induction to prove $\sum_{i=1}^n (2i - 1) = n^2$.

Solution

1. [1] Base case $n = 1$: both sides equal 1.
2. [2] Assume the statement holds for $n = k$: $\sum_{i=1}^k (2i - 1) = k^2$.
3. For $n = k + 1$, add the next term $2k + 1$ to the left side and use the hypothesis.
4. $k^2 + 2k + 1 = (k + 1)^2$, which is the right side at $k + 1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Final answer

$$n^2$$

Worked Solutions

Recurrence and Induction

Q104 Written

Use mathematical induction to prove $\sum_{i=1}^n i(i+1) = \frac{n(n+1)(n+2)}{3}$.

Solution

1. [1] Base case $n = 1$: both sides equal 2.
2. [2] Assume the statement holds for $n = k$: $\sum_{i=1}^k i(i+1) = \frac{k(k+1)(k+2)}{3}$.
3. For $n = k + 1$, add the next term $(k+1)(k+2)$ to the left side and use the hypothesis.
4. $\frac{k(k+1)(k+2)}{3} + (k+1)(k+2) = \frac{(k+1)(k+2)(k+3)}{3}$, which is the right side at $k + 1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Final answer

$$\frac{n(n+1)(n+2)}{3}$$

Worked Solutions

Recurrence and Induction

Q105 Written

Use mathematical induction to prove $\sum_{i=1}^n i(i+2) = \frac{n(n+1)(2n+7)}{6}$.

Solution

1. [1] Base case $n = 1$: both sides equal 3.
2. [2] Assume the statement holds for $n = k$: $\sum_{i=1}^k i(i+2) = \frac{k(k+1)(2k+7)}{6}$.
3. For $n = k + 1$, add the next term $(k+1)(k+3)$ to the left side and use the hypothesis.
4. $\frac{k(k+1)(2k+7)}{6} + (k+1)(k+3) = \frac{(k+1)(k+2)(2k+9)}{6}$, which is the right side at $k + 1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Final answer

$$\frac{n(n+1)(2n+7)}{6}$$

Worked Solutions

Recurrence and Induction

Q106 Written

Use mathematical induction to prove $\sum_{i=1}^n 3 \cdot 4^{i-1} = 4^n - 1$.

Solution

1. [1] Base case $n = 1$: both sides equal 3.
2. [2] Assume the statement holds for $n = k$: $\sum_{i=1}^k 3 \cdot 4^{i-1} = 4^k - 1$.
3. For $n = k + 1$, add the next term $3 \cdot 4^k$ to the left side and use the hypothesis.
4. $4^k - 1 + 3 \cdot 4^k = (2 \cdot 2^k - 1)(2 \cdot 2^k + 1)$, which is the right side at $k + 1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Final answer

$$4^n - 1$$

Worked Solutions

Recurrence and Induction

Q107 Written

Use mathematical induction to prove $\sum_{i=0}^n 10^i = \frac{10 \cdot 10^n - 1}{9}$.

Solution

1. [1] Base case $n = 1$: both sides equal 11.
2. [2] Assume the statement holds for $n = k$: $\sum_{i=0}^k 10^i = \frac{10 \cdot 10^k - 1}{9}$.
3. For $n = k + 1$, add the next term 10^{k+1} to the left side and use the hypothesis.
4. $\frac{10 \cdot 10^k - 1}{9} + 10^{k+1} = \frac{100 \cdot 10^k - 1}{9}$, which is the right side at $k + 1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Final answer

$$\frac{10 \cdot 10^n - 1}{9}$$

Worked Solutions

Recurrence and Induction

Q108 Challenge

Use mathematical induction to prove $\sum_{i=1}^n (i^2 + i + 1) = \frac{n(n^2 + 3n + 5)}{3}$.

Solution

1. [1] Base case $n = 1$: both sides equal 3.
2. [2] Assume the statement holds for $n = k$: $\sum_{i=1}^k (i^2 + i + 1) = \frac{k(k^2 + 3k + 5)}{3}$.
3. For $n = k + 1$, add the next term $k^2 + 3k + 3$ to the left side and use the hypothesis.
4. $\frac{k(k^2 + 3k + 5)}{3} + k^2 + 3k + 3 = \frac{(k+1)(k^2 + 5k + 9)}{3}$, which is the right side at $k + 1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Final answer

$$\frac{n(n^2 + 3n + 5)}{3}$$

Worked Solutions

Recurrence and Induction

Q109 Written

Prove by mathematical induction that $2^n > 3n$ for $n \geq 4$.

Solution

1. $n = 4$: $2^4 = 16 > 12 = 3 \cdot 4$.
2. Assume the inequality holds for $n = k$ with $k \geq 4$.
3. $2^{k+1} > 2(3k + 0)$ after multiplying by the positive base.
4. $2(3k + 0) - (3(k + 1) + 0) = 3(k - 1)$ is positive for the allowed k .
5. Hence the inequality holds for $k+1$ and therefore for every allowed natural number.

Final answer

$$2^n > 3n$$

Worked Solutions

Recurrence and Induction

Q110 Written

Prove by mathematical induction that $3^n > 5n + 1$ for $n \geq 3$.

Solution

1. $n = 3$: $3^3 = 27 > 16 = 5 \cdot 3 + 1$.
2. Assume the inequality holds for $n = k$ with $k \geq 3$.
3. $3^k + 1 > 3(5k + 1)$ after multiplying by the positive base.
4. $3(5k + 1) - (5(k + 1) + 1) = 10k - 3$ is positive for the allowed k .
5. Hence the inequality holds for $k+1$ and therefore for every allowed natural number.

Final answer

$$3^n > 5n + 1$$

Worked Solutions

Recurrence and Induction

Q111 Written

Prove by mathematical induction that $3^n > 2n$ for $n \geq 1$.

Solution

1. $n = 1$: $3^1 = 3 > 2 = 2$.
2. Assume the inequality holds for $n = k$ with $k \geq 1$.
3. $3^k + 1 > 3(2k + 0)$ after multiplying by the positive base.
4. $3(2k + 0) - (2(k + 1) + 0) = 2(2k - 1)$ is positive for the allowed k .
5. Hence the inequality holds for $k+1$ and therefore for every allowed natural number.

Final answer

$$3^n > 2n$$

Worked Solutions

Recurrence and Induction

Q112 Written

Prove by mathematical induction that $2^n > 1n + 2$ for $n \geq 3$.

Solution

1. $n = 3$: $2^3 = 8 > 5 = 1 \cdot 3 + 2$.
2. Assume the inequality holds for $n = k$ with $k \geq 3$.
3. $2^k + 1 > 2(1k + 2)$ after multiplying by the positive base.
4. $2(1k + 2) - (1(k + 1) + 2) = k + 1$ is positive for the allowed k .
5. Hence the inequality holds for $k+1$ and therefore for every allowed natural number.

Final answer

$$2^n > 1n + 2$$

Worked Solutions

Recurrence and Induction

Q113 Written

Prove by mathematical induction that $3^n > 5n + 1$ for $n \geq 4$.

Solution

1. $n = 4$: $3^4 = 81 > 21 = 5 \cdot 4 + 1$.
2. Assume the inequality holds for $n = k$ with $k \geq 4$.
3. $3^k + 1 > 3(5k + 1)$ after multiplying by the positive base.
4. $3(5k + 1) - (5(k + 1) + 1) = 10k - 3$ is positive for the allowed k .
5. Hence the inequality holds for $k+1$ and therefore for every allowed natural number.

Final answer

$$3^n > 5n + 1$$

Worked Solutions

Recurrence and Induction

Q114 Written

Prove by mathematical induction that $3^n > 5n + 2$ for $n \geq 4$.

Solution

1. $n = 4$: $3^4 = 81 > 22 = 5 \cdot 4 + 2$.
2. Assume the inequality holds for $n = k$ with $k \geq 4$.
3. $3^k + 1 > 3(5k + 2)$ after multiplying by the positive base.
4. $3(5k + 2) - (5(k + 1) + 2) = 10k - 1$ is positive for the allowed k .
5. Hence the inequality holds for $k+1$ and therefore for every allowed natural number.

Final answer

$$3^n > 5n + 2$$

Worked Solutions

Recurrence and Induction

Q115 Written

Prove by mathematical induction that $4^n > 2n + 3$ for $n \geq 4$.

Solution

1. $n = 4$: $4^4 = 256 > 11 = 2 \cdot 4 + 3$.
2. Assume the inequality holds for $n = k$ with $k \geq 4$.
3. $4^k + 1 > 4(2k + 3)$ after multiplying by the positive base.
4. $4(2k + 3) - (2(k + 1) + 3) = 6k + 7$ is positive for the allowed k .
5. Hence the inequality holds for $k+1$ and therefore for every allowed natural number.

Final answer

$$4^n > 2n + 3$$

Worked Solutions

Recurrence and Induction

Q116 MCQ

Select the correct option: which inequality is proved for $n \geq 3$?:**Solution**

1. $n = 3$: $2^3 = 8 > 5 = 5$.
2. Assume the inequality holds for $n = k$ with $k \geq 3$.
3. $2^k + 1 > 2(1k + 2)$ after multiplying by the positive base.
4. $2(1k + 2) - (1(k + 1) + 2) = k + 1$ is positive for the allowed k .
5. Hence the inequality holds for $k+1$ and therefore for every allowed natural number.

Correct option: D**Final answer**

$$2^n > 1n + 2$$

Worked Solutions

Recurrence and Induction

Q117 MCQ

Select the correct option: which inequality is proved for $n \geq 2$?

Solution

1. $n = 2$: $3^2 = 9 > 4 = 2n$.
2. Assume the inequality holds for $n = k$ with $k \geq 2$.
3. $3^k + 1 > 3(2k + 0)$ after multiplying by the positive base.
4. $3(2k + 0) - (2(k + 1) + 0) = 2(2k - 1)$ is positive for the allowed k .
5. Hence the inequality holds for $k+1$ and therefore for every allowed natural number.

Correct option: A

Final answer

$$3^n > 2n$$

Worked Solutions

Recurrence and Induction

Q118 MCQ

Select the correct option: which inequality is proved for $n \geq 4$?:**Solution**

1. $n = 4$: $3^4 = 81 > 21 = 21$.
2. Assume the inequality holds for $n = k$ with $k \geq 4$.
3. $3^k + 1 > 3(5k + 1)$ after multiplying by the positive base.
4. $3(5k + 1) - (5(k + 1) + 1) = 10k - 3$ is positive for the allowed k .
5. Hence the inequality holds for $k+1$ and therefore for every allowed natural number.

Correct option: B**Final answer**

$$3^n > 5n + 1$$

Worked Solutions

Recurrence and Induction

Q119 MCQ

Select the correct option: which inequality is proved for $n \geq 3$?

Solution

1. $n = 3$: $3^3 = 27 > 9 = 3n$.
2. Assume the inequality holds for $n = k$ with $k \geq 3$.
3. $3^k + 1 > 3(3k + 0)$ after multiplying by the positive base.
4. $3(3k + 0) - (3(k + 1) + 0) = 3(2k - 1)$ is positive for the allowed k .
5. Hence the inequality holds for $k+1$ and therefore for every allowed natural number.

Correct option: D

Final answer

$$3^n > 3n$$

Worked Solutions

Recurrence and Induction

Q120 MCQ

Select the correct option: which inequality is proved for $n \geq 4$?:

Solution

1. $n = 4$: $3^4 = 81 > 22 = 22$.
2. Assume the inequality holds for $n = k$ with $k \geq 4$.
3. $3^k + 1 > 3(5k + 2)$ after multiplying by the positive base.
4. $3(5k + 2) - (5(k + 1) + 2) = 10k - 1$ is positive for the allowed k .
5. Hence the inequality holds for $k+1$ and therefore for every allowed natural number.

Correct option: A

Final answer

$$3^n > 5n + 2$$

Worked Solutions

Recurrence and Induction

Q121 MCQ

Select the correct option: which inequality is proved for $n \geq 3$?

Solution

1. $n = 3$: $4^3 = 64 > 10 = 10$.
2. Assume the inequality holds for $n = k$ with $k \geq 3$.
3. $4^k + 1 > 4(3k + 1)$ after multiplying by the positive base.
4. $4(3k + 1) - (3(k + 1) + 1) = 9k$ is positive for the allowed k .
5. Hence the inequality holds for $k+1$ and therefore for every allowed natural number.

Correct option: B

Final answer

$$4^n > 3n + 1$$

Worked Solutions

Recurrence and Induction

Q122 MCQ

Select the correct option: which inequality is proved for $n \geq 4$?:

Solution

1. $n = 4$: $4^4 = 256 > 11 = 11$.
2. Assume the inequality holds for $n = k$ with $k \geq 4$.
3. $4^k + 1 > 4(2k + 3)$ after multiplying by the positive base.
4. $4(2k + 3) - (2(k + 1) + 3) = 6k + 7$ is positive for the allowed k .
5. Hence the inequality holds for $k+1$ and therefore for every allowed natural number.

Correct option: D

Final answer

$$4^n > 2n + 3$$

Worked Solutions

Recurrence and Induction

Q123 MCQ

Select the correct option: which inequality is proved for $n \geq 4$?

Solution

1. $n = 4$: $4^4 = 256 > 17 = 17$.
2. Assume the inequality holds for $n = k$ with $k \geq 4$.
3. $4^k + 1 > 4(4k + 1)$ after multiplying by the positive base.
4. $4(4k + 1) - (4(k + 1) + 1) = 12k - 1$ is positive for the allowed k .
5. Hence the inequality holds for $k+1$ and therefore for every allowed natural number.

Correct option: A

Final answer

$$4^n > 4n + 1$$

Worked Solutions

Recurrence and Induction

Q124 Written

Prove by mathematical induction that $2^n > 3n + 1$ for $n \geq 4$.

Solution

1. $n = 4$: $2^4 = 16 > 13 = 3 \cdot 4 + 1$.
2. Assume the inequality holds for $n = k$ with $k \geq 4$.
3. $2^k + 1 > 2(3k + 1)$ after multiplying by the positive base.
4. $2(3k + 1) - (3(k + 1) + 1) = 3k - 2$ is positive for the allowed k .
5. Hence the inequality holds for $k+1$ and therefore for every allowed natural number.

Final answer

$$2^n > 3n + 1$$

Worked Solutions

Recurrence and Induction

Q125 MCQ

Select the correct option: which inequality is proved for $n \geq 3$?

Solution

1. $n = 3$: $4^3 = 64 > 9 = 9$.
2. Assume the inequality holds for $n = k$ with $k \geq 3$.
3. $4^k + 1 > 4(3k + 0)$ after multiplying by the positive base.
4. $4(3k + 0) - (3(k + 1) + 0) = 3(3k - 1)$ is positive for the allowed k .
5. Hence the inequality holds for $k+1$ and therefore for every allowed natural number.

Correct option: B

Final answer

$$4^n > 3n$$

Worked Solutions

Recurrence and Induction

Q126 Challenge

When $h > 0$ and $n \geq 3$, prove $(1 + h)^n > 1 + nh^2$.

Solution

1. $n = 3$: $(1 + h)^3 = 1 + 3h + 3h^2 + h^3 > 1 + 3h^2$ because $h > 0$ and $3h + h^3 > 0$.
2. Assume $(1 + h)^k > 1 + kh^2$ for $k \geq 3$.
3. Multiplying by the positive factor $1 + h$ gives a lower bound $(1 + h)(1 + kh^2)$.
4. $(1 + h)(1 + kh^2) - [1 + (k + 1)h^2] = h(1 - h + kh^2)$. The quadratic $1 - h + kh^2$ is positive for $k \geq 3$ because its discriminant is $1 - 4k < 0$.
5. Therefore the inequality holds for every $n \geq 3$ and $h > 0$.

Final answer

$$(1 + h)^n > 1 + nh^2$$

Worked Solutions

Recurrence and Induction

Q127 Written

Prove that $n^3 + (n + 1)^3 + (n + 2)^3$ is a multiple of 9 for every natural number n , using mathematical induction.

Solution

1. [1] Base case: $n = 1$, $f(1) = 36 = 9(4)$, so the statement holds.
2. [2] Assume $f(k) = 9m$ for an integer m .
3. [3] The induction difference is $f(k + 1) - f(k) = 9(k^2 + 3k + 3) = 9(k^2 + 3k + 3)$.
4. [4] Therefore $f(k + 1) = 9[m + k^2 + 3k + 3]$ is a multiple of the divisor.
5. The result follows for every natural number by mathematical induction.

Final answer

$$9 \mid [n^3 + (n + 1)^3 + (n + 2)^3]$$

Worked Solutions

Recurrence and Induction

Q128 Written

Prove that $n^3 + 2n$ is a multiple of 3 for every natural number n , using mathematical induction.

Solution

1. [1] Base case: $n = 1$, $f(1) = 3 = 3(1)$, so the statement holds.
2. [2] Assume $f(k) = 3m$ for an integer m .
3. [3] The induction difference is $f(k+1) - f(k) = 3(k^2 + k + 1) = 3(k^2 + k + 1)$.
4. [4] Therefore $f(k+1) = 3[m + k^2 + k + 1]$ is a multiple of the divisor.
5. The result follows for every natural number by mathematical induction.

Final answer

$$3 \mid (n^3 + 2n)$$

Worked Solutions

Recurrence and Induction

Q129 Written

Prove by mathematical induction that $n^2 + n$ is a multiple of 2.

Solution

1. [1] Base case: $n = 1$, $f(1) = 2 = 2(1)$.
2. [2] Assume $f(k) = 2m$ for an integer m .
3. [3] Consecutive-product shift: $f(k+1) - f(k) = 2 \prod_{j=1}^1 (k+0+j)$.
4. [4] Since the remaining 1 consecutive integers contain a multiple of 1 factorial, the difference is a multiple of 2.
5. Thus $f(k+1) = 2[m + \text{an integer}]$ and the result follows by induction.

Final answer

$$2 \mid (n^2 + n)$$

Worked Solutions

Recurrence and Induction

Q130 Written

Prove by mathematical induction that $n^2 + 5n + 8$ is a multiple of 2.

Solution

1. [1] Base case: $n = 1$, $f(1) = 14 = 2(7)$.
2. [2] Assume $f(k) = 2m$ for an integer m .
3. [3] Consecutive-product shift: $f(k+1) - f(k) = 2 \prod_{j=1}^1 (k+2+j)$.
4. [4] Since the remaining 1 consecutive integers contain a multiple of 1 factorial, the difference is a multiple of 2.
5. Thus $f(k+1) = 2[m + \text{an integer}]$ and the result follows by induction.

Final answer

$$2 \mid (n^2 + 5n + 8)$$

Worked Solutions

Recurrence and Induction

Q131 Written

Prove by mathematical induction that $n^2 + 7n + 12$ is a multiple of 2.

Solution

1. [1] Base case: $n = 1$, $f(1) = 20 = 2(10)$.
2. [2] Assume $f(k) = 2m$ for an integer m .
3. [3] Consecutive-product shift: $f(k+1) - f(k) = 2 \prod_{j=1}^1 (k+3+j)$.
4. [4] Since the remaining 1 consecutive integers contain a multiple of 1 factorial, the difference is a multiple of 2.
5. Thus $f(k+1) = 2[m + \text{an integer}]$ and the result follows by induction.

Final answer

$$2 \mid (n^2 + 7n + 12)$$

Worked Solutions

Recurrence and Induction

Q132 Written

Prove by mathematical induction that $n^2 + 11n + 32$ is a multiple of 2.

Solution

1. [1] Base case: $n = 1$, $f(1) = 44 = 2(22)$.
2. [2] Assume $f(k) = 2m$ for an integer m .
3. [3] Consecutive-product shift: $f(k+1) - f(k) = 2 \prod_{j=1}^1 (k+5+j)$.
4. [4] Since the remaining 1 consecutive integers contain a multiple of 1 factorial, the difference is a multiple of 2.
5. Thus $f(k+1) = 2[m + \text{an integer}]$ and the result follows by induction.

Final answer

$$2 \mid (n^2 + 11n + 32)$$

Worked Solutions

Recurrence and Induction

Q133 MCQ

Select the correct option: which positive integer is guaranteed to divide $n^2 + n$ for every natural number n ?:

Solution

1. [1] Base case: $n = 1$, $f(1) = 2 = 2(1)$.
2. [2] Assume $f(k) = 2m$ for an integer m .
3. [3] Consecutive-product shift: $f(k + 1) - f(k) = 2 \prod_{j=1}^1 (k + 0 + j)$.
4. [4] Since the remaining 1 consecutive integers contain a multiple of 1 factorial, the difference is a multiple of 2.
5. Thus $f(k + 1) = 2[m + \text{an integer}]$ and the result follows by induction.

Correct option: A

Final answer

2

Worked Solutions

Recurrence and Induction

Q134 MCQ

Select the correct option: which positive integer is guaranteed to divide $n^2 + 5n + 8$ for every natural number n ?:

Solution

1. [1] Base case: $n = 1$, $f(1) = 14 = 2(7)$.
2. [2] Assume $f(k) = 2m$ for an integer m .
3. [3] Consecutive-product shift: $f(k+1) - f(k) = 2 \prod_{j=1}^1 (k+2+j)$.
4. [4] Since the remaining 1 consecutive integers contain a multiple of 1 factorial, the difference is a multiple of 2.
5. Thus $f(k+1) = 2[m + \text{an integer}]$ and the result follows by induction.

Correct option: D

Final answer

2

Worked Solutions

Recurrence and Induction

Q135 MCQ

Select the correct option: which positive integer is guaranteed to divide $n^2 + 7n + 12$ for every natural number n ?:

Solution

1. [1] Base case: $n = 1$, $f(1) = 20 = 2(10)$.
2. [2] Assume $f(k) = 2m$ for an integer m .
3. [3] Consecutive-product shift: $f(k+1) - f(k) = 2 \prod_{j=1}^1 (k+3+j)$.
4. [4] Since the remaining 1 consecutive integers contain a multiple of 1 factorial, the difference is a multiple of 2.
5. Thus $f(k+1) = 2[m + \text{an integer}]$ and the result follows by induction.

Correct option: C

Final answer

2

Worked Solutions

Recurrence and Induction

Q136 MCQ

Select the correct option: which positive integer is guaranteed to divide $n^2 + 11n + 32$ for every natural number n ?:

Solution

1. [1] Base case: $n = 1$, $f(1) = 44 = 2(22)$.
2. [2] Assume $f(k) = 2m$ for an integer m .
3. [3] Consecutive-product shift: $f(k+1) - f(k) = 2 \prod_{j=1}^1 (k+5+j)$.
4. [4] Since the remaining 1 consecutive integers contain a multiple of 1 factorial, the difference is a multiple of 2.
5. Thus $f(k+1) = 2[m + \text{an integer}]$ and the result follows by induction.

Correct option: C

Final answer

2

Worked Solutions

Recurrence and Induction

Q137 MCQ

Select the correct option: which positive integer is guaranteed to divide $n^3 + 3n^2 + 2n$ for every natural number n ?:

Solution

1. [1] Base case: $n = 1$, $f(1) = 6 = 6(1)$.
2. [2] Assume $f(k) = 6m$ for an integer m .
3. [3] Consecutive-product shift: $f(k+1) - f(k) = 3 \prod_{j=1}^2 (k+0+j)$.
4. [4] Since the remaining 2 consecutive integers contain a multiple of 2 factorial, the difference is a multiple of 6.
5. Thus $f(k+1) = 6[m + \text{an integer}]$ and the result follows by induction.

Correct option: B

Final answer

6

Worked Solutions

Recurrence and Induction

Q138 MCQ

Select the correct option: which positive integer is guaranteed to divide $n^3 + 9n^2 + 26n + 30$ for every natural number n ?:

Solution

1. [1] Base case: $n = 1$, $f(1) = 66 = 6(11)$.
2. [2] Assume $f(k) = 6m$ for an integer m .
3. [3] Consecutive-product shift: $f(k+1) - f(k) = 3 \prod_{j=1}^2 (k+2+j)$.
4. [4] Since the remaining 2 consecutive integers contain a multiple of 2 factorial, the difference is a multiple of 6.
5. Thus $f(k+1) = 6[m + \text{an integer}]$ and the result follows by induction.

Correct option: A

Final answer

6

Worked Solutions

Recurrence and Induction

Q139 MCQ

Select the correct option: which positive integer is guaranteed to divide $n^3 + 12n^2 + 47n + 60$ for every natural number n ?:

Solution

1. [1] Base case: $n = 1$, $f(1) = 120 = 6(20)$.
2. [2] Assume $f(k) = 6m$ for an integer m .
3. [3] Consecutive-product shift: $f(k+1) - f(k) = 3 \prod_{j=1}^2 (k+3+j)$.
4. [4] Since the remaining 2 consecutive integers contain a multiple of 2 factorial, the difference is a multiple of 6.
5. Thus $f(k+1) = 6[m + \text{an integer}]$ and the result follows by induction.

Correct option: D

Final answer

6

Worked Solutions

Recurrence and Induction

Q140 MCQ

Select the correct option: which positive integer is guaranteed to divide $n^3 + 18n^2 + 107n + 216$ for every natural number n ?:

Solution

1. [1] Base case: $n = 1$, $f(1) = 342 = 6(57)$.
2. [2] Assume $f(k) = 6m$ for an integer m .
3. [3] Consecutive-product shift: $f(k+1) - f(k) = 3 \prod_{j=1}^2 (k+5+j)$.
4. [4] Since the remaining 2 consecutive integers contain a multiple of 2 factorial, the difference is a multiple of 6.
5. Thus $f(k+1) = 6[m + \text{an integer}]$ and the result follows by induction.

Correct option: D

Final answer

6

Worked Solutions

Recurrence and Induction

Q141 MCQ

Select the correct option: which positive integer is guaranteed to divide $n^2 + n + 2$ for every natural number n ?:

Solution

1. [1] Base case: $n = 1$, $f(1) = 4 = 2(2)$.
2. [2] Assume $f(k) = 2m$ for an integer m .
3. [3] Consecutive-product shift: $f(k+1) - f(k) = 2 \prod_{j=1}^1 (k+0+j)$.
4. [4] Since the remaining 1 consecutive integers contain a multiple of 1 factorial, the difference is a multiple of 2.
5. Thus $f(k+1) = 2[m + \text{an integer}]$ and the result follows by induction.

Correct option: D

Final answer

2

Worked Solutions

Recurrence and Induction

Q142 Written

Prove by mathematical induction that $2n^3 - 3n^2 + n$ is a multiple of 6.

Solution

1. [1] Base case: $n = 1$, $f(1) = 0 = 6(0)$, so the statement holds.
2. [2] Assume $f(k) = 6m$ for an integer m .
3. [3] The induction difference is $f(k+1) - f(k) = 6k^2 = 6(k^2)$.
4. [4] Therefore $f(k+1) = 6[m + k^2]$ is a multiple of the divisor.
5. The result follows for every natural number by mathematical induction.

Final answer

$$6 \mid (2n^3 - 3n^2 + n)$$

Worked Solutions

Recurrence and Induction

Q143 Written

Prove by mathematical induction that $4n^3 - n$ is a multiple of 3.

Solution

1. [1] Base case: $n = 1$, $f(1) = 3 = 3(1)$, so the statement holds.
2. [2] Assume $f(k) = 3m$ for an integer m .
3. [3] The induction difference is $f(k+1) - f(k) = 3(2k+1)^2 = 3((2k+1)^2)$.
4. [4] Therefore $f(k+1) = 3[m + (2k+1)^2]$ is a multiple of the divisor.
5. The result follows for every natural number by mathematical induction.

Final answer

$$3 \mid (4n^3 - n)$$

Worked Solutions

Recurrence and Induction

Q144 Challenge

For any integer $n \geq 1$, prove that $(n + 1)^n - 1$ is a multiple of n^2 .

Solution

1. Use the binomial theorem on $(n + 1)^n = (1 + n)^n$.
2. $(1 + n)^n - 1 = \binom{n}{1}n + \binom{n}{2}n^2 + \cdots + n^n$.
3. $\binom{n}{1}n = n^2$, and every remaining term also contains n^2 .
4. $(n + 1)^n - 1 = n^2[1 + \binom{n}{2} + \binom{n}{3}n + \cdots + n^{n-2}]$. The bracket is an integer for every integer $n \geq 1$.
5. Hence the expression is a multiple of n^2 .

Final answer

$$n^2 \mid [(n + 1)^n - 1]$$

Worked Solutions

Recurrence and Induction

Q145 Written

For the sequence determined by $a_1 = 3$ and $a_{n+1} = \frac{a_n^2 - 1}{n+1}$, find a_2, a_3, a_4 and estimate the general term.

Solution

1. $a_2 = 4, \quad a_3 = 5, \quad a_4 = 6.$
2. The pattern suggests $a_n = n + 2.$

Final answer

$$a_2 = 4, \quad a_3 = 5, \quad a_4 = 6; \quad a_n = n + 2$$

Q146 Written

For the sequence determined by $a_1 = -1$ and $a_{n+1} = a_n^2 + 2na_n - 2$, find a_2, a_3, a_4 and estimate the general term.

Solution

1. $a_2 = -3, \quad a_3 = -5, \quad a_4 = -7.$
2. The pattern suggests $a_n = 1 - 2n.$

Final answer

$$a_2 = -3, \quad a_3 = -5, \quad a_4 = -7; \quad a_n = 1 - 2n$$

Worked Solutions

Recurrence and Induction

Q147 Written

For the sequence determined by $a_1 = 1$ and $a_{n+1} = a_n^2 - n^2 a_n + (n+1)^2$, find a_2, a_3, a_4 and estimate the general term.

Solution

1. $a_2 = 4, \quad a_3 = 9, \quad a_4 = 16.$
2. The pattern suggests $a_n = n^2.$

Final answer

$$a_2 = 4, \quad a_3 = 9, \quad a_4 = 16; \quad a_n = n^2$$

Q148 Written

For the sequence determined by $a_1 = 1$ and $a_{n+1} = \frac{a_n - 4}{a_n - 3}$, find a_2, a_3, a_4 and estimate the general term.

Solution

1. $a_2 = \frac{3}{2}, \quad a_3 = \frac{5}{3}, \quad a_4 = \frac{7}{4}.$
2. The pattern suggests $a_n = \frac{2n-1}{n}.$

Final answer

$$a_2 = \frac{3}{2}, \quad a_3 = \frac{5}{3}, \quad a_4 = \frac{7}{4}; \quad a_n = \frac{2n-1}{n}$$

Worked Solutions

Recurrence and Induction

Q149 Written

For a sequence with $a_1 = 3$ and $a_{n+1} = \frac{a_n^2 - 1}{n+1}$, prove by mathematical induction that $a_n = n + 2$.

Solution

1. [1] Base case: $a_1 = 3 = 3$.
2. [2] Assume $a_k = k + 2$ for some positive integer k .
3. [3] Substitute into $a_{k+1} = \frac{a_k^2 - 1}{k+1}$: $a_{k+1} = k + 3$.
4. [4] This is $k + 3$, the proposed formula with $n = k + 1$.
5. Therefore the formula holds for every natural number by mathematical induction.

Final answer

$$a_n = n + 2$$

Worked Solutions

Recurrence and Induction

Q150 MCQ

Select the correct option: for $a_1 = 1$ and $a_{n+1} = a_n + 2$, the general term is:

Solution

1. [1] Base case: $a_1 = 1 = 1$.
2. [2] Assume $a_k = 2k - 1$ for some positive integer k .
3. [3] Substitute into $a_{k+1} = a_k + 2$: $a_{k+1} = 2k + 1$.
4. [4] This is $2k + 1$, the proposed formula with $n = k + 1$.
5. Therefore the formula holds for every natural number by mathematical induction.

Correct option: D

Final answer

$$a_n = 2n - 1$$

Worked Solutions

Recurrence and Induction

Q151 MCQ

Select the correct option: for $a_1 = 2$ and $a_{n+1} = a_n + 2$, the general term is:

Solution

1. [1] Base case: $a_1 = 2 = 2$.
2. [2] Assume $a_k = 2k$ for some positive integer k .
3. [3] Substitute into $a_{k+1} = a_k + 2$: $a_{k+1} = 2(k + 1)$.
4. [4] This is $2(k + 1)$, the proposed formula with $n = k + 1$.
5. Therefore the formula holds for every natural number by mathematical induction.

Correct option: C

Final answer

$$a_n = 2n$$

Worked Solutions

Recurrence and Induction

Q152 MCQ

Select the correct option: for $a_1 = 3$ and $a_{n+1} = a_n + 2$, the general term is:

Solution

1. [1] Base case: $a_1 = 3 = 3$.
2. [2] Assume $a_k = 2k + 1$ for some positive integer k .
3. [3] Substitute into $a_{k+1} = a_k + 2$: $a_{k+1} = 2k + 3$.
4. [4] This is $2k + 3$, the proposed formula with $n = k + 1$.
5. Therefore the formula holds for every natural number by mathematical induction.

Correct option: B

Final answer

$$a_n = 2n + 1$$

Worked Solutions

Recurrence and Induction

Q153 MCQ

Select the correct option: for $a_1 = 4$ and $a_{n+1} = a_n + 2$, the general term is:

Solution

1. [1] Base case: $a_1 = 4 = 4$.
2. [2] Assume $a_k = 2(k + 1)$ for some positive integer k .
3. [3] Substitute into $a_{k+1} = a_k + 2$: $a_{k+1} = 2(k + 2)$.
4. [4] This is $2(k + 2)$, the proposed formula with $n = k + 1$.
5. Therefore the formula holds for every natural number by mathematical induction.

Correct option: A

Final answer

$$a_n = 2(n + 1)$$

Worked Solutions

Recurrence and Induction

Q154 MCQ

Select the correct option: for $a_1 = 1$ and $a_{n+1} = a_n + 3$, the general term is:

Solution

1. [1] Base case: $a_1 = 1 = 1$.
2. [2] Assume $a_k = 3k - 2$ for some positive integer k .
3. [3] Substitute into $a_{k+1} = a_k + 3$: $a_{k+1} = 3k + 1$.
4. [4] This is $3k + 1$, the proposed formula with $n = k + 1$.
5. Therefore the formula holds for every natural number by mathematical induction.

Correct option: D

Final answer

$$a_n = 3n - 2$$

Worked Solutions

Recurrence and Induction

Q155 MCQ

Select the correct option: for $a_1 = 2$ and $a_{n+1} = a_n + 3$, the general term is:

Solution

1. [1] Base case: $a_1 = 2 = 2$.
2. [2] Assume $a_k = 3k - 1$ for some positive integer k .
3. [3] Substitute into $a_{k+1} = a_k + 3$: $a_{k+1} = 3k + 2$.
4. [4] This is $3k + 2$, the proposed formula with $n = k + 1$.
5. Therefore the formula holds for every natural number by mathematical induction.

Correct option: C

Final answer

$$a_n = 3n - 1$$

Worked Solutions

Recurrence and Induction

Q156 MCQ

Select the correct option: for $a_1 = 3$ and $a_{n+1} = a_n + 3$, the general term is:

Solution

1. [1] Base case: $a_1 = 3 = 3$.
2. [2] Assume $a_k = 3k$ for some positive integer k .
3. [3] Substitute into $a_{k+1} = a_k + 3$: $a_{k+1} = 3(k + 1)$.
4. [4] This is $3(k + 1)$, the proposed formula with $n = k + 1$.
5. Therefore the formula holds for every natural number by mathematical induction.

Correct option: B

Final answer

$$a_n = 3n$$

Worked Solutions

Recurrence and Induction

Q157 MCQ

Select the correct option: for $a_1 = 4$ and $a_{n+1} = a_n + 3$, the general term is:

Solution

1. [1] Base case: $a_1 = 4 = 4$.
2. [2] Assume $a_k = 3k + 1$ for some positive integer k .
3. [3] Substitute into $a_{k+1} = a_k + 3$: $a_{k+1} = 3k + 4$.
4. [4] This is $3k + 4$, the proposed formula with $n = k + 1$.
5. Therefore the formula holds for every natural number by mathematical induction.

Correct option: A

Final answer

$$a_n = 3n + 1$$

Worked Solutions

Recurrence and Induction

Q158 Written

For a sequence with $a_1 = 1$ and $a_{n+1} = a_n^2 - n^2 a_n + (n+1)^2$, prove by mathematical induction that $a_n = n^2$.

Solution

1. [1] Base case: $a_1 = 1 = 1$.
2. [2] Assume $a_k = k^2$ for some positive integer k .
3. [3] Substitute into $a_{k+1} = a_k^2 - k^2 a_k + (k+1)^2$: $a_{k+1} = (k+1)^2$.
4. [4] This is $(k+1)^2$, the proposed formula with $n = k+1$.
5. Therefore the formula holds for every natural number by mathematical induction.

Final answer

$$a_n = n^2$$

Worked Solutions

Recurrence and Induction

Q159 Written

For a sequence with $a_1 = -1$ and $a_{n+1} = a_n^2 + 2na_n - 2$, prove by mathematical induction that $a_n = 1 - 2n$.

Solution

1. [1] Base case: $a_1 = -1 = -1$.
2. [2] Assume $a_k = 1 - 2k$ for some positive integer k .
3. [3] Substitute into $a_{k+1} = a_k^2 + 2ka_k - 2$: $a_{k+1} = -2k - 1$.
4. [4] This is $-2k - 1$, the proposed formula with $n = k + 1$.
5. Therefore the formula holds for every natural number by mathematical induction.

Final answer

$$a_n = 1 - 2n$$

Worked Solutions

Recurrence and Induction

Q160 MCQ

Select the correct option: for $a_1 = 1$ and $a_{n+1} = 2a_n + 0$, the general term is:

Solution

1. [1] Base case: $a_1 = 1 = 1$.
2. [2] Assume $a_k = 2^{k-1}$ for some positive integer k .
3. [3] Substitute into $a_{k+1} = 2a_k + 0$: $a_{k+1} = 2^k$.
4. [4] This is 2^k , the proposed formula with $n = k + 1$.
5. Therefore the formula holds for every natural number by mathematical induction.

Correct option: C

Final answer

$$a_n = 2^{n-1}$$

Worked Solutions

Recurrence and Induction

Q161 Written

For a sequence with $a_1 = 1$ and $a_{n+1} = \frac{a_n - 4}{a_n - 3}$, prove by mathematical induction that $a_n = \frac{2n-1}{n}$.

Solution

1. [1] Base case: $a_1 = 1 = 1$.
2. [2] Assume $a_k = \frac{2k-1}{k}$ for some positive integer k .
3. [3] Substitute into $a_{k+1} = \frac{a_k - 4}{a_k - 3}$: $a_{k+1} = \frac{2k+1}{k+1}$.
4. [4] This is $\frac{2k+1}{k+1}$, the proposed formula with $n = k + 1$.
5. Therefore the formula holds for every natural number by mathematical induction.

Final answer

$$a_n = \frac{2n-1}{n}$$

Worked Solutions

Recurrence and Induction

Q162 Challenge

Three numbers A, B, C in that order form a geometric sequence with common ratio r . If $A, 2B, C$ in that order form an arithmetic sequence, determine the possible values of r .

Solution

1. Let the geometric sequence be A, Ar, Ar^2 .
2. Since $A, 2B, C$ is arithmetic, its middle term is the average of the outer terms: $4B = A + C$.
3. Substitute $B = Ar, C = Ar^2$ and cancel the non-zero scale factor A : $r^2 - 4r + 1 = 0$.
4. $r = \frac{4 \pm \sqrt{16-4}}{2} = 2 \pm \sqrt{3}$.

Final answer

$$r = 2 \pm \sqrt{3}$$

Worked Solutions

Recurrence and Induction

Quiz MCQ 1 **Concept Quiz · MCQ**

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 1a_n + (4)$. The general term is:

Solution

1. The rule adds the difference term, so it is arithmetic or a finite sum.
2. $a_n = 2(2n - 1)$.

Correct option: D

Final answer

$$a_n = 2(2n - 1)$$

Worked Solutions

Recurrence and Induction

Quiz MCQ 2 Concept Quiz · MCQ

Select the correct option: If $a_1 = 2$ and $a_{n+1} = -2a_n + (3)$, the general term is:

Solution

1. $c = 1$ from $c = -2c + (3)$.
2. $a_{n+1} - c = -2(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = -\frac{(-2)^n - 2}{2}$.
5. Substitute $n = 1$ to verify the given first term.

Correct option: A

Final answer

$$a_n = -\frac{(-2)^n - 2}{2}$$

Worked Solutions

Recurrence and Induction

Quiz MCQ 3 Concept Quiz · MCQ

Select the correct option: which inequality is proved for $n \geq 4$?

Solution

1. $n = 4$: $2^4 = 16 > 13 = 13$.
2. Assume the inequality holds for $n = k$ with $k \geq 4$.
3. $2^k + 1 > 2(3k + 1)$ after multiplying by the positive base.
4. $2(3k + 1) - (3(k + 1) + 1) = 3k - 2$ is positive for the allowed k .
5. Hence the inequality holds for $k+1$ and therefore for every allowed natural number.

Correct option: B

Final answer

$$2^n > 3n + 1$$

Worked Solutions

Recurrence and Induction

Quiz MCQ 4 **Concept Quiz · MCQ**

Select the correct option: for $a_1 = 4$ and $a_{n+1} = a_n + 2$, the general term is:

Solution

1. [1] Base case: $a_1 = 4 = 4$.
2. [2] Assume $a_k = 2(k + 1)$ for some positive integer k .
3. [3] Substitute into $a_{k+1} = a_k + 2$: $a_{k+1} = 2(k + 2)$.
4. [4] This is $2(k + 2)$, the proposed formula with $n = k + 1$.
5. Therefore the formula holds for every natural number by mathematical induction.

Correct option: C

Final answer

$$a_n = 2(n + 1)$$

Worked Solutions

Recurrence and Induction

Quiz WRITTEN 5 Concept Quiz · Written

For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 1a_n + (5)$. Find the general term.

Solution

1. The rule adds the difference term, so it is arithmetic or a finite sum.
2. $a_n = 5n - 3$.

Final answer

$$a_n = 5n - 3$$

Worked Solutions

Recurrence and Induction

Quiz WRITTEN 6 Concept Quiz · Written

Find the general term of $\{a_n\}$ given $a_1 = 1$ and $a_{n+1} = 5a_n + (8)$.

Solution

1. $c = -2$ from $c = 5c + (8)$.
2. $a_{n+1} - c = 5(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = \frac{3 \cdot 5^n - 10}{5}$.
5. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = \frac{3 \cdot 5^n - 10}{5}$$

Worked Solutions

Recurrence and Induction

Quiz WRITTEN 7 Concept Quiz · Written

Prove by mathematical induction that $2^n > 4n + 1$ for $n \geq 4$.

Solution

1. $n = 4$: $2^4 = 16 > 17 = 4 \cdot 4 + 1$.
2. Assume the inequality holds for $n = k$ with $k \geq 4$.
3. $2^k + 1 > 2(4k + 1)$ after multiplying by the positive base.
4. $2(4k + 1) - (4(k + 1) + 1) = 4k - 3$ is positive for the allowed k .
5. Hence the inequality holds for $k+1$ and therefore for every allowed natural number.

Final answer

$$2^n > 4n + 1$$

Worked Solutions

Recurrence and Induction

Quiz WRITTEN 8 Concept Quiz · Written

For a sequence with $a_1 = 5$ and $a_{n+1} = a_n + 1(2n + 1) + 2$, prove by mathematical induction that $a_n = n^2 + 2n + 2$.

Solution

1. [1] Base case: $a_1 = 5 = 5$.
2. [2] Assume $a_k = k^2 + 2k + 2$ for some positive integer k .
3. [3] Substitute into $a_{k+1} = a_k + 1(2k + 1) + 2$: $a_{k+1} = k^2 + 4k + 5$.
4. [4] This is $k^2 + 4k + 5$, the proposed formula with $n = k + 1$.
5. Therefore the formula holds for every natural number by mathematical induction.

Final answer

$$a_n = n^2 + 2n + 2$$

Answer key

Use the question number shown on each page.

Q001 $a_2 = 8, a_3 = 26, a_4 = 80$

Q002 $a_2 = -9, a_3 = -15, a_4 = -27$

Q003 $a_2 = -7, a_3 = -7, a_4 = -4$

Q004 $a_2 = -3, a_3 = -5, a_4 = -8$

Q005 $a_2 = 4, a_3 = 8, a_4 = 14$

Q006 $a_2 = 1, a_3 = 10, a_4 = 37$

Q007 B

Q008 D

Q009 A

Q010 B

Q011 D

Q012 A

Q013 C

Q014 D

Q015 C

Q016 D

Q017 B

Q018 C

Q019 $a_n = 5n - 6$

Q020 $a_n = 3n - 2$

Q021 $a_n = -2(n - 3)$

Q022 $a_n = 6n - 5$

Q023 $a_n = 8 - 5n$

Q024 $a_n = 3n - 5$

Q025 $a_n = -4(n - 2)$

Q026 D

Q027 A

Q028 A

Q029 A

Q030 $a_n = 2 \cdot 3^{n-1}$

Q031 $a_n = 3 \cdot 5^{n-1}$

Q032 $a_n = 4 \cdot 3^{n-1}$

Q033 $a_n = 2^{2n-2}$

Q034 B

Q035 B

Q036 D

Q037 A

Q038 $a_n = \frac{3(-2)^n}{2}$

Q039 $a_n = 2(-5)^{n-1}$

Q040 $a_n = \frac{(-4)^n}{2}$

Q041 B

Q042 C

Q043 B

Q044 C

Q045 $a_2 = 5, a_3 = 13, a_4 = 36$

Q046 $a_2 = 3, a_3 = -1, a_4 = 7$

Q047 $a_2 = 8, a_3 = 22, a_4 = 63$

Q048 $a_n = -\frac{4 \cdot 2^n - 3 \cdot 3^n - 1}{2}$

Q049 $a_n = (n - 2)^2$

Q050 $a_n = \frac{3n^2 - 3n + 2}{2}$

Q051 $a_n = n^2 - n + 2$

Q052 $a_n = \frac{2n^3 - 3n^2 + n + 4}{2}$

Q053 $a_n = \frac{3n^2 - n + 2}{2}$

Q054 $a_n = \frac{4n^3 - 3n^2 - n + 12}{6}$

Q055 C

Q056 A

Q057 C

Q058 D

Q059 C

Q060 A

Q061 D

Q062 A

Q063 B

Q064 A

Q065 $a_n = \frac{2n^3 - 3n^2 + n + 12}{6}$

Q066 A

Q067 $a_n = 2^n - 1$

Q068 $a_n = -\frac{3^n - 7}{2}$

Q069 $a_n = -\frac{4^n - 16}{3}$

Q070 $a_n = -\frac{n^2 - n - 200}{50}$

Q071 $a_n = 3^n - 2$

Q072 $a_n = \frac{3 \cdot 5^n - 10}{5}$

Q073 $a_n = \frac{2^n + 2}{2}$

Q074 $a_n = -\frac{4^n + 2}{2}$

Q075 $a_n = \frac{2 \cdot 3^n + 3}{3}$

Q076 $a_n = (-3)^n + 1$

Q077 $a_n = 3 \cdot 3^{-n} (3^n - 2)$

Q078 $a_n = \frac{3 \cdot 3^{-n} (7 \cdot 2^n - 2 \cdot 3^n)}{4}$

Q079 $a_n = -\frac{3(-2)^n - 2}{2}$

Q080 B

Q081 C

Q082 D

Q083 B

Q084 C

Q085 D

Q086 B

Q087 C

Q088 D

Q089 D

Q090 $c = 3, a_n = -\frac{2^{2n} - 12}{4}, a_5 = -253$

Q091 $\frac{n(n+1)}{2}$

Q092 $\frac{n(2n-1)(2n+1)}{3}$

Q093 $\frac{n(n+1)(n+2)}{3}$

Q094 A

Q095 B	Q113 $3^n > 5n + 1$	Q133 A	Q152 B
Q096 C	Q114 $3^n > 5n + 2$	Q134 D	Q153 A
Q097 B	Q115 $4^n > 2n + 3$	Q135 C	Q154 D
Q098 C	Q116 D	Q136 C	Q155 C
Q099 A	Q117 A	Q137 B	Q156 B
Q100 C	Q118 B	Q138 A	Q157 A
Q101 A	Q119 D	Q139 D	Q158 $a_n = n^2$
Q102 B	Q120 A	Q140 D	Q159 $a_n = 1 - 2n$
Q103 n^2	Q121 B	Q141 D	Q160 C
Q104 $\frac{n(n+1)(n+2)}{3}$	Q122 D	Q142 $6 \mid (2n^3 - 3n^2 + n)$	Q161 $a_n = \frac{2n-1}{n}$
Q105 $\frac{n(n+1)(2n+7)}{6}$	Q123 A	Q143 $3 \mid (4n^3 - n)$	Q162 $r = 2 \pm \sqrt{3}$
Q106 $4^n - 1$	Q124 $2^n > 3n + 1$	Q144 $n^2 \mid [(n+1)^n - 1]$	Quiz MCQ 1 D
Q107 $\frac{10 \cdot 10^n - 1}{9}$	Q125 B	Q145 $a_2 = 4, a_3 = 5, a_4 = 6; a_n = n + 2$	Quiz MCQ 2 A
Q108 $\frac{n(n^2 + 3n + 5)}{3}$	Q126 $(1+h)^n > 1 + nh^2$	Q146 $a_2 = -3, a_3 = -5, a_4 = -7; a_n = 1 - 2n$	Quiz MCQ 3 B
Q109 $2^n > 3n$	Q127 $9 \mid [n^3 + (n+1)^3 + (n+2)^3]$	Q147 $a_2 = 4, a_3 = 9, a_4 = 16; a_n = n^2$	Quiz MCQ 4 C
Q110 $3^n > 5n + 1$	Q128 $3 \mid (n^3 + 2n)$	Q148 $a_2 = \frac{3}{2}, a_3 = \frac{5}{3}, a_4 = \frac{7}{4}; a_n = \frac{2n-1}{n}$	Quiz WRITTEN 5 $a_n = 5n - 3$
Q111 $3^n > 2n$	Q129 $2 \mid (n^2 + n)$	Q149 $a_n = n + 2$	Quiz WRITTEN 6 $a_n = \frac{3 \cdot 5^n - 10}{5}$
Q112 $2^n > 1n + 2$	Q130 $2 \mid (n^2 + 5n + 8)$	Q150 D	Quiz WRITTEN 7 $2^n > 4n + 1$
	Q131 $2 \mid (n^2 + 7n + 12)$	Q151 C	Quiz WRITTEN 8 $a_n = n^2 + 2n + 2$
	Q132 $2 \mid (n^2 + 11n + 32)$		